Contents

Thesis

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June 10, 2026

Abstract

We study the stationary queue-length behaviour of a 2-class $M/M/1$ queueing system with non-preemptive priority with two added mechanisms: *jockeying*, in which a custonmer in line switches to the other queue; and *abandonment*, in which a waiting customer may abandon the system entirely. While the former preserves the total number of customers in the system, it is not the case for the latter. Since combining both jockeying and abandonments leads to a very complex model (Model-X), we analyse a series of less mathematically involved submodels: Model-A (baseline), Model-B (bidirectional jockeying), Model-B_2 (one-way jockeying Class-1 to Class-2), and Model-C_2 (class-1 abandonment). We further study two *head-of-line* variants, Model-B_2^1 and Model-C_2^1, in which the jockeying or abandonment acts only on the customer at the head of Queue-1; this flat-rate restriction keeps the functional equation algebraic and yields rational generating functions in closed form.

The derivation of an expression for the probability generation function (PGF) $P(x, y)$ of each of these models is carried out using purely analytical techniques and, where possible, by means of probabilistic arguments. Model-A, widely studied, reduces to an algebraic expression solved by the kernel method; Model-B is left open, with a Volterra-type integral equation when it comes to finding the boundary for the queue length of Class-2 that falls beyond the scope of this thesis; However, Model-B_2 reduces to a first-order linear ODE in x, where we use the analyticity of the PGF to derive the boundary function of Class-2 at an interior point; and Model-C_2 reduces to another first-order linear ODE in x where we use the boundedness of our PGF at the boundary of its domain to determine the boundary function.

Lastly, the findings are validated numerically through simulations, showing how the models reduce to the baseline Model-A as the jockeying and/or abandonment parameters approximate 0.

1 Introduction

Motivation

Priority queues arise naturally in a wide range of contexts where a common server must satisfy heterogeneous demand. The primary motivation for this thesis comes from the context of healthcare operations. In particular, the management of waiting lists for access to elderly care facilities, where the medical severity dictates a user's —hereafter referred to as *customer*— priority class. In this context, prolonged waiting times may alter the system's dynamics, given that the lack of appropriate care may deteriorate the health condition of a waiting potential user, leading to a change in their priority status (jockeying) or leaving the system (abandonment), be it due to seeking alternative care or passing away. Despite the initial motivation, this work aims to address the problem from a general modeling, which may be applicable to a broad class of systems: characterising the long-run behaviour of queueing systems when phenomenons such as jockeying and/or abandonments play a role.

In this thesis, an analysis of a 2-class $M/M/1$ queueing system with non-preemptive priority is carried out. In short, this means that arrivals from each priority class are driven by a Poisson process with arrival rates λ_1 and λ_2. exponentially distributed service times at a rate μ —common to both priority classes—, and a single server that completes the current service despite the arrivals of higher-priority customers; with the addition of the mechanisms of jockeying and abandonments. The model and its mechanisms are described in more depth in Section 4.

The interplay of these two mechanisms within a priority discipline and how they affect the difficulty of obtaining a closed-form expression for their Probability Generating Function (PGF) are the pivotal objects of this thesis.

The analytical challenge

The combination of jockeying and abandonment in full generality defines Model-X. The bivariate PGF satisfies a functional equation that involves terms of three kinds: algebraic coefficients, a partial derivative term in x —from jockeying and abandonments in Class-1— and a partial derivative term in y —from the same mechanisms in Class-2. When derivative terms of the two kinds are present, the equation belongs to a family of integro-differential equations for which no close-form solution is known, and it becomes a substantially challenging problem.

The strategy adopted here is to explore less complex models that require a use of less advanced mathematical tools by setting some of the jockeying and/or abandonment parameters equal to zero. The table below (Table 1) lists the models studied in this work and provide some insight of the challenge they imply:

Table 1: Models studied in this thesis, with the active mechanism, the resulting functional-equation type, and the solution status. The head-of-line variants B_2^1 and C_2^1 restrict the mechanism to the customer at the head of Queue 1, which collapses the derivative term to an algebraic one and renders the model solvable in closed form.

Model	**Active mechanism**	**Equation type**	**Status**
A	none	algebraic PGF equation	solved
B	bidirectional jockeying	PDE, Volterra family	open
B_2	one-way jockeying $(1 \to 2)$	ODE in x, Kummer	solved
B_2^1	one-way jockeying, head-of-line	algebraic PGF equation	solved
C_2	class-1 abandonment	ODE in x, Kummer	solved
C_2^1	class-1 abandonment, head-of-line	algebraic PGF equation	solved

As one can see, some models —in particular, Model-X and Model-B— have been left incomplete, as they involve derivative terms in y, and some other models such a potential Model-B_1, Model-C or Model-C_1 have not even been considered for the same reason: they all reduce to a Volterra integral equation that falls beyond the scope of the work carried out in this thesis. Here Model-B serves as an example of how this sort of problems reduce to such an integral equation, whereas the other models correspond to problems we can solve. The last two entries, Models B_2^1 and C_2^1 (Sections 10 and 11), are *head-of-line* simplifications in which only the customer at the head of Queue 1 may jockey or abandon: the mechanism then acts at the constant rate $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ or $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$ rather than the length-proportional $\gamma_1 n_1$ or $\theta_1 n_1$, so the fundamental equation stays algebraic and is solved by the Kernel Method exactly as in Model-A.

Methods

Three complementary proof strategies are used throughout.

Kernel method / ODE with integrating factor. For Model-A, which does not involve derivative terms whatsoever, the algebraic coefficient of $P(x, y)$ vanishes on a curve (i.e. the *kernel*). Evaluating the equation there lets us determine the rest of the unknowns. Model-B_2 and Model-C_2, which do involve derivative terms in x, reduce to a first-order linear ODE in x and admit an explicit integrating factor. The unknowns there are obtained my means of a regilarity argument at a singularity point of the integrating factor. Both strategies are here explained as one, as the underlying idea is the same. One could say that the latter is the extension of the former when the unknowns are independent from the derivative terms and can be taken outside of the integral.

Probabilistic argument via Pollaczek–Khinchine and the effective service time. For Model-A and Model-C_2, we have Poisson arrivals for Class-2 customers —and therefore the PASTA property holds—, which allows us to utilize the Pollaczek–Khinchine formula, combined with the *effective service time* seen by Class-2 customers: an $M/G/1$ queueing system where the general service times are equivalent to the busy period of the Class-1 subsystem, which is an $M/M/1$ queueing system for Model-A and an $M/M/1+M$ system for Model-B. In both cases, the probabilistic argument confirms the results obtained analytically, allowing to determine the unknown in an independent way. Since jockeying destroys the Poisson structure, this approach is not valid for the other models we studied.

Partial PGF (PPGF) in the alternative state space $\widetilde{S}$. For the models with no abandonments —Model-A, Model-B and Model-B_2—, we create an alternative space state that looks at these models by taking the number of Class-2 customers —difficult distribution— and the total amount of customers —easy, since the total number of customers is preserved and is basically a regular $M/M/1$ queueing system. This yields a PPGF with a non-homogeneous term. For Model-A, a recursion argument derived by Cohen [Coh56] applied to the regular state space. We explore, without much success, the challenges of such an approach on this alternative space and evaluate whether a rough approximation is feasible using Cohen's trick.

Structure of the thesis

Section 2 describes the relevant literature. Section 3 collects the probabilistic and analytic background used throughout this work. Section 4 defines Model-X and derives the general balance equations and the fundamental PGF equations, both for the regular and alternative state space. Sections 6–9 analyse Models A, B, B_2, and C_2 in turn. Section 12 compares the three solved models. Section 13 presents the numerical validation and performance metrics. Sections 10 and 11 introduce and solve the head-of-line variants, Models B_2^1 and C_2^1.

2 Literature

The study of two-class priority queues with departure mechanisms sits at the intersection of three well-developed streams: exact PGF analysis of non-preemptive priority queues, the theory of queues with customer impatience (abandonment), and the theory of jockeying or priority changes. This thesis contributes an exact, single-server treatment that combines jockeying and abandonment within a unified PGF framework, filling a gap that the multi-server and approximate literature has so far left open.

Non-preemptive priority queues: exact queue-length analysis

Priority queues with abandonment

Jockeying and priority changes

Multi-server extensions and performance design

Position of this thesis

Table 2: Literature overview: models and mechanisms. ✓ = present, – = absent, ≈ = approximate only.

Reference	Servers	Non-preem.	Jockeying	Aband.	Queue-len.	Exact
Cohen (1956)	1	✓	–	–	✓	✓
de Waal (1992)	1	✓	✓	–	–	≈
Jouini & Roubos (2014)	c	✓	–	✓	–	✓
Wang et al. (2015)	c	–	–	–	✓	✓
Stanford et al. (2014)	1	✓	✓	–	–	✓
Zuk & Kirszenblat (2023)	c	✓	–	–	✓	✓
Baron et al. (2025)	c	✓	–	–	–	✓
Hu, Chan & Dong (2022)	c	✓	✓	✓	–	≈
Shmelev & Zychlinski (2025)	c	✓	–	✓	–	≈
This thesis (Model-A)	**1**	✓	–	–	✓	✓
This thesis (Model-B_2)	**1**	✓	✓	–	✓	✓
This thesis (Model-C_2)	**1**	✓	–	✓	✓	✓

3 Preliminaries

This section collects background material referenced throughout the derivations. Introducing these building blocks here allows the subsequent arguments to remain focused on their primary objectives.

The central mathematical object in our derivations will be the (P)PGF of the model in question, as well as the idle probability of the system for some corollaries.

3.1 $M/M/1$ queue

The model described here is the canonical $M/M/1$ queue, well studied in multiple textbooks and manuals and considered the absolute foundation of queueing theory. Here, we will use [AR02] as our reference, but there is a lot of good quality literature and sources about it. Customers arrive at the system according to a Poisson process —i.e., the inter-arrival times are exponentially distributed with parameter λ—, service times are exponentially distributed with parameter μ, and a single server operates in first-come-first-serve (FCFS) —also known as first-in-first-out (FIFO)— order. The system is stable if and only if the system's load $\rho = \lambda/\mu < 1$.

Let π_n be the limiting probability of finding n customers in the system. Under stability, the detailed balance equations $\lambda\,\pi_n = \mu\,\pi_{n+1}$ $(n \geq 0)$ together with the normalisation $\sum_{n=0}^{\infty} \pi_n = 1$ give us the geometric steady-state distribution of the number of customers in the system:

$$\pi_n = (1-\rho)\,\rho^n, \quad n \geq 0. \tag{1}$$

By the PASTA property (§3.2), this also corresponds to the distribution seen by the arriving customers. Let L be the random variable that tells us how many customers—including the one that might be in service—are in the system. The expected number of customers in the system is the following:

$$\mathbb{E}[L] = \frac{\rho}{1-\rho}.$$

From this, the expected sojourn time W—the time that elapses between a customer's arrival and their departure after service completion—is derived by Little's Law ($\mathbb{E}[L] = \lambda\mathbb{E}[W]$):

$$\mathbb{E}[W] = \frac{\mathbb{E}[L]}{\lambda} = \frac{1/\mu}{1-\rho}.$$

Another concept, more rare but highly relevant for the content that will be discussed in this thesis, is the Busy Period (BP) of the system, i.e., the time elapsed between the arrival of a customer to an idle system and the departure of a customer who leaves the system empty again after service completion. The Laplace-Stieltjes Transform (LST) and the expectation of this system's busy period are known:

$$\widetilde{BP}(s) = \frac{(\mu+\lambda+s) - \sqrt{(\mu+\lambda+s)^2 - 4\mu\lambda}}{2\lambda}, \tag{2}$$

$$\mathbb{E}[BP] = \frac{1}{\mu(1-\rho)}. \tag{3}$$

3.2 Non-preemptive priority and the PASTA property

Throughout this thesis the two customer classes are served under a *non-preemptive* (head-of-line) priority discipline: whenever the server becomes free it admits a waiting class-1 customer if any are present, and a class-2 customer only otherwise, but a customer already in service is *never* interrupted—a class-1 arrival that finds a class-2 customer in service waits until that service completes. Priority therefore governs only the order in which the queues are drained, not the service in progress. Because the service rate μ is common to both classes, the *memorylessness* of the exponential service time makes the residual service of the customer in progress $\mathrm{Exp}(\mu)$ regardless of its class; hence the class of the in-service customer does not influence the future evolution of the queue lengths, which is why a single in-service customer of either class suffices to mark the server busy.

We also invoke repeatedly the *PASTA* property (*Poisson Arrivals See Time Averages*) [AR02]: if a system is fed by a Poisson arrival process that satisfies the *lack-of-anticipation assumption* (at each instant the future increments of the arrival process are independent of the current system state), then the long-run fraction of arrivals that find the system in a given state equals the stationary probability of that state, so the distribution seen by an arriving customer coincides with the time-stationary one. Both hypotheses hold here: the exogenous class-i arrivals are independent Poisson processes at rates λ_i, and they are independent of the system's internal exponential service, jockeying and abandonment clocks, so they cannot anticipate the future. By Poisson *superposition* the merged arrival stream is itself Poisson at rate $\lambda = \lambda_1 + \lambda_2$, so PASTA applies to the aggregate state; and since each class stream is an independent Poisson process—equivalently, an i.i.d. λ_i/λ *thinning* of the merged stream—PASTA applies to each class separately.

3.3 $M/M/1{+}M$ queue (Erlang-A)

Another relevant queueing model for this thesis is the so-called $M/M/1+M$ model, also known as the *Erlang A* model. It's based on the $M/M/1$ queue —Poisson arrivals at a rate λ and exponentially distributed waiting times with parameter μ-, but adds customer impatience to it: each waiting customer can independently abandon at an exponentially distributed waiting time with parameter θ. The key difference of this model with respect to the $M/M/1$ is that the $M/M/1+M$ does not require $\rho = \lambda/\mu \leq 1$ to be stable: it is positive recurrent for all $\lambda, \mu, \theta > 0$.

The steady state distribution follows, naturally, from its balance equations. At state $n \geq 0$ —for n being the number of customers in the *system*— we have

$$\lambda\, \pi_n = (\mu + n\theta)\, \pi_{n+1}, \qquad n \geq 0. \tag{4}$$

Iterating yields

$$\pi_n = \pi_0\, \frac{(\lambda/\theta)^n}{(\mu/\theta)^{(n)}} \quad n \geq 0; \quad \text{where} \quad a^{(n)} = a(a+1)\cdots(a+n-1) \tag{5}$$

Normalisation gives the non-trivial idle probability $\pi_0 = [{}_1F_1(1;\, \mu/\theta;\, \lambda/\theta)]^{-1}$ (see §3.5 for the Kummer function ${}_1F_1$).

The *Busy Period* of the $M/M/1+M$ defined here is far more complex than the one expressed in the $M/M/1$ queueing system. We record it for the class-1 subsystem used in Model-C_2, taking arrival rate λ_1, service rate μ and abandonment rate θ_1, so that the relevant idle probability is $\pi_0 = [{}_1F_1(1;\, \mu/\theta_1;\, \lambda_1/\theta_1)]^{-1}$. The LST and the expectation are given by

$$\widetilde{B}(s) = \cfrac{\mu}{\lambda_1 + \mu + s - \cfrac{\lambda_1(\mu+\theta_1)}{\lambda_1 + \mu + \theta_1 + s - \cfrac{\lambda_1(\mu + 2\theta_1)}{\lambda_1 + \mu + 2\theta_1 + s - \cdots}}}, \tag{6}$$

$$\mathbb{E}[B] = \frac{1}{\lambda_1}\Big({}_1F_1\big(1;\, \mu/\theta_1;\, \lambda_1/\theta_1\big) - 1\Big) \tag{7}$$

The continued fraction (6) is generated by a first-passage recursion. Let $\phi_n(s)$ denote the LST of the first-passage (sub-busy-period) time from n customers in the system down to $n-1$. Conditioning on the first transition out of state n—an arrival (rate λ_1) to state $n+1$, or a departure by service completion or abandonment (combined rate $\mu + (n-1)\theta_1$: one customer in service plus $n-1$ waiting, each impatient at rate θ_1) to state $n-1$—the strong Markov property gives

$$\phi_n(s) \;=\; \frac{\mu + (n-1)\theta_1}{\lambda_1 + \mu + (n-1)\theta_1 + s - \lambda_1\, \phi_{n+1}(s)}, \qquad n \geq 1, \qquad \widetilde{B}(s) = \phi_1(s). \tag{8}$$

Unrolling (8) from $n = 1$ reproduces the continued fraction (6) term by term. The continued fraction also sums in closed form: *linearising* the Riccati-type recursion (8) yields the three-term contiguous recurrence for ${}_1F_1$ [DLM, §13.3], and by Pincherle's theorem the convergent continued fraction selects its minimal (recessive) solution, so $\phi_1(s)$ equals a ratio of contiguous ${}_1F_1$ functions of the form (13). The explicit parameter identification ($a = \mu/\theta_1$, $b = s/\theta_1$, $c = \lambda_1/\theta_1$) is carried out in the boundedness route for Model-C_2 (§9, Equation (257)).

3.4 Pollaczek–Khinchine formula

The $M/G/1$ queue —explained in [AR02]— is a single-server queue with Poisson arrivals at a rate λ but with generally distributed i.i.d. service times B with mean $\mathbb{E}[B]$

The $M/G/1$ queue [AR02] is a single-server queue with Poisson arrivals at rate λ and generally distributed i.i.d. service times B with mean and LST $\widetilde{B}(s) = \mathbb{E}[e^{-sB}]$. For stability purposes, the system load $\rho = \lambda\, \mathbb{E}[B] < 1$ is required. Under these two conditions —Poisson arrivals and stability— the Pollaczek–Khinchine (PK) formula gives the PGF of the stationary number of customers in the system:

$$L(z) \;=\; \frac{(1-\rho_B)(1-z)\, \widetilde{B}(\lambda(1-z))}{\widetilde{B}(\lambda(1-z)) - z}. \tag{9}$$

3.5 Confluent hypergeometric function

The *Kummer confluent hypergeometric function of the first kind* function will come up in some of the derivations of this thesis, and it reads as follows:

$$ {}_1F_1(a;\, b;\, z) \;=\; \sum_{n=0}^{\infty} \frac{a^{(n)}}{b^{(n)}\, n!}\, z^n, \qquad a^{(n)} = \prod_{k=0}^{n-1} (a+k), \quad a^{(0)} = 1, \tag{10} $$

where b is not 0 or a negative integer —i.e. $b \notin \{0, -1, -2, \dots\}$— and the series converges for all $z \in \mathbb{C}$. The function ${}_1F_1(a;\, b;\, z)$ is the unique solution of *Kummer's equation* [DLM]:

$$ z\, u'' + (b - z)\, u' - a\, u = 0, \tag{11} $$

normalized by ${}_1F_1(a;\, b;\, 0) = 1$. The *Kummer series* (Equation (10)) converges for all $z \in \mathbb{C}$ and $b \notin \{0, -1, -2, \dots\}$.

For this thesis, two identities related to it are used. First, the *Euler integral representation*, for $a, b > 0$ we have:

$$ {}_1F_1(a;\, a+b;\, z) \;=\; \frac{1}{B(a,b)} \int_0^1 t^{a-1} (1-t)^{b-1} e^{zt}\, dt, \tag{12} $$

where B is the *Beta function* $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, and Γ is the *Gamma function*. Secondly, the *Contiguous-function ratio*:

$$ \frac{{}_1F_1(a+1;\, b;\, z)}{{}_1F_1(a;\, b;\, z)}. \tag{13} $$

3.6 First-order linear ODEs and PDEs

Since jockeying and abandonment add state-dependent rates to the balance equations, derivative terms end showing up in the derivations and they require solving first-order linear ODEs and PDEs for the PGF. The three methods below are invoked at specific steps of the analysis.

3.6.1 Integrating factor

This method is standard to solve —for each fixed y— a *first-order linear ODE in x*. Such an equation has standard form

$$ \frac{dP}{dx} + p(x;\, y)\, P \;=\; q(x;\, y). \tag{14} $$

The *integrating factor* $\mu_I(x; y) = \exp\!\big(\int_0^x p(t; y)\, dt\big)$ converts (14) to $\frac{d}{dx}[P \cdot \mu_I] = q \cdot \mu_I$. Integrating from 0 to x gives the general solution

$$ P(x, y) \;=\; \frac{1}{\mu_I(x;\, y)} \left[C_0(y) + \int_0^x q(t;\, y)\, \mu_I(t;\, y)\, dt \right], \tag{15} $$

where $C_0(y)$ is an arbitrary function of y, determined by a boundary condition.

3.6.2 Variation of parameters

Variation of parameters provides an alternative derivation of (15) that makes the role of the free function explicit. The homogeneous equation $\frac{dP}{dx} + p(x;\, y)\, P = 0$ has general solution $P_h(x; y) = C/\mu_I(x; y)$ for an arbitrary constant C. To solve the full equation, one *varies* that constant, setting $P(x, y) = C(x;\, y)/\mu_I(x;\, y)$ and substituting into (14). Since the homogeneous part cancels, one obtains the following.

$$ \frac{dC}{dx}(x;\, y) \;=\; q(x;\, y)\, \mu_I(x;\, y), \tag{16} $$

and integrating recovers (15) with $C_0(y)$.

3.6.3 Method of Characteristics

This method is standard for solving a *first-order linear PDE in both x and y* simultaneously. It reduces the PDE to a family of ODEs along the so-called *characteristic curves* in the (x, y)-plane; along each curve, the PDE collapses to a single-variable ODE.

A PDE of the form

$$ a\frac{\partial P}{\partial x} + b\frac{\partial P}{\partial y} = cP + d \tag{17} $$

is solved by tracing the characteristic curve $(x(t), y(t))$ satisfying

$$\frac{dx}{dt} = a, \qquad \frac{dy}{dt} = b. \tag{18}$$

By the chain rule, $\frac{d}{dt}P(x(t), y(t)) = a\,P_x + b\,P_y$, so (17) reduces to

$$\frac{dP}{dt} = cP + d, \tag{19}$$

a first-order linear ODE in t, solved by the integrating factor (§3.6.1). When a and b are constants, the characteristics are straight lines and the quantity

$$\xi \ := \ b\,x - a\,y \tag{20}$$

is preserved along each curve —it is the *first integral* that labels each characteristic. The integration constant of (19) therefore varies between characteristics: it is an arbitrary function $F(\xi)$. Re-expressing the solution in the original variables (x, y) via (18) gives the general solution of (17). In the PGF setting, F is identified with a boundary generating function, and analyticity of $P(x, y)$ for $|x|, |y| \leq 1$ uniquely determines it.

4 Model description: Model-X

This section defines the general two-class system with both jockeying (γ_i) and abandonments (θ_i). We refer to this full model as *Model-X*. The simpler variants studied are recovered as special cases by zeroing the appropriate parameters, as summarised in the table below (Table 3):

Model	Jockeying	Abandonment	Parameter restriction
X	both $(1\leftrightarrow 2)$	both $(1, 2)$	$\gamma_i \geq 0,\ \theta_i \geq 0$ (general)
A	none	none	$\gamma_1 = \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
B	both $(1\leftrightarrow 2)$	none	$\gamma_i \geq 0,\ \theta_1 = \theta_2 = 0$
B_2	$1 \to 2$ only	none	$\gamma_1 \geq 0,\ \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
C_2	none	class 1 only	$\theta_1 \geq 0,\ \theta_2 = 0,\ \gamma_1 = \gamma_2 = 0$

Table 3: Nested family of models. Model-X is the parent; Models A, B, B_2 and C_2 are obtained by restricting the jockeying directions and the abandoning classes.

Figure 1 provides a visual overview of Model-X and each of the variants that we will consider:

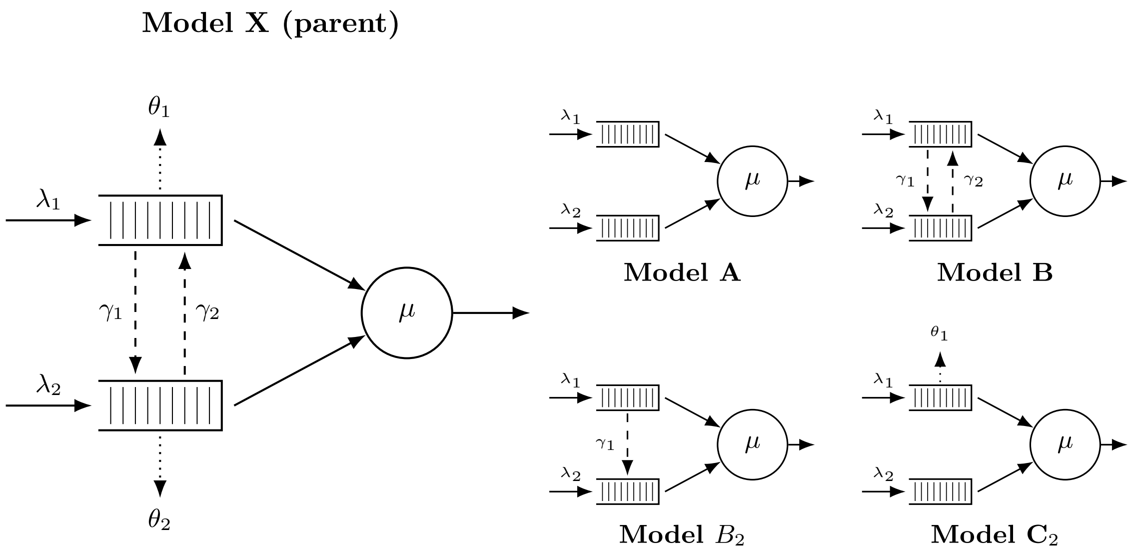

Figure 1: Physical queue layout for Model-X (left) and its four specializations (right, 2×2 grid). Queue 1 (top buffer) has non-preemptive priority over Queue 2. Solid arrows: Poisson arrivals (λ_i) and exponential service (μ). Dashed arrows: jockeying (γ_i), which transfers a waiting customer between queues and conserves the total n. Dotted arrows: abandonment (θ_i), a true departure that reduces n.

The model discussed in this section is an $M/M/1$ queueing system with 2 priority classes, where customers are served on a First-Come, First-Served (FCFS) basis within their own class and those coming from the priority class (Class-1) have non-preemptive priority over those of the non-priority class (Class-2); that is, when a Class-2 customer is in service, that service will not be interrupted by a potential arrival of a Class-1 customer. Beyond basic dynamics, Model-X incorporates two distinct queue-leaving mechanisms. First, *jockeying*: a waiting customer can abandon its own queue and *join the other queue*, at a per-customer rate γ_1 for direction $1 \to 2$ and γ_2 for direction $2 \to 1$, so the total number of customers is preserved. Second, *abandonment* (reneging): a waiting customer may completely leave the *system* at a rate per-customer θ_i, so the total number of customers decreases by one. Concretely, each Class-i customer that is waiting in line (i.e. not the one in service) holds an independent $\mathrm{Exp}(\theta_i)$ patience clock and abandons the system when it expires; consequently, the aggregate Class-i abandonment rate in a state with n_i waiting customers is the linear term $\theta_i n_i$ that appears in the balance equations. This is precisely the linear-reneging mechanism of the $M/M/1+M$ queue introduced in Section 3.3.

This model, being an $M/M/1$ queueing model according to Kendall's notation, features arrivals driven by a Poisson process with arrival rates λ_i for Class-i, exponentially distributed service times with rate μ, and a service capacity of 1 customer at a time. In addition, a waiting customer of Class-1 jockeys in the class-2 queue at the per-customer rate γ_1 and a waiting customer of Class-2 jockeys in the class-1 queue at the per-customer rate γ_2 (transfers that conserve the total), while a waiting customer of Class-i abandons the system at the per-customer rate θ_i (a true departure that reduces the total).

In order to define the stochastic processes that drive this queueing system, it is necessary to introduce the following random variables:

- $\mathcal{B}$: binary indicator for the server being idle or busy, $\mathcal{B} \in \{0, 1\}$;
- N_1, N_2 : number of customers of class 1 and 2, respectively, in their respective queues;
- N : total number of customers in the queues.

Note that the variables N_1, N_2 and N count the number of customers in the queues, as opposed to the customers in the system. This definition is convenient given that we have a common service rate for both priority classes, and hence the states of the system are equivalent regardless of the priority class the customer in service comes from. Since having customers in line directly implies that someone is in service, we will denote the idle state as (0) and the rest of the states by the number of people in each queue as the tuple (n_1, n_2), where n_i denotes the number of customers in the queue for class-i. This is the most conventional way to define the state space, namely S, in such a queueing system. However, we can also look at the total number of customers in the queues. We can define an alternative state space $\widetilde{S}$ using the idle state (0) and the pair (n_2, n) —we could also choose (n_1, n)—, which gives that we have $(n - n_2)$ Class-1 customers in the priority queue. Note that, since n is the total number of customers, we need the constraint $n \geq n_2$. Therefore, we define our state spaces as follows:

$$S := \{(0)\} \cup \{(n_1, n_2) \mid n_1, n_2 \geq 0\}, \tag{21}$$

$$\widetilde{S} := \{(0)\} \cup \{(n_2, n) \mid n \geq n_2 \geq 0\}. \tag{22}$$

For these state spaces, we define the limiting —as opposed to time-dependent— probabilities of finding the system in that state in the long run as follows:

- π_0 for the idle state (0), which belongs to both spaces S and $\widetilde{S}$;
- $\pi(n_1, n_2)$ for states $(n_1, n_2) \in S$, where $n_1, n_2 \geq 0$;
- $\widetilde{\pi}(n_2, n)$ for states $(n_2, n) \in \widetilde{S}$, where $n \geq n_2 \geq 0$.

Regarding the parameters of the system, the arrival process for each class follows a Poisson process (i.e., the interarrival times are exponentially distributed). By the properties of Poisson processes, the total number of arrivals will also form a Poisson process. Furthermore, we have a common service rate for exponentially distributed service times. We can organize our notation into class-specific parameters (alongside our common service rate) and class-blind totals. For the class-specific metrics and service rate, we define:

- λ_1, λ_2: arrival rates of classes 1 and 2, respectively;
- μ: service rate of a customer, regardless of class;

- $\rho_i = \frac{\lambda_i}{\mu}$: system load of class $i \in \{1, 2\}$;
- γ_1, γ_2: per-customer jockeying rates from Class $1 \to 2$ and $2 \to 1$, respectively; jockeying transfers a customer between queues, preserving the total number of customers;
- θ_1, θ_2: per-customer abandonment (reneging) rates from classes 1 and 2, respectively; abandonment removes a customer from the system, reducing the total n by one.

For the aggregate system parameters, we define:

- $\lambda = \lambda_1 + \lambda_2$: total arrival rate;
- $\rho = \frac{\lambda}{\mu}$: total system load.

Stability. The stability conditions depend on the abandonments; jockeying does not affect this condition, since customers are merely relocated. Model-X, having abandonments from both classes, is always stable, as the total number of customers behaves like some sort of $M/M/1+M$ —which is always stable, as stated before 3.3. The rate out is $\mu + \theta_1 n_1 + \theta_2 n_2$, which grows to infinity in every direction of the (n_1, n_2)-plane, so the system is always positive-recurrent. In contrast, Model-A, having no abandonments whatsoever, requires $\rho \leq 1$, since the total number of customers behaves exactly like an $M/M/1$ queueing system (3.3). It is more complex for models where only one type of abandonment is possible, this will be discussed in the dedicated section for Model-C_2 (9) .

Computation of π_0 and $\pi(0, 0)$.

The following relation between π_0 and $\pi(0, 0)$ holds no only for Model-X but also for every other model we will see in this work. The only transition into state (0) is a service completion from $(0, 0)$, and the only transition out from $(0, 0)$ is an arrival of a customer from any class to an idle system, leaving us whe following balance equation: $\lambda\pi_0 = \mu\,\pi(0, 0)$. Therefore

$$\pi(0,0) \;=\; \frac{\lambda}{\mu}\,\pi_0 \;=\; \rho\,\pi_0. \tag{23}$$

Note that this holds regardless jockeying and abandonments, as in neither of these states we have waiting customers. However, what does depend on the abandonment parameters —not on the jockeying ones— is the value of π_0 itself. For models *without* abandonment, the total number of customers in the system behaves like that of an ordinary $M/M/1$ single-class system, giving

$$\pi_0 = 1 - \rho = 1 - \frac{\lambda_1 + \lambda_2}{\mu}, \qquad \pi(0,0) = \rho\,\pi_0 = (1-\rho)\rho. \tag{24}$$

However, for models *with* abandonments, the total number of customers is no longer conserved. For a busy state, the departure rate is $\mu + \theta_1 n_1 + \theta_2 n_2$, so the rate of state (n_1, n_2) depends on the split between n_1 and n_2. Indeed, the states $(n, 0)$ and $(0, n)$ carry departure rates $\mu + \theta_1 n$ and $\mu + \theta_2 n$, which differ whenever for assymetric abandonment rates $\theta_1 \neq \theta_2$. To determine π_0, we will need our bivariate PGF, which will be defined in Section 4.2.

4.1 Balance equations

This subsection presents the state-transition diagrams of Model-X on both state spaces S and $\widetilde{S}$, and derives the corresponding sets of balance equations.

4.1.1 Balance equations for state space S

The state-transition diagram of Model-X on S is given in Figure 2.

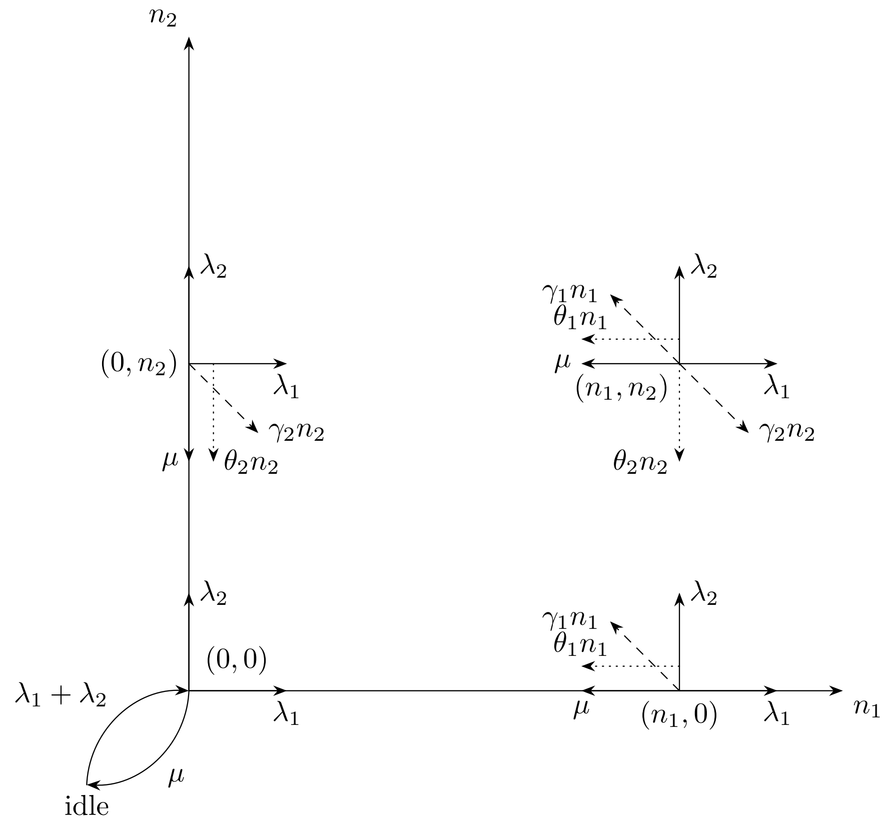

Figure 2: State transition diagram on state space S for the model with jockeying ($\gamma_i n_i$, dashed) and abandonments ($\theta_i n_i$, dotted).

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)n_1 + (\gamma_2 + \theta_2)n_2]\,\pi(n_1, n_2) \\
&\quad = \quad \mu\pi(n_1 + 1, n_2) \\
&\qquad + \lambda_1\pi(n_1 - 1, n_2) \\
&\qquad + \lambda_2\pi(n_1, n_2 - 1) \\
&\qquad + \gamma_1(n_1 + 1)\pi(n_1 + 1, n_2 - 1) \qquad\qquad \text{for} \quad n_1 \geq 1, n_2 \geq 1 \\
&\qquad + \gamma_2(n_2 + 1)\pi(n_1 - 1, n_2 + 1) \\
&\qquad + \theta_1(n_1 + 1)\pi(n_1 + 1, n_2) \\
&\qquad + \theta_2(n_2 + 1)\pi(n_1, n_2 + 1)
\end{aligned} \tag{25}$$

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2)n_2]\,\pi(0, n_2) \\
&\quad = \quad \lambda_2\pi(0, n_2 - 1) \\
&\qquad + \mu\pi(1, n_2) \\
&\qquad + \mu\pi(0, n_2 + 1) \qquad\qquad \text{for} \quad n_1 = 0, n_2 \geq 1 \\
&\qquad + \gamma_1\pi(1, n_2 - 1) \\
&\qquad + \theta_1\pi(1, n_2) \\
&\qquad + \theta_2(n_2 + 1)\pi(0, n_2 + 1)
\end{aligned} \tag{26}$$

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)n_1]\,\pi(n_1, 0) \\
&\quad = \quad \lambda_1\pi(n_1 - 1, 0) \\
&\qquad + \mu\pi(n_1 + 1, 0) \\
&\qquad + \gamma_2\pi(n_1 - 1, 1) \qquad\qquad \text{for} \quad n_1 \geq 1, n_2 = 0 \\
&\qquad + \theta_1(n_1 + 1)\pi(n_1 + 1, 0) \\
&\qquad + \theta_2\pi(n_1, 1)
\end{aligned} \tag{27}$$

$$[\lambda_1 + \lambda_2]\,\pi(0, 0) = (\mu + \theta_1)\pi(1, 0) + (\mu + \theta_2)\pi(0, 1) \tag{28}$$

$$(\lambda_1 + \lambda_2)\pi_0 = \mu\pi(0, 0) \tag{29}$$

4.1.2 Balance equations for state space $\widetilde{S}$

Recall the state space $\widetilde{S}$ defined in (22).

For the sake of readability, we will denote the limiting probabilities associated to this state space with $\widetilde{\pi}(n_2, n)$, for state $(n_2, n) \in \widetilde{S}$.

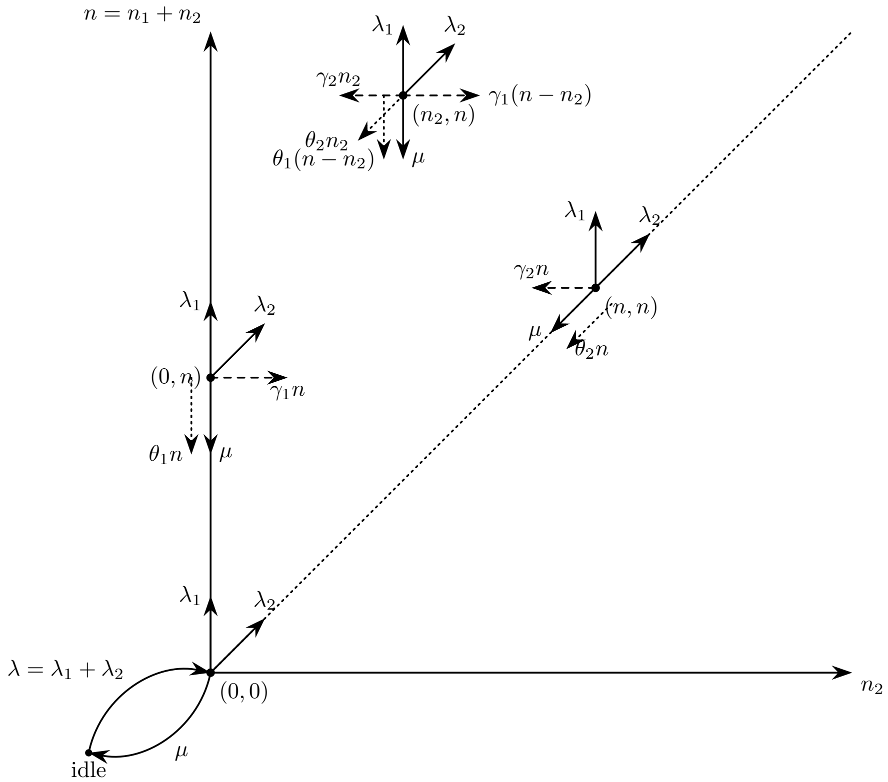

Figure 3: State transition diagram on state space $\widetilde{S}$.

The balance equations, after combining the equations where $\widetilde{\pi}_0$ shows up, are:

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)(n - n_2) + (\gamma_2 + \theta_2)n_2]\, \widetilde{\pi}(n_2, n) \\
&\quad = \quad \lambda_1 \widetilde{\pi}(n_2, n - 1) \\
&\qquad + \lambda_2 \widetilde{\pi}(n_2 - 1, n - 1) \\
&\qquad + \mu \widetilde{\pi}(n_2, n + 1) \\
&\qquad + \gamma_1(n - n_2 + 1)\widetilde{\pi}(n_2 - 1, n) \\
&\qquad + \gamma_2(n_2 + 1)\widetilde{\pi}(n_2 + 1, n) \\
&\qquad + \theta_1(n + 1 - n_2)\widetilde{\pi}(n_2, n + 1) \\
&\qquad + \theta_2(n_2 + 1)\widetilde{\pi}(n_2 + 1, n + 1)
\end{aligned} \qquad \text{for} \quad 1 \le n_2 \le n - 1,\ n \ge 2 \tag{30}$$

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2)n]\, \widetilde{\pi}(n, n) \\
&\quad = \quad \lambda_2 \widetilde{\pi}(n - 1, n - 1) \\
&\qquad + \mu \widetilde{\pi}(n + 1, n + 1) \\
&\qquad + \mu \widetilde{\pi}(n, n + 1) \\
&\qquad + \gamma_1 \widetilde{\pi}(n - 1, n) \\
&\qquad + \theta_1 \widetilde{\pi}(n, n + 1) \\
&\qquad + \theta_2(n + 1)\widetilde{\pi}(n + 1, n + 1)
\end{aligned} \qquad \text{for} \quad n_2 = n,\ n \ge 1 \tag{31}$$

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)n]\, \widetilde{\pi}(0, n) \\
&\quad = \quad \lambda_1 \widetilde{\pi}(0, n - 1) \\
&\qquad + \mu \widetilde{\pi}(0, n + 1) \\
&\qquad + \gamma_2 \widetilde{\pi}(1, n) \\
&\qquad + \theta_1(n + 1)\widetilde{\pi}(0, n + 1) \\
&\qquad + \theta_2 \widetilde{\pi}(1, n + 1)
\end{aligned} \qquad \text{for} \quad n_2 = 0,\ n \ge 1 \tag{32}$$

$$\begin{aligned}&[\lambda_1+\lambda_2]\,\widetilde{\pi}(0,0)\\&\quad=\quad(\mu+\theta_1)\widetilde{\pi}(0,1)+(\mu+\theta_2)\widetilde{\pi}(1,1)\end{aligned}\tag{33}$$

$$(\lambda_1+\lambda_2)\widetilde{\pi}_0=\mu\widetilde{\pi}(0,0)\tag{34}$$

4.2 Derivation of equations for (Partial) Probability Generating functions

In order to study the limiting behaviour of this two-class system, we define the bivariate Probability Generating Function (PGF) of the joint number of customers in the queue for each class, N_1 and N_2:

$$P(x,y)=\mathbb{E}[x^{N_1}y^{N_2}]=\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}.\tag{35}$$

The derivation that will be carried out here includes the boundary terms, where at least one of the two queues is empty:

$$P_x(x)=P(x,0)=\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}\tag{36}$$

$$P_y(y)=P(0,y)=\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}.\tag{37}$$

Note that both of these terms include the state where the server is busy but both queues are empty, which is represented by $\pi(0,0)$, while the state where the entire system is empty (the idle state) is treated separately.

Another term that arises during the application of the balance equations characterizes the boundary dynamics when the class-1 queue contains exactly one waiting customer:

$$P_1(y)=\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}.\tag{38}$$

This term serves as an intermediate step; as the derivation proceeds, it will be possible to express it in terms of the primary boundary functions $P_x(x)$ and $P_y(y)$.

In the same fashion, we introduce the concept of the **Partial Probability Generating Function (PPGF)**. This technique, notably utilized by Cohen [Coh56], consists of transforming only a subset of the random variables while maintaining the discrete indices of the remainder. In the context of a priority queue, this method is particularly advantageous: it allows us to exploit the well-known properties of the priority class (often a standard $M/M/1$ system) and focus the analytical effort on the non-trivial PGF of the non-priority class.

While Cohen originally developed this framework for multi-server systems, we apply it here to the single-server case. To avoid the potential confusion stemming from mid-20th-century notation, we provide a modern formulation consistent with the variables defined above.

We define the PPGF with respect to the second variable N_2, conditioned on the first queue being in state n_1, as:

$$\widetilde{P}(n_1,y)=\mathbb{E}\left[y^{N_2}\mathbb{1}_{\{N_1=n_1\}}\right]=\sum_{n_2=0}^{\infty}\pi(n_1,n_2)y^{n_2}.\tag{39}$$

Similarly, the transform with respect to the first variable N_1, while maintaining the discrete state n_2 for the second queue, is defined as:

$$\widetilde{P}(x,n_2)=\mathbb{E}\left[x^{N_1}\mathbb{1}_{\{N_2=n_2\}}\right]=\sum_{n_1=0}^{\infty}\pi(n_1,n_2)x^{n_1}.\tag{40}$$

These partial transforms are linked to the joint PGF by the following relations:

$$P(x,y)=\sum_{n_1=0}^{\infty}\widetilde{P}(n_1,y)x^{n_1}=\sum_{n_2=0}^{\infty}\widetilde{P}(x,n_2)y^{n_2}.\tag{41}$$

4.2.1 Equation for PGF in state space S

Lemma 1 (Fundamental equation for the PGF on S)**.** *Consider the two-class non-preemptive priority queue on the state space $S = \{(n_1, n_2) : n_1, n_2 \geq 0\}$, with Poisson arrival rates λ_1, λ_2, exponential service rate μ, jockeying rates γ_1, γ_2 and abandonment rates θ_1, θ_2, where n_1 and n_2 count the class-1 and class-2 customers* waiting *in the queues. Assume the associated CTMC is positive recurrent, so that the stationary distribution $\{\pi(n_1, n_2)\}$ exists and the joint PGF*

$$P(x, y) = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(n_1, n_2)\, x^{n_1} y^{n_2}$$

is analytic on the closed unit polydisc $|x| \leq 1,\ |y| \leq 1$. Then $P(x, y)$, together with the boundary function $P_y(y) = P(0, y) = \sum_{n_2=0}^{\infty} \pi(0, n_2) y^{n_2}$, satisfies the first-order linear partial differential equation

$$\begin{aligned}
&\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 xy^2\right] P(x, y) \\
&+ xy\left[\gamma_1(x - y) + \theta_1(x - 1)\right] \frac{\partial}{\partial x} P(x, y) \\
&+ xy\left[\gamma_2(y - x) + \theta_2(y - 1)\right] \frac{\partial}{\partial y} P(x, y) \\
&\quad = \quad \mu(x - y)P_y(y) - \mu x(1 - y)\, \pi(0, 0).
\end{aligned} \tag{42}$$

Proof. We multiply each balance equation by $x^{n_1} y^{n_2}$ and sum over its domain.

From the interior equation (25):

$$\begin{aligned}
&(\lambda_1 + \lambda_2 + \mu) \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&+ \gamma_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_1 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&+ \gamma_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&+ \theta_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_1 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&+ \theta_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&\quad = \quad \mu \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1 + 1, n_2) x^{n_1} y^{n_2} \\
&\qquad + \lambda_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1 - 1, n_2) x^{n_1} y^{n_2} \\
&\qquad + \lambda_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1, n_2 - 1) x^{n_1} y^{n_2} \\
&\qquad + \gamma_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_1 + 1)\pi(n_1 + 1, n_2 - 1) x^{n_1} y^{n_2} \\
&\qquad + \gamma_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_2 + 1)\pi(n_1 - 1, n_2 + 1) x^{n_1} y^{n_2} \\
&\qquad + \theta_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_1 + 1)\pi(n_1 + 1, n_2) x^{n_1} y^{n_2} \\
&\qquad + \theta_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_2 + 1)\pi(n_1, n_2 + 1) x^{n_1} y^{n_2}
\end{aligned} \tag{43}$$

$$\begin{aligned}
&LHS_1 + LHS_2 + LHS_3 + LHS_4 + LHS_5 \\
&\quad = \quad RHS_1 + RHS_2 + RHS_3 + RHS_4 + RHS_5 + RHS_6 + RHS_7
\end{aligned} \tag{44}$$

Evaluating each summand:

$$
\begin{aligned}
& LHS_1 \\
= \quad & (\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2} \\
= \quad & (\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\left[\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\pi(n_1,0)x^{n_1}y^{0}\right] \\
= \quad & (\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}\right] \\
= \quad & (\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}\right] \\
& -\ (\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right] \\
= \quad & (\lambda_1+\lambda_2+\mu)\left[P(x,y)-P_y(y)-P_x(x)+\pi(0,0)\right]
\end{aligned}
\tag{45}
$$

$$
\begin{aligned}
& LHS_2 \\
= \quad & \gamma_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2} \\
= \quad & \gamma_1 x\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} \\
= \quad & \gamma_1 x\left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty}n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2}-\sum_{n_1=1}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\right] \\
= \quad & \gamma_1 x\left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2}-\sum_{n_1=0}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\right] \\
= \quad & \gamma_1 x\left[\frac{\partial}{\partial x}P(x,y)-\frac{d}{dx}P_x(x)\right]
\end{aligned}
\tag{46}
$$

$$
\begin{aligned}
& LHS_3 \\
= \quad & \gamma_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2} \\
= \quad & \gamma_2\sum_{n_1=1}^{\infty}\left[\sum_{n_2=0}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2}-0\cdot\pi(n_1,0)x^{n_1}\right] \\
= \quad & \gamma_2 y\left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2-1}-\sum_{n_2=0}^{\infty}n_2\pi(0,n_2)y^{n_2-1}\right] \\
= \quad & \gamma_2 y\left[\frac{\partial}{\partial y}P(x,y)-\frac{d}{dy}P_y(y)\right]
\end{aligned}
\tag{47}
$$

$$
\begin{aligned}
& LHS_4 \\
= \quad & \theta_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2} \\
= \quad & \theta_1 x\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} \\
= \quad & \theta_1 x\left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty}n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2}-\sum_{n_1=1}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\right] \\
= \quad & \theta_1 x\left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2}-\sum_{n_1=0}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\right] \\
= \quad & \theta_1 x\left[\frac{\partial}{\partial x}P(x,y)-\frac{d}{dx}P_x(x)\right]
\end{aligned}
\tag{48}
$$

$$
\begin{aligned}
&LHS_5 \\
&= \quad \theta_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2} \\
&= \quad \theta_2 y \left[\sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} n_2 \pi(n_1, n_2) x^{n_1} y^{n_2-1} - \sum_{n_2=0}^{\infty} n_2 \pi(0, n_2) y^{n_2-1} \right] \\
&= \quad \theta_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right]
\end{aligned}
\tag{49}
$$

$$
\begin{aligned}
&RHS_1 \\
&= \quad \mu \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1+1, n_2) x^{n_1} y^{n_2} \\
&= \quad \mu x^{-1} \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1+1, n_2) x^{n_1+1} y^{n_2} \\
&= \quad \mu x^{-1} \sum_{m=2}^{\infty} \sum_{n_2=1}^{\infty} \pi(m, n_2) x^{m} y^{n_2} \\
&= \quad \mu x^{-1} \sum_{m=2}^{\infty} \left[\sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \pi(m,0) x^m \right] \\
&= \quad \mu x^{-1} \left[\sum_{m=2}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \sum_{m=2}^{\infty} \pi(m,0) x^m \right] \\
&= \quad \mu x^{-1} \left[\sum_{m=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - x \sum_{n_2=0}^{\infty} \pi(1, n_2) y^{n_2} - \sum_{n_2=0}^{\infty} \pi(0, n_2) y^{n_2} \right] \\
&\quad - \; \mu x^{-1} \left[\sum_{m=0}^{\infty} \pi(m,0) x^m - x\pi(1,0) - \pi(0,0) \right] \\
&= \quad \mu x^{-1} \left[P(x,y) - x P_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0) \right]
\end{aligned}
\tag{50}
$$

$$
\begin{aligned}
&RHS_2 \\
&= \quad \lambda_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1-1, n_2) x^{n_1} y^{n_2} \\
&= \quad \lambda_1 x \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1-1, n_2) x^{n_1-1} y^{n_2} \\
&= \quad \lambda_1 x \sum_{m=0}^{\infty} \sum_{n_2=1}^{\infty} \pi(m, n_2) x^{m} y^{n_2} \\
&= \quad \lambda_1 x \sum_{m=0}^{\infty} \left[\sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \pi(m,0) x^m \right] \\
&= \quad \lambda_1 x \left[\sum_{m=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(m, n_2) x^{m} y^{n_2} - \sum_{m=0}^{\infty} \pi(m,0) x^m \right] \\
&= \quad \lambda_1 x \left[P(x,y) - P_x(x) \right]
\end{aligned}
\tag{51}
$$

$$
\begin{aligned}
& RHS_3 \\
= \quad & \lambda_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1, n_2-1) x^{n_1} y^{n_2} \\
= \quad & \lambda_2 y \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} \pi(n_1, n_2-1) x^{n_1} y^{n_2-1} \\
= \quad & \lambda_2 y \sum_{n_1=1}^{\infty} \sum_{m=0}^{\infty} \pi(n_1, m) x^{n_1} y^{m} \\
= \quad & \lambda_2 y \left[\sum_{n_1=0}^{\infty} \sum_{m=0}^{\infty} \pi(n_1, m) x^{n_1} y^{m} - \sum_{m=0}^{\infty} \pi(0, m) y^{m} \right] \\
= \quad & \lambda_2 y \left[P(x,y) - P_y(y) \right]
\end{aligned} \tag{52}
$$

$$
\begin{aligned}
& RHS_4 \\
= \quad & \gamma_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_1+1) \pi(n_1+1, n_2-1) x^{n_1} y^{n_2} \\
= \quad & \gamma_1 y \sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} (n_1+1) \pi(n_1+1, m_2) x^{n_1} y^{m_2} \\
= \quad & \gamma_1 y \sum_{m_1=2}^{\infty} \sum_{m_2=0}^{\infty} m_1 \pi(m_1, m_2) x^{m_1-1} y^{m_2} \\
= \quad & \gamma_1 y \left[\sum_{m_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_1 \pi(m_1, m_2) x^{m_1-1} y^{m_2} - \sum_{m_2=0}^{\infty} \pi(1, m_2) y^{m_2} \right] \\
= \quad & \gamma_1 y \left[\frac{\partial}{\partial x} P(x,y) - P_y(y|n_1=1) \right]
\end{aligned} \tag{53}
$$

$$
\begin{aligned}
& RHS_5 \\
= \quad & \gamma_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_2+1) \pi(n_1-1, n_2+1) x^{n_1} y^{n_2} \\
= \quad & \gamma_2 \sum_{n_1=1}^{\infty} \sum_{m_2=2}^{\infty} m_2 \pi(n_1-1, m_2) x^{n_1} y^{m_2-1} \\
= \quad & \gamma_2 x \left[\sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1-1, m_2) x^{n_1-1} y^{m_2-1} - \sum_{n_1=1}^{\infty} \pi(n_1-1, 1) x^{n_1-1} \right] \\
= \quad & \gamma_2 x \left[\sum_{m_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(m_1, m_2) x^{m_1} y^{m_2-1} - \sum_{m_1=0}^{\infty} \pi(m_1, 1) x^{m_1} \right] \\
= \quad & \gamma_2 x \left[\frac{\partial}{\partial y} P(x,y) - P_x(x|n_2=1) \right]
\end{aligned} \tag{54}
$$

$$
\begin{aligned}
&RHS_6 \\
&= \theta_1 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_1+1)\pi(n_1+1,n_2)x^{n_1}y^{n_2} \\
&= \theta_1 \sum_{m_1=2}^{\infty} \sum_{n_2=1}^{\infty} m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2} \\
&= \theta_1 \left[\sum_{m_1=0}^{\infty} \sum_{n_2=1}^{\infty} m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2} - \sum_{n_2=1}^{\infty} \pi(1,n_2)y^{n_2} \right] \\
&= \theta_1 \left[\sum_{m_1=0}^{\infty} \sum_{n_2=0}^{\infty} m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2} - \sum_{m_1=0}^{\infty} m_1\pi(m_1,0)x^{m_1-1} \right] \\
&\quad - \theta_1 \left[\sum_{n_2=0}^{\infty} \pi(1,n_2)y^{n_2} - \pi(1,0) \right] \\
&= \theta_1 \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x) - P_y(y|n_1=1) + \pi(1,0) \right]
\end{aligned} \tag{55}
$$

$$
\begin{aligned}
&RHS_7 \\
&= \theta_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_2+1)\pi(n_1,n_2+1)x^{n_1}y^{n_2} \\
&= \theta_2 \sum_{n_1=1}^{\infty} \sum_{m_2=2}^{\infty} m_2\pi(n_1,m_2)x^{n_1}y^{m_2-1} \\
&= \theta_2 \left[\sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} m_2\pi(n_1,m_2)x^{n_1}y^{m_2-1} - \sum_{n_1=1}^{\infty} \pi(n_1,1)x^{n_1} \right] \\
&= \theta_2 \left[\sum_{n_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_2\pi(n_1,m_2)x^{n_1}y^{m_2-1} - \sum_{m_2=0}^{\infty} m_2\pi(0,m_2)y^{m_2-1} \right] \\
&\quad - \theta_2 \left[\sum_{n_1=0}^{\infty} \pi(n_1,1)x^{n_1} - \pi(0,1) \right] \\
&= \theta_2 \left[\frac{\partial}{\partial y}P(x,y) - \frac{d}{dy}P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned} \tag{56}
$$

Collecting and multiplying through by xy to clear the x^{-1} factor:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[P(x,y) - P_y(y) - P_x(x) + \pi(0,0)\right] \\
&+ \gamma_1 x \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x) \right] \\
&+ \gamma_2 y \left[\frac{\partial}{\partial y}P(x,y) - \frac{d}{dy}P_y(y) \right] \\
&+ \theta_1 x \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x) \right] \\
&+ \theta_2 y \left[\frac{\partial}{\partial y}P(x,y) - \frac{d}{dy}P_y(y) \right] \\
&\quad = \mu x^{-1}\left[P(x,y) - xP_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0)\right] \\
&\qquad + \lambda_1 x\left[P(x,y) - P_x(x)\right] \\
&\qquad + \lambda_2 y\left[P(x,y) - P_y(y)\right] \\
&\qquad + \gamma_1 y \left[\frac{\partial}{\partial x}P(x,y) - P_y(y|n_1=1) \right] \\
&\qquad + \gamma_2 x \left[\frac{\partial}{\partial y}P(x,y) - P_x(x|n_2=1) \right] \\
&\qquad + \theta_1 \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x) - P_y(y|n_1=1) + \pi(1,0) \right] \\
&\qquad + \theta_2 \left[\frac{\partial}{\partial y}P(x,y) - \frac{d}{dy}P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned}
$$

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x-\lambda_2 y\right] P(x,y)\\
&+\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right] P_x(x)\\
&\quad\quad+\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_2 y\right] P_y(y)\\
&\quad\quad+\left[-(\lambda_1+\lambda_2+\mu)+\mu x^{-1}\right]\pi(0,0)\\
&\quad\quad+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad\quad+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad\quad-\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\quad\quad-\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\quad\quad+\mu\pi(1,0)\\
&\quad\quad+\theta_1\pi(1,0)\\
&\quad\quad+\theta_2\pi(0,1)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 x y^2\right] P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right] P_x(x)\\
&\quad\quad+\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 x y^2\right] P_y(y)\\
&\quad\quad-\left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\quad\quad+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad\quad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad\quad-xy\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\quad\quad-xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\quad\quad+(\mu+\theta_1)xy\,\pi(1,0)\\
&\quad\quad+\theta_2 xy\,\pi(0,1)
\end{aligned}
\tag{57}
$$

The x-boundary equation (27), treated analogously, gives:

$$
\begin{aligned}
&\sum_{n_1=1}^{\infty}\left[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n_1\right]\pi(n_1,0)x^{n_1}\\
&\quad=\quad \lambda_1\sum_{n_1=1}^{\infty}\pi(n_1-1,0)x^{n_1}+\mu\sum_{n_1=1}^{\infty}\pi(n_1+1,0)x^{n_1}\\
&\quad\quad+\gamma_2\sum_{n_1=1}^{\infty}\pi(n_1-1,1)x^{n_1}+\theta_1\sum_{n_1=1}^{\infty}(n_1+1)\pi(n_1+1,0)x^{n_1}+\theta_2\sum_{n_1=1}^{\infty}\pi(n_1,1)x^{n_1}\\
&(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}+(\gamma_1+\theta_1)x\sum_{n_1=1}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad=\quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^{m}+\mu x^{-1}\sum_{m=2}^{\infty}\pi(m,0)x^{m}\\
&\quad\quad+\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^{m}+\theta_1\sum_{m=2}^{\infty}m\pi(m,0)x^{m-1}+\theta_2\sum_{m=1}^{\infty}\pi(m,1)x^{m}
\end{aligned}
$$

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right]+(\gamma_1+\theta_1)x\sum_{n_1=0}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad=\quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^m\\
&\qquad+\mu x^{-1}\left[\sum_{m=0}^{\infty}\pi(m,0)x^m-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad+\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^m\\
&\qquad+\theta_1\left[\sum_{m=0}^{\infty}m\pi(m,0)x^{m-1}-\pi(1,0)\right]\\
&\qquad+\theta_2\left[\sum_{m=0}^{\infty}\pi(m,1)x^m-\pi(0,1)\right]\\
&(\lambda_1+\lambda_2+\mu)\left[P_x(x)-\pi(0,0)\right]+(\gamma_1+\theta_1)x\frac{d}{dx}P_x(x)\\
&\quad=\quad \lambda_1 xP_x(x)+\mu x^{-1}\left[P_x(x)-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad+\gamma_2 xP_x(x|n_2=1)+\theta_1\left[\frac{d}{dx}P_x(x)-\pi(1,0)\right]+\theta_2\left[P_x(x|n_2=1)-\pi(0,1)\right]\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right]P_x(x)+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}\right]\pi(0,0)-(\mu+\theta_1)\pi(1,0)-\theta_2\pi(0,1)\\
&\qquad+\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right]P_x(x)+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\qquad-(\mu+\theta_1)xy\,\pi(1,0)\\
&\qquad-\theta_2 xy\,\pi(0,1)\\
&\qquad+xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)
\end{aligned}
\tag{58}
$$

The y-boundary equation (26) gives:

$$
\begin{aligned}
&\sum_{n_2=1}^{\infty}\left[\lambda_1+\lambda_2+\mu+(\gamma_2+\theta_2)n_2\right]\pi(0,n_2)y^{n_2}\\
&\quad=\quad \lambda_2\sum_{n_2=1}^{\infty}\pi(0,n_2-1)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{\infty}\pi(0,n_2+1)y^{n_2}\\
&\qquad+\gamma_1\sum_{n_2=1}^{\infty}\pi(1,n_2-1)y^{n_2}\\
&\qquad+\theta_1\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\theta_2\sum_{n_2=1}^{\infty}(n_2+1)\pi(0,n_2+1)y^{n_2}
\end{aligned}
$$

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(0,n_2)y^{n_2}\\
&+(\gamma_2+\theta_2)y\sum_{n_2=1}^{\infty}n_2\pi(0,n_2)y^{n_2-1}\\
&\quad=\quad \lambda_2 y\sum_{m=0}^{\infty}\pi(0,m)y^m\\
&\qquad+\mu\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\mu y^{-1}\sum_{m=2}^{\infty}\pi(0,m)y^m\\
&\qquad+\gamma_1 y\sum_{m=0}^{\infty}\pi(1,m)y^m\\
&\qquad+\theta_1\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\theta_2\sum_{m=2}^{\infty}m\pi(0,m)y^{m-1}\\
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}-\pi(0,0)\right]\\
&+(\gamma_2+\theta_2)y\sum_{n_2=0}^{\infty}n_2\pi(0,n_2)y^{n_2-1}\\
&\quad=\quad \lambda_2 y\sum_{m=0}^{\infty}\pi(0,m)y^m\\
&\qquad+\mu\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[\sum_{m=0}^{\infty}\pi(0,m)y^m-y\pi(0,1)-\pi(0,0)\right]\\
&\qquad+\gamma_1 y\sum_{m=0}^{\infty}\pi(1,m)y^m\\
&\qquad+\theta_1\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\theta_2\left[\sum_{m=0}^{\infty}m\pi(0,m)y^{m-1}-\pi(0,1)\right]\\
&(\lambda_1+\lambda_2+\mu)\left[P_y(y)-\pi(0,0)\right]+(\gamma_2+\theta_2)y\frac{d}{dy}P_y(y)\\
&\quad=\quad \lambda_2 yP_y(y)+\mu\left[P_y(y|n_1=1)-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[P_y(y)-y\pi(0,1)-\pi(0,0)\right]+\gamma_1 yP_y(y|n_1=1)\\
&\qquad+\theta_1\left[P_y(y|n_1=1)-\pi(1,0)\right]+\theta_2\left[\frac{d}{dy}P_y(y)-\pi(0,1)\right]\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-(\mu+\theta_1)\pi(1,0)-(\mu+\theta_2)\pi(0,1)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}\right]\pi(0,0)
\end{aligned}
$$

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad[\mu+\gamma_1 y+\theta_1]\,P_y(y|n_1=1)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-(\lambda_1+\lambda_2)\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad[\mu+\gamma_1 y+\theta_1]\,P_y(y|n_1=1)+\left[\mu-\mu y^{-1}\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2 xy^2\right]P_y(y)+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad[\mu+\gamma_1 y+\theta_1]\,xy\,P_y(y|n_1=1)-\mu x(1-y)\,\pi(0,0)
\end{aligned}
\tag{59}
$$

Substituting (58) into (57) eliminates all P_x terms:

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right]P_y(y)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-xy\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)
\end{aligned}
\tag{60}
$$

Isolating $[\mu+\gamma_1 y+\theta_1]\,xy\,P_y(y|n_1=1)$ from (59) and substituting into (60):

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right]P_y(y)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-\Big\{\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2 xy^2\right]P_y(y)\\
&\qquad\quad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)+\mu x(1-y)\,\pi(0,0)\Big\}\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad\mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0)
\end{aligned}
\quad . \tag{61}
$$

□

4.2.2 Equation for PPGF in state space $\widetilde{S}$

Lemma 2 (Fundamental equation for the PPGF on $\widetilde{S}$)**.** *Consider the same system on the state space $\widetilde{S}=\{(n_2,n):0\le n_2\le n\}$, where n is the total number of customers* waiting *in the queues and n_2 the number of waiting class-2 customers. Assume positive recurrence, so that the stationary distribution $\{\widetilde{\pi}(n_2,n)\}$ exists, and define the partial PGF (transforming only n_2, keeping n discrete)*

$$
\widetilde{P}(y,n)=\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)\,y^{n_2},
$$

a polynomial of degree n in y. Then, for every $n \geq 1$, the family $\{\widetilde{P}(y,n)\}_{n\geq 0}$ satisfies the differential–difference equation

$$\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n)\\
&\quad= \quad (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1) \qquad (62)\\
&\qquad +(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1)+\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1).
\end{aligned}$$

Proof. Multiply the interior equation (30) by y^{n_2} and sum over $1 \leq n_2 \leq n-1$:

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_1\sum_{n_2=1}^{n-1}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_2\sum_{n_2=1}^{n-1}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_1\sum_{n_2=1}^{n-1}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_2\sum_{n_2=1}^{n-1}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&\quad= \quad \lambda_1\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n-1)y^{n_2}\\
&\qquad+\lambda_2\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2-1,n-1)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\gamma_1\sum_{n_2=1}^{n-1}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2}\\
&\qquad+\gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2}\\
&\qquad+\theta_1\sum_{n_2=1}^{n-1}(n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\theta_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2}
\end{aligned}$$

Adding the diagonal equation (31) multiplied by y^n extends most summation limits to n; the λ_1- and γ_2-terms on the RHS remain restricted to $n_2 \leq n-1$, and a spare $\mu\widetilde{\pi}(n+1,n+1)y^n$

survives:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&= \quad \lambda_1\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n-1)y^{n_2}\\
&\quad +\lambda_2\sum_{n_2=1}^{n}\widetilde{\pi}(n_2-1,n-1)y^{n_2}\\
&\quad +\mu\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\quad +\mu\widetilde{\pi}(n+1,n+1)y^{n}\\
&\quad +\gamma_1\sum_{n_2=1}^{n}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2}\\
&\quad +\gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2}\\
&\quad +\theta_1\sum_{n_2=1}^{n}(n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\quad +\theta_2\sum_{n_2=1}^{n}(n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2}.
\end{aligned}
$$

Evaluating each summand:

$$
\begin{aligned}
&LHS_1\\
&= (\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&= (\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\widetilde{\pi}(0,n)y^0\right]\\
&= (\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\right]
\end{aligned}
\tag{63}
$$

$$
\begin{aligned}
&LHS_2\\
&= \gamma_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&= \gamma_1\left[n\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\right]\\
&= \gamma_1\left[n\left(\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\widetilde{\pi}(0,n)y^0\right)-y\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2-1}\right]\\
&= \gamma_1\left[n\big(\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\big)-y\frac{d}{dy}\widetilde{P}(y,n)\right]
\end{aligned}
\tag{64}
$$

$$
\begin{aligned}
&LHS_3\\
&= \gamma_2 \sum_{n_2=1}^{n} n_2 \widetilde{\pi}(n_2,n) y^{n_2}\\
&= \gamma_2 \left[y \sum_{n_2=1}^{n} n_2 \widetilde{\pi}(n_2,n) y^{n_2-1} \right] \\
&= \gamma_2 y \frac{d}{dy} \widetilde{P}(y,n)
\end{aligned} \tag{65}
$$

$$
\begin{aligned}
&LHS_4\\
&= \theta_1 \sum_{n_2=1}^{n} (n-n_2) \widetilde{\pi}(n_2,n) y^{n_2}\\
&= \theta_1 \left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y \frac{d}{dy} \widetilde{P}(y,n) \right]
\end{aligned} \tag{66}
$$

$$
\begin{aligned}
&LHS_5\\
&= \theta_2 \sum_{n_2=1}^{n} n_2 \widetilde{\pi}(n_2,n) y^{n_2}\\
&= \theta_2 y \frac{d}{dy} \widetilde{P}(y,n)
\end{aligned} \tag{67}
$$

$$
\begin{aligned}
&RHS_1\\
&= \lambda_1 \sum_{n_2=1}^{n-1} \widetilde{\pi}(n_2,n-1) y^{n_2}\\
&= \lambda_1 \left[\sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n-1) y^{n_2} - \widetilde{\pi}(n,n-1) y^{n} \right]\\
&= \lambda_1 \left[\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1) y^{n_2} - \widetilde{\pi}(n,n-1) y^{n} - \widetilde{\pi}(0,n-1) y^{0} \right]\\
&= \lambda_1 \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1) y^{n} - \widetilde{\pi}(0,n-1) \right]
\end{aligned} \tag{68}
$$

$$
\begin{aligned}
&RHS_2\\
&= \lambda_2 \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1) y^{n_2}\\
&= \lambda_2 \left[y \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1) y^{n_2-1} \right]\\
&= \lambda_2 \left[y \sum_{n_2=0}^{n-1} \widetilde{\pi}(n_2,n-1) y^{n_2} \right]\\
&= \lambda_2 \left[y \left(\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1) y^{n_2} - \widetilde{\pi}(n,n-1) y^{n} \right) \right]\\
&= \lambda_2 y \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1) y^{n} \right]
\end{aligned} \tag{69}
$$

$$
\begin{aligned}
&RHS_3\\
&= \mu \widetilde{\pi}(n+1,n+1) y^{n} + \mu \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1) y^{n_2}\\
&= \mu \left[\widetilde{\pi}(n+1,n+1) y^{n} + \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1) y^{n_2} \right]\\
&= \mu \left[\widetilde{\pi}(n+1,n+1) y^{n} + \sum_{n_2=0}^{n+1} \widetilde{\pi}(n_2,n+1) y^{n_2} - \widetilde{\pi}(0,n+1) y^{0} - \widetilde{\pi}(n+1,n+1) y^{n+1} \right]\\
&= \mu \left[\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1) + \widetilde{\pi}(n+1,n+1) y^{n} (1-y) \right]
\end{aligned} \tag{70}
$$

$$
\begin{aligned}
&RHS_4 \\
= \quad & \gamma_1 \sum_{n_2=1}^{n} (n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2} \\
= \quad & \gamma_1 \left[y \sum_{n_2=1}^{n} (n-(n_2-1))\widetilde{\pi}(n_2-1,n)y^{n_2-1} \right] \\
= \quad & \gamma_1 \left[y \sum_{k=0}^{n-1} (n-k)\widetilde{\pi}(k,n)y^{k} \right] \\
= \quad & \gamma_1 \left[y \sum_{k=0}^{n} (n-k)\widetilde{\pi}(k,n)y^{k} \right] \quad \text{(since the } k=n \text{ term is 0)} \\
= \quad & \gamma_1 y \left[n \sum_{k=0}^{n} \widetilde{\pi}(k,n)y^{k} - y \sum_{k=1}^{n} k\widetilde{\pi}(k,n)y^{k-1} \right] \\
= \quad & \gamma_1 y \left[n\widetilde{P}(y,n) - y\frac{d}{dy}\widetilde{P}(y,n) \right]
\end{aligned} \tag{71}
$$

$$
\begin{aligned}
&RHS_5 \\
= \quad & \gamma_2 \sum_{n_2=1}^{n-1} (n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2} \\
= \quad & \gamma_2 \sum_{k=2}^{n} k\widetilde{\pi}(k,n)y^{k-1} \quad \text{(letting } k=n_2+1\text{)} \\
= \quad & \gamma_2 \left[\sum_{k=1}^{n} k\widetilde{\pi}(k,n)y^{k-1} - 1\cdot\widetilde{\pi}(1,n)y^{0} \right] \\
= \quad & \gamma_2 \left[\frac{d}{dy}\widetilde{P}(y,n) - \widetilde{\pi}(1,n) \right]
\end{aligned} \tag{72}
$$

$$
\begin{aligned}
&RHS_6 \\
= \quad & \theta_1 \sum_{n_2=1}^{n} (n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2} \\
= \quad & \theta_1 \left[(n+1) \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1)y^{n_2} - \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n+1)y^{n_2} \right] \\
= \quad & \theta_1 \left[(n+1) \left(\sum_{n_2=0}^{n+1} \widetilde{\pi}(n_2,n+1)y^{n_2} - \widetilde{\pi}(0,n+1) - \widetilde{\pi}(n+1,n+1)y^{n+1} \right) \right] \\
- \, & \theta_1 \left[y \left(\sum_{n_2=0}^{n+1} n_2\widetilde{\pi}(n_2,n+1)y^{n_2-1} - (n+1)\widetilde{\pi}(n+1,n+1)y^{n} \right) \right] \\
= \quad & \theta_1 \left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - (n+1)\widetilde{\pi}(n+1,n+1)y^{n+1} \right] \\
- \, & \theta_1 \left[y\frac{d}{dy}\widetilde{P}(y,n+1) - (n+1)\widetilde{\pi}(n+1,n+1)y^{n+1} \right] \\
= \quad & \theta_1 \left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - y\frac{d}{dy}\widetilde{P}(y,n+1) \right]
\end{aligned} \tag{73}
$$

RHS_7

$$
\begin{aligned}
&= \theta_2 \sum_{n_2=1}^{n} (n_2+1)\widetilde{\pi}(n_2+1, n+1) y^{n_2} \\
&= \theta_2 \sum_{k=2}^{n+1} k\widetilde{\pi}(k, n+1) y^{k-1} \quad (\text{letting } k = n_2+1) \\
&= \theta_2 \left[\sum_{k=1}^{n+1} k\widetilde{\pi}(k, n+1) y^{k-1} - 1 \cdot \widetilde{\pi}(1, n+1) y^0 \right] \\
&= \theta_2 \left[\frac{d}{dy} \widetilde{P}(y, n+1) - \widetilde{\pi}(1, n+1) \right]
\end{aligned}
\tag{74}
$$

Collecting:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\right] \\
&+\gamma_1\left[n\big(\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\big)-y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+\gamma_2 y\frac{d}{dy}\widetilde{P}(y,n) \\
&+\theta_1\left[n\big(\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\big)-y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+\theta_2 y\frac{d}{dy}\widetilde{P}(y,n) \\
&\quad = \lambda_1\left[\widetilde{P}(y,n-1)-\widetilde{\pi}(n,n-1)y^n-\widetilde{\pi}(0,n-1)\right] \\
&\quad +\lambda_2 y\left[\widetilde{P}(y,n-1)-\widetilde{\pi}(n,n-1)y^n\right] \\
&\quad +\mu\left[\widetilde{P}(y,n+1)-\widetilde{\pi}(0,n+1)+\widetilde{\pi}(n+1,n+1)y^n(1-y)\right] \\
&\quad +\gamma_1 y\left[n\widetilde{P}(y,n)-y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&\quad +\gamma_2\left[\frac{d}{dy}\widetilde{P}(y,n)-\widetilde{\pi}(1,n)\right] \\
&\quad +\theta_1\left[(n+1)\big(\widetilde{P}(y,n+1)-\widetilde{\pi}(0,n+1)\big)-y\frac{d}{dy}\widetilde{P}(y,n+1)\right] \\
&\quad +\theta_2\left[\frac{d}{dy}\widetilde{P}(y,n+1)-\widetilde{\pi}(1,n+1)\right]
\end{aligned}
$$

Regrouping:

$$
\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n) \\
&\quad = (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)+(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1) \\
&\quad + \{[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n]\,\widetilde{\pi}(0,n)-\lambda_1\widetilde{\pi}(0,n-1)-[\mu+\theta_1(n+1)]\,\widetilde{\pi}(0,n+1) \\
&\quad\quad -\gamma_2\widetilde{\pi}(1,n)-\theta_2\widetilde{\pi}(1,n+1)\} \\
&\quad - y^n(\lambda_1+\lambda_2 y)\widetilde{\pi}(n,n-1)+\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
$$

The y-boundary equation (32) makes the bracketed terms vanish; since $\widetilde{\pi}(n,n-1)=0$ (outside $\widetilde{S}$):

$$
\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n) \\
&\quad = (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)+(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1) \\
&\quad + \mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
\tag{75}
$$

□

5 Analysis of Model-X $(\gamma_1, \gamma_2 > 0,\ \theta_1, \theta_2 > 0)$

Model-X is the full model: it retains *both* bidirectional jockeying ($\gamma_1, \gamma_2 > 0$) *and* two-class abandonment ($\theta_1, \theta_2 > 0$). It is the parent of the hierarchy, and every other model analysed in this work is recovered from it by zeroing parameters. Jockeying conserves the total number of customers while abandonment removes them; the second mechanism makes the chain positive recurrent for all loads, so the stationary distribution always exists. As in Model-B, jockeying breaks the per-class Poisson/renewal structure — a Class-2 customer can no longer identify its effective service time with a Class-1 busy period — and abandonment, although it leaves each arrival stream Poisson, couples the two queues through the state-dependent rates $\theta_i n_i$; in either case the Pollaczek–Khinchine probabilistic route is unavailable. Applying the method of characteristics to the fundamental equation still yields a general solution, but closing it — i.e., determining the unknown boundary function $P_y(y)$ — requires solving a Volterra-type integral equation that does not admit a closed form. In fact the obstruction is strictly more severe than in Model-B, on two counts developed below: the characteristics are no longer straight lines but integral curves of a nonlinear ODE, and the diagonal value $P(z, z)$ that anchored the reduction in Model-B is itself no longer known. Model-X is therefore intractable and is left open; its role is to expose the precise obstructions that its tractable specialisations A, B, B_2 and C_2 each circumvent. The queue layout is the left panel of Figure 1.

Stability. Model-X has abandonment from both classes, so the total departure rate in state (n_1, n_2) is $\mu + \theta_1 n_1 + \theta_2 n_2$, which grows without bound in every direction of the (n_1, n_2)-plane. The chain is therefore positive recurrent for *all* $\lambda_1, \lambda_2, \mu > 0$ and all $\theta_1, \theta_2 > 0$; no traffic-intensity restriction is required (cf. the Erlang-A queue, §3.3). Setting $\theta_1 = \theta_2 = 0$ removes this and returns the Model-B/Model-A condition $\rho = (\lambda_1 + \lambda_2)/\mu < 1$.

Corollary 1. *The empty-system probability π_0 (server idle, no one waiting) and the both-queues-empty busy probability $\pi(0, 0)$ (server busy, $N_1 = N_2 = 0$) of Model-X satisfy the universal idle balance*

$$\pi(0,0) = \rho\,\pi_0, \qquad \rho = \frac{\lambda_1 + \lambda_2}{\mu}, \tag{76}$$

as in every model of this work (23). Unlike Models A and B, however, abandonment makes the total-customer process a state-dependent birth–death chain rather than an $M/M/1$, so $\pi_0 \neq 1 - \rho$: the idle probability depends on θ_1, θ_2 and is fixed only by the global normalisation $P(1, 1) = 1 - \pi_0$ once P is known. Correspondingly, the diagonal of the PGF no longer collapses to $\pi(0, 0)/(1-\rho z)$; setting $x = y = z$ in the fundamental equation leaves the abandonment-corrected relation

$$(\mu - \lambda z)\, P(z, z) + z\big[\theta_1\, \partial_x P + \theta_2\, \partial_y P\big]_{(z,z)} = \mu\, \pi(0, 0), \qquad \lambda = \lambda_1 + \lambda_2. \tag{77}$$

As $\theta_1, \theta_2 \to 0^+$ the gradient terms drop out and (77) returns the Model-B identities $P(z, z) = \pi(0, 0)/(1 - \rho z)$, $\pi_0 = 1 - \rho$ and $\pi(0, 0) = \rho(1 - \rho)$ (151), (109).

Proof. Setting $x = y$ in the fundamental equation (42), both jockeying terms carry the factor $(x - y)$ and vanish, as does the boundary term $\mu(x - y)P_y(y)$; the abandonment terms, carrying instead the factor $(x-1)$, survive. The coefficient of P becomes $(\lambda_1+\lambda_2+\mu)z^2-\mu z-(\lambda_1+\lambda_2)z^3 = z(z - 1)(\mu - \lambda z)$, and the right-hand side becomes $-\mu z(1 - z)\pi(0, 0) = \mu z(z - 1)\pi(0, 0)$; dividing the surviving terms by the common factor $z(z - 1)$ (for $z \in (0, 1)$) leaves

$$z(z - 1)(\mu - \lambda z)P(z, z) + z^2(z - 1)\big[\theta_1\, \partial_x P + \theta_2\, \partial_y P\big]_{(z,z)} = \mu z(z - 1)\, \pi(0, 0), \tag{78}$$

which is (77). The idle state (0) exchanges probability flux only with the both-queues-empty busy state, so global balance across the idle–busy cut reads $\lambda\pi_0 = \mu\pi(0, 0)$ (29) (an arrival at total rate λ leaves (0); a service completion at rate μ returns to it), whence $\pi(0, 0) = \rho\pi_0$, which is (76). In Models A and B the gradient terms in (77) are absent ($\theta_1 = \theta_2 = 0$), so $P(z, z) = \pi(0, 0)/(1 - \rho z)$; evaluating at $z = 1$ with $P(1, 1) = 1 - \pi_0$ then forces $(1 - \pi_0)(1 - \rho) = \rho\pi_0$, i.e. $\pi_0 = 1 - \rho$. With abandonment those gradient terms are present and *not* known — exactly the phenomenon isolated for Model-C_2 (269) — so the diagonal no longer determines $P(z, z)$, and π_0 cannot be read off in closed form: it follows only from the full solution via normalisation. □

Claim 1. *Under positive recurrence, the bivariate PGF of Model-X satisfies, on each characteristic curve $\Phi(x, y) = C_1$ of the fundamental equation, the integral representation*

$$P(x, y(x)) \;=\; P_h(x, y(x)) \int_0^x \frac{h\big(t;\, C_1\big)}{P_h\big(t,\, y(t)\big)}\, dt, \tag{79}$$

where P_h is the homogeneous solution (91)*, $y(t) = y(t; C_1)$ is the characteristic parametrisation, Φ is the first integral* (89) *(a transcendental function of (x, y), not the linear form of Model-B), and $h(t; C_1)$ is a forcing function depending on the unknown boundary trace $P_y(y) = P(0, y)$. Evaluating* (79) *along the characteristic through a diagonal point (z, z) reduces the problem to a* Volterra integral equation of the first kind *for P_y:*

$$\mathcal{F}(z) \;=\; \int_0^z \mathcal{K}(z,t)\, P_y\big(y(t;z)\big)\, dt, \tag{80}$$

whose kernel $\mathcal{K}$ and right-hand side $\mathcal{F}$ are defined in (102)*–*(104) *below. The kernel $\mathcal{K}$ is built from the non-elementary homogeneous solution P_h and does not simplify for general $(\gamma_1, \gamma_2, \theta_1, \theta_2)$; moreover, in contrast to Model-B, the right-hand side $\mathcal{F}$ is* not *fully known — it is coupled through* (77) *to the diagonal gradient $[\theta_1 \partial_x P + \theta_2 \partial_y P]_{(z,z)}$. Equation* (80) *therefore does not reduce to a closed-form solution for P_y, and Model-X is left open.*

Proof of Claim 1. The proof follows the five steps of Model-B: (i) homogeneous solution via characteristics, (ii) behaviour of P_h at the boundaries, (iii) particular solution via variation of parameters, (iv) pinning the free function from the boundary condition, and (v) diagonal evaluation yielding the Volterra equation. At each step we point out where the abandonment terms θ_i obstruct the closed-form algebra available in Model-B.

Step 1: Homogeneous solution via the method of characteristics.

The fundamental equation (42) is a first-order linear PDE in $P(x, y)$. Setting the right-hand side to zero and dividing through by y yields the homogeneous PDE

$$x\big[\gamma_1(x-y) + \theta_1(x-1)\big]\,\frac{\partial P}{\partial x} + x\big[\gamma_2(y-x) + \theta_2(y-1)\big]\,\frac{\partial P}{\partial y} = -A(x,y)\,P, \tag{81}$$

with the same coefficient polynomial as Model-B,

$$A(x,y) := (\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy; \tag{82}$$

abandonment changes only the two convection (drift) coefficients, adding the terms $\theta_1(x-1)$ and $\theta_2(y-1)$. Following the method of characteristics (§3.6.3), we associate to (81) the characteristic system

$$\frac{dx}{x\big[\gamma_1(x-y) + \theta_1(x-1)\big]} = \frac{dy}{x\big[\gamma_2(y-x) + \theta_2(y-1)\big]} = \frac{dP}{-A(x,y)\,P}. \tag{83}$$

Taking the ratio of the first two differentials, the common factor x cancels:

$$\frac{dy}{dx} = \frac{\gamma_2(y-x) + \theta_2(y-1)}{\gamma_1(x-y) + \theta_1(x-1)} = \frac{-\gamma_2 x + (\gamma_2+\theta_2)y - \theta_2}{(\gamma_1+\theta_1)x - \gamma_1 y - \theta_1}. \tag{84}$$

In Model-B ($\theta_1 = \theta_2 = 0$) both affine forms shared the factor $(x - y)$ and the ratio collapsed to the constant $-\gamma_2/\gamma_1$, integrating to the straight lines $\gamma_2 x + \gamma_1 y = \text{const}$. With abandonment this collapse no longer occurs: (84) is a genuine first-order ODE whose right-hand side is a ratio of two affine forms, and its integral curves are not straight lines.

Both affine forms in (84) vanish simultaneously only at the single point $(x, y) = (1, 1)$ (the unique intersection of the lines $(\gamma_1 + \theta_1)x - \gamma_1 y - \theta_1 = 0$ and $-\gamma_2 x + (\gamma_2 + \theta_2)y - \theta_2 = 0$), which is therefore the lone singular point of the characteristic field. Shifting to it via $X = x - 1$, $Y = y - 1$ removes the constant terms and renders (84) homogeneous:

$$\frac{dY}{dX} = \frac{-\gamma_2 X + (\gamma_2+\theta_2)Y}{(\gamma_1+\theta_1)X - \gamma_1 Y}. \tag{85}$$

The substitution $v = Y/X$ then separates the variables,

$$\frac{(\gamma_1+\theta_1) - \gamma_1 v}{Q(v)}\, dv = \frac{dX}{X}, \qquad Q(v) := \gamma_1 v^2 + (\gamma_2 + \theta_2 - \gamma_1 - \theta_1)\, v - \gamma_2. \tag{86}$$

The quadratic Q has discriminant $D = (\gamma_2 + \theta_2 - \gamma_1 - \theta_1)^2 + 4\gamma_1\gamma_2 > 0$ and real roots of opposite sign (their product is $-\gamma_2/\gamma_1 < 0$),

$$v_\pm = \frac{-(\gamma_2 + \theta_2 - \gamma_1 - \theta_1) \pm \sqrt{D}}{2\gamma_1}, \qquad v_- < 0 < v_+. \tag{87}$$

Decomposing the left-hand side of (86) by partial fractions over $Q(v) = \gamma_1(v - v_+)(v - v_-)$,

$$\frac{(\gamma_1 + \theta_1) - \gamma_1 v}{\gamma_1(v - v_+)(v - v_-)} = \frac{a_+}{v - v_+} + \frac{a_-}{v - v_-}, \qquad a_\pm = \frac{(\gamma_1 + \theta_1) - \gamma_1 v_\pm}{\gamma_1\,(v_\pm - v_\mp)}, \qquad a_+ + a_- = -1, \tag{88}$$

(the relation $a_+ + a_- = -1$ is the residue of the integrand at $v = \infty$), and integrating gives $a_+ \ln(v - v_+) + a_- \ln(v - v_-) = \ln X + \text{const}$. Exponentiating and returning to (x, y) via $v - v_\pm = (Y - v_\pm X)/X$ and $a_+ + a_- = -1$, the powers of X cancel and we obtain the first integral

$$\Phi(x, y) := \big[(y - 1) - v_+(x - 1)\big]^{a_+}\,\big[(y - 1) - v_-(x - 1)\big]^{a_-} = C_1, \tag{89}$$

constant along each characteristic. Unlike Model-B's linear constant $\gamma_2 x + \gamma_1 y$, the first integral (89) is a product of non-integer powers of two linear forms — a transcendental level curve. As a consistency check, letting $\theta_1, \theta_2 \to 0^+$ gives $Q(v) \to (\gamma_1 v + \gamma_2)(v - 1)$, so $v_+ \to 1$, $v_- \to -\gamma_2/\gamma_1$, and $(a_+, a_-) \to (0, -1)$; then $\Phi(x, y) \to \big[(y-1) + (\gamma_2/\gamma_1)(x-1)\big]^{-1}$, whose level curves are exactly the straight lines $\gamma_2 x + \gamma_1 y = \text{const}$ of Model-B.

To find P along a characteristic we take the ratio of the first and third differentials in (83),

$$\frac{dP}{P} = \frac{-A\big(x, y(x)\big)}{x\big[\gamma_1(x - y(x)) + \theta_1(x - 1)\big]}\,dx, \tag{90}$$

where $y(x) = y(x; C_1)$ is the characteristic (89). Integrating yields the general homogeneous solution

$$P_h(x, y) = F(C_1)\,\exp\!\left(-\int_{x_\star}^{x} \frac{A\big(s, y(s)\big)}{s\big[\gamma_1(s - y(s)) + \theta_1(s - 1)\big]}\,ds\right), \qquad C_1 = \Phi(x, y), \tag{91}$$

where F is an arbitrary differentiable function of the characteristic constant and $x_\star$ is an arbitrary base point (its choice is absorbed into F). Here the parallel with Model-B ends in closed form: because $y(s)$ is the transcendental characteristic (89), the integrand in (91) is not a rational function of s, and the quadrature is not elementary. Model-B, by contrast, could substitute the linear $y(s)$, partial-fraction the rational integrand, and read off the explicit factors $x^{-\mu/C_1}(x - y)^E$ with a closed-form exponent E. No such explicit P_h exists for Model-X.

Step 2: Boundary behaviour of P_h.

Although the quadrature (91) is not elementary, the two qualitative features needed downstream survive, and a third feature of Model-B is lost.

(a) Blow-up at $x = 0$. A characteristic meets the axis $x = 0$ at a finite height $y_0 := y(0; C_1) \in (0, 1]$, where $A(0, y_0) = -\mu$ and the convection coefficient is $\gamma_1(0 - y_0) + \theta_1(0 - 1) = -(\gamma_1 y_0 + \theta_1) < 0$. Hence the integrand in (91) behaves like $-\mu/[(\gamma_1 y_0 + \theta_1)\,s]$ as $s \to 0^+$, so

$$P_h\big(x, y(x)\big) \;\sim\; x^{-\mu/(\gamma_1 y_0 + \theta_1)} \;\longrightarrow\; +\infty \qquad (x \to 0^+), \tag{92}$$

with a strictly negative exponent because $\gamma_1 y_0 + \theta_1 > 0$. This is exactly the factor $x^{-\mu/C_1}$ of Model-B, now with the denominator shifted by θ_1; it drives the pinning of F in Step 4.

(b) Regularity on the diagonal. In Model-B the convection coefficient was $\gamma_1 x(x - y)$, which *vanishes* on the diagonal $x = y$; the resulting factor $(x - y)^E$ with $E > 0$ made $P_h \to 0$ there and forced the delicate Hadamard finite-part regularisation of Model-B's Step 5. For Model-X the coefficient is $x[\gamma_1(x - y) + \theta_1(x - 1)]$, which at $x = y = z$ equals $z\,\theta_1(z - 1) \neq 0$ for $z \in (0, 1)$: the abandonment term $\theta_1(x - 1)$ keeps the characteristic field away from zero on the diagonal. Consequently the integrand in (91) is regular as $x \to z$ along the diagonal-crossing characteristic, and

$$P_h^{\text{diag}}(z) := \lim_{x \to z} P_h\big(x, y(x; z)\big) \quad \text{is finite and non-zero.} \tag{93}$$

The diagonal singularity of Model-B thus disappears; as we will see in Step 5, the obstruction migrates instead to the right-hand side of the Volterra equation.

Branch convention. As in Model-B, the non-integer powers in (89) and the quadrature in (91) may carry a complex phase off the real region; this phase is absorbed into the arbitrary function $F(C_1)$ and cancels exactly in the ratio $P_h(x, y(x))/P_h(t, y(t))$ appearing in (79), since numerator and denominator lie on the same characteristic (same C_1).

Step 3: Particular solution via variation of parameters.

We now restrict the full (inhomogeneous) fundamental equation (42) to a fixed characteristic. Along $y = y(x; C_1)$ the total derivative is $\frac{d}{dx}P(x, y(x)) = \partial_x P + y'(x)\,\partial_y P$, and by the characteristic ODE (84) the bracket multiplying $\partial_y P$ equals $[\gamma_1(x-y)+\theta_1(x-1)]\,y'(x)$, so the two convection terms combine:

$$xy\big[\gamma_1(x-y)+\theta_1(x-1)\big]\partial_x P + xy\big[\gamma_2(y-x)+\theta_2(y-1)\big]\partial_y P = xy\big[\gamma_1(x-y)+\theta_1(x-1)\big]\frac{d}{dx}P(x, y(x)).$$

Substituting into (42) and dividing by $xy\big[\gamma_1(x-y)+\theta_1(x-1)\big]$ reduces the PDE to the first-order linear ODE (for fixed C_1)

$$\frac{dP}{dx} = f(x; C_1)\,P + h(x; C_1), \tag{94}$$

with coefficient and forcing

$$f(x; C_1) = -\frac{A\big(x, y(x)\big)}{x\big[\gamma_1(x-y(x))+\theta_1(x-1)\big]}, \tag{95}$$

$$h(x; C_1) = \frac{G\big(x, y(x)\big)}{x\,y(x)\big[\gamma_1(x-y(x))+\theta_1(x-1)\big]}, \tag{96}$$

where A is as in (82) and the right-hand side of the fundamental equation is, as in Model-B,

$$G(x, y) := \mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0). \tag{97}$$

Comparing (95)–(96) with their Model-B counterparts, the denominator $\gamma_1 x(x-y)$ is replaced throughout by $x[\gamma_1(x-y)+\theta_1(x-1)]$ — the only footprint of abandonment, and the source of all the difficulty. Since P_h from (91) satisfies the homogeneous equation $\frac{d}{dx}P_h = f(x; C_1)P_h$, it is the fundamental solution of (94); applying variation of parameters (§3.6.2), the general solution of the inhomogeneous ODE is

$$P(x, y(x)) = P_h(x, y(x))\left[F(C_1) + \int_{x_0}^{x}\frac{h(t; C_1)}{P_h(t, y(t))}\,dt\right]. \tag{98}$$

Step 4: Pinning the free function from the boundary condition at $x = 0$.

Along any characteristic the endpoint at $x = 0$ is $y(0) = y_0$, giving $P(0, y_0) = P_y(y_0)$, which is finite. However, by (92) the homogeneous factor blows up, $P_h(x, y(x)) \to +\infty$ as $x \to 0^+$, while the integral in (98) vanishes as $x_0 \to 0$ (the integration interval collapses). Therefore

$$\lim_{x\to 0^+} P(x, y(x)) = \lim_{x\to 0^+} P_h(x, y(x)) \cdot F(C_1) = +\infty \cdot F(C_1),$$

which can be finite only if $F(C_1) = 0$. Setting $x_0 = 0$ and $F(C_1) = 0$ in (98) gives the integral representation of Claim 1,

$$P(x, y(x)) = P_h(x, y(x)) \int_0^x \frac{h(t; C_1)}{P_h(t, y(t))}\,dt. \tag{99}$$

This argument is identical to Model-B's; abandonment shifts the blow-up exponent in (92) from μ/C_1 to $\mu/(\gamma_1 y_0 + \theta_1)$ but leaves it strictly positive, so the conclusion $F \equiv 0$ is unchanged.

Step 5: Reduction to the Volterra equation.

We simplify the forcing h on the characteristic. Substituting G from (97) into (96) and cancelling one factor of t in the second term,

$$h(t; C_1) = \frac{\mu\,(t-y(t))}{t\,y(t)\big[\gamma_1(t-y(t))+\theta_1(t-1)\big]}\,P_y\big(y(t)\big) - \frac{\mu\,(1-y(t))}{y(t)\big[\gamma_1(t-y(t))+\theta_1(t-1)\big]}\,\pi(0,0). \tag{100}$$

We evaluate (99) along the characteristic through a diagonal point (z, z), that is, the curve with $C_1 = \Phi(z, z)$; its parametrisation $y(t; z)$ satisfies $y(z; z) = z$. Setting $x = z$,

$$P(z, z) = P_h(z, z) \int_0^z \frac{h\big(t; \Phi(z, z)\big)}{P_h\big(t, y(t; z)\big)}\,dt. \tag{101}$$

By the regularity (93), $P_h(z, z) = P_h^{\mathrm{diag}}(z)$ is finite and non-zero and the integrand is regular as $t \to z$ (its numerator $t - y(t; z) \to 0$), so (101) is a *proper* integral — no finite-part regularisation is needed, in contrast to Model-B. Substituting the two pieces of (100) and separating the term containing the unknown P_y from the known remainder, define the kernel and remainder

$$\mathcal{K}(z,t) := \frac{\mu\,\big(t - y(t;z)\big)\,P_h^{\mathrm{diag}}(z)}{t\,y(t;z)\big[\gamma_1(t - y(t;z)) + \theta_1(t-1)\big]\,P_h\big(t, y(t;z)\big)}, \tag{102}$$

$$\mathcal{R}(z) := -\,\mu\,\pi(0,0)\,P_h^{\mathrm{diag}}(z)\int_0^z \frac{1 - y(t;z)}{y(t;z)\big[\gamma_1(t - y(t;z)) + \theta_1(t-1)\big]\,P_h\big(t, y(t;z)\big)}\,dt, \tag{103}$$

so that (101) reads $P(z,z) = \int_0^z \mathcal{K}(z,t)\,P_y(y(t;z))\,dt + \mathcal{R}(z)$. Setting

$$\mathcal{F}(z) := P(z,z) - \mathcal{R}(z) = \underbrace{\frac{\pi(0,0)}{1-\rho z}}_{\text{Model-}B\text{ value}} - \underbrace{\frac{z}{\mu - \lambda z}\big[\theta_1\partial_x P + \theta_2\partial_y P\big]_{(z,z)}}_{\text{abandonment coupling}} - \mathcal{R}(z), \tag{104}$$

where the expression for $P(z,z)$ is the abandonment-corrected diagonal (77), equation (101) becomes the Volterra integral equation of the first kind (80), confirming Claim 1. Two features distinguish (80) from its Model-B analogue, and both make Model-X strictly less tractable. First, the kernel (102) is built from the non-elementary homogeneous solution P_h along transcendental characteristics, so it has no closed form for general parameters (in Model-B, P_h and the straight-line characteristics were explicit). Second, and decisively, the right-hand side (104) is *not* known data: through (77) it is coupled to the diagonal gradient $[\theta_1\partial_x P + \theta_2\partial_y P]_{(z,z)}$, which is precisely the quantity that abandonment prevents the diagonal from determining (269). Where Model-B left a clean first-kind Volterra equation with explicit kernel and known forcing (itself already beyond closed-form inversion), Model-X leaves one whose kernel is implicit and whose forcing is entangled with the unknown solution. Resolving it lies well beyond the scope of this thesis. □

Model-X is therefore left open at this point. Its role in the hierarchy is to make the two obstructions explicit and to show how each tractable specialisation removes them. The limit $\theta_1 = \theta_2 \to 0$ (Model-B) restores both at once: the characteristics (89) straighten to the lines $\gamma_2 x + \gamma_1 y = \text{const}$, so the kernel (102) becomes explicit, and the gradient coupling in (104) vanishes, so $\mathcal{F}(z) = \pi(0,0)/(1-\rho z) - \mathcal{R}(z)$ becomes known data — leaving the clean (still unsolved) Volterra equation of Section 7. Removing one mechanism at a time then closes the problem entirely: setting in addition $\gamma_2 = 0$ (Model-B_2) collapses the characteristic family to vertical lines and degenerates the PDE to an ODE in x alone, whose boundary function $P_y(y)$ is pinned by an analyticity argument at the interior singular point $x = y$ (Section 8); removing jockeying entirely while keeping only class-1 abandonment (Model-C_2) likewise degenerates the PDE to an ODE in x, whose $P_y(y)$ is pinned by a boundedness argument at $x = 1$ (Section 9); and zeroing all four parameters returns the fully solvable baseline Model-A (Section 6). The full model X, retaining all mechanisms, retains both obstructions simultaneously and is the one case that does not close.

6 Analysis of Model-A ($\gamma_1 = \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$)

Model-A is the baseline of this collection of models: all jockeying and abandonment parameters are set to zero, leaving an ordinary two-class $M/M/1$ queueing model with non-preemptive priority. The analysis is carried out in both state spaces S and $\widetilde{S}$: on the former, two complete proofs are presented; while on the latter, an approximate solution and its verification are provided.

The balance equations are obtained by plugging $\gamma_1 = \gamma_2 = 0$ and $\theta_1 = \theta_2 = 0$ into Equations (25), (27), (26), (28), and (29) for state space S, and Equations (30), (31), (32), (33), and (34) for state space $\widetilde{S}$. The corresponding starting points are Lemma 1 for S and Lemma 2 for $\widetilde{S}$.

Stability. Since there is no abandonment, the system is positive recurrent if and only if the total traffic intensity satisfies

$$\rho = \frac{\lambda_1 + \lambda_2}{\mu} < 1. \tag{105}$$

The figure below corresponds to the queueing diagram of Model-A.

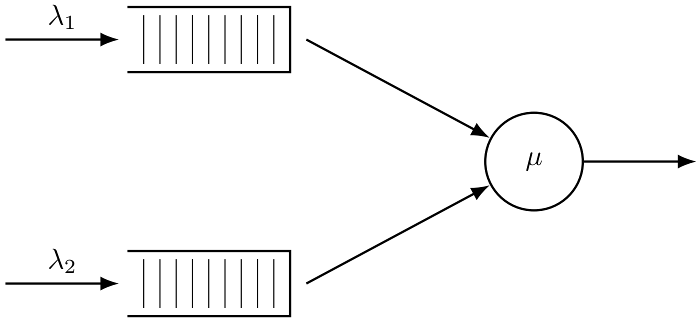

Figure 4: Queue layout for Model-A. Only Poisson arrivals and exponential service are present; all jockeying and abandonment rates are zero.

6.1 Analysis of Model-A: state space S

We begin from Equation (42) with $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$, in which the marginal $P_y(y) = P(0, y)$ is still to be determined:

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x, y) = \mu(x - y)P_y(y) - \mu x(1 - y)\pi(0, 0). \tag{106}$$

The following theorem summarises the main result of this subsection; two standalone proofs are given, one analytical and one probabilistic.

Theorem 1. *Consider a queueing system with two priority classes with Poisson arrival rates λ_1 and λ_2, where class-1 customers have non-preemptive priority over those of class 2, and exponentially distributed service times with rate μ independently of the class. Under the stability condition (105), the joint PGF $P(x, y) = \mathbb{E}[x^{N_1} y^{N_2}]$ is*

$$P(x, y) = \left[(1 + \rho_1 + \rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1} \cdot y(1 - y) \left[\frac{x - x^*(y)}{x^*(y) - y}\right] \cdot \rho(1 - \rho), \tag{107}$$

where

$$x^*(y) = \frac{[\mu + \lambda_1 + \lambda_2(1 - y)] - \sqrt{[\mu + \lambda_1 + \lambda_2(1 - y)]^2 - 4\lambda_1\mu}}{2\lambda_1}. \tag{108}$$

Corollary 2. *Under the stability condition (105), the empty-state probabilities for Model-A are*

$$\pi_0 = 1 - \rho, \qquad \pi(0, 0) = \rho(1 - \rho). \tag{109}$$

In particular, $P(1, 1) = 1 - \pi_0 = \rho$ and the diagonal generating function satisfies $P(z, z) = \pi(0, 0)/(1 - \rho z)$.

Proof of Corollary 2. Setting $x = y = z$ in the fundamental equation (106), the boundary term $\mu(x - y)P_y(y)$ vanishes:

$$\left[(\lambda_1 + \lambda_2 + \mu)z^2 - \mu z - \lambda_1 z^3 - \lambda_2 z^3\right] P(z, z) = -\mu z(1 - z)\pi(0, 0).$$

Dividing both sides by z and factoring the bracket:

$$-(1 - z)\left[\mu - (\lambda_1 + \lambda_2)z\right] P(z, z) = -\mu(1 - z)\pi(0, 0).$$

Cancelling $(1 - z) \neq 0$ for $z \in [0, 1)$ and rearranging:

$$P(z, z) = \frac{\mu\, \pi(0, 0)}{\mu - \lambda z} = \frac{\pi(0, 0)}{1 - \rho z}, \tag{110}$$

where $\lambda = \lambda_1 + \lambda_2$ and the second equality follows by dividing numerator and denominator by μ.

Setting $z = 1$ and using the normalisation $P(1, 1) = 1 - \pi_0$:

$$1 - \pi_0 = \frac{\pi(0, 0)}{1 - \rho}. \tag{111}$$

The global balance equation for the idle state gives $\lambda\pi_0 = \mu\pi(0, 0)$, i.e. $\pi(0, 0) = \rho\pi_0$. Substituting into (111):

$$1 - \pi_0 = \frac{\rho\pi_0}{1 - \rho} \implies (1 - \pi_0)(1 - \rho) = \rho\pi_0 \implies \pi_0 = 1 - \rho,$$

and therefore $\pi(0, 0) = \rho(1 - \rho)$. $\square$

6.1.1 Analytical approach for Model-A in state space S

Proof. We begin from the fundamental equation for Model-A (106):

$$\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right] P(x,y)=\mu(x-y)P_y(y)-\mu x(1-y)\pi(0,0).$$

We apply the *Kernel Method*: fix $y \in (0,1]$ and find $x^*(y)$ such that the LHS of the expression above is zero. Since $P(x,y)$ is a PGF, it is bounded. At any such root the LHS vanishes, so the RHS must also vanish:

$$\mu\left[(x^*(y)-y)\,P_y(y)-x^*(y)(1-y)\,\pi(0,0)\right]=0.$$

To find the roots of the coefficient of $P(x,y)$, divide by $y\neq 0$ and set

$$\begin{aligned}0&=(\lambda_1+\lambda_2+\mu)x-\mu-\lambda_1x^2-\lambda_2xy\\0&=\lambda_1x^2-[\mu+\lambda_1+\lambda_2(1-y)]\,x+\mu,\end{aligned}$$

whose roots are

$$x^{\pm}(y)=\frac{[\mu+\lambda_1+\lambda_2(1-y)]\pm\sqrt{[\mu+\lambda_1+\lambda_2(1-y)]^2-4\mu\lambda_1}}{2\lambda_1}.$$

We label the root with the negative sign $x^*(y)$ and the root with the positive sign $x^+(y)$. At $y=1$:

$$x^*(1)=1 \quad\text{and}\quad x^+(1)=\mu/\lambda_1>1.$$

We require our root to lie inside the closed unit disk for $y\in[0,1)$. Computing the derivative of $x^*(y)$:

$$\frac{d}{dy}x^*(y)=\frac{\lambda_2}{2\lambda_1}\left[-1+\frac{\mu+\lambda_1+\lambda_2(1-y)}{\sqrt{(\mu+\lambda_1+\lambda_2(1-y))^2-4\lambda_1\mu}}\right]>0, \tag{112}$$

since $(\mu+\lambda_1+\lambda_2(1-y))^2>(\mu+\lambda_1+\lambda_2(1-y))^2-4\lambda_1\mu$ whenever $\lambda_1\mu>0$. Therefore $x^*(y)$ is strictly increasing on $[0,1]$, and since $x^*(1)=1$ we have $x^*(y)<1$ for $y\in[0,1)$. By Vieta's formulas for the kernel polynomial $\lambda_1x^2-[\mu+\lambda_1+\lambda_2(1-y)]x+\mu=0$, the product of the two roots satisfies $x^*(y)\,x^+(y)=\mu/\lambda_1>1$, so $x^+(y)=\mu/(\lambda_1x^*(y))>1$ throughout $[0,1)$. Hence the only root strictly inside the unit disk for $y\in[0,1)$ is $x^*(y)$:

$$x^*(y)=\frac{(\mu+\lambda_1+\lambda_2(1-y))-\sqrt{(\mu+\lambda_1+\lambda_2(1-y))^2-4\lambda_1\mu}}{2\lambda_1}. \tag{113}$$

Since subsequent steps involve L'Hôpital's rule at $y\to1$, we evaluate $\frac{d}{dy}x^*(y)$ from (112) at $y=1$. Under stability, $\rho_1=\lambda_1/\mu<\rho<1$ implies $\mu>\lambda_1$, so $\sqrt{(\mu-\lambda_1)^2}=\mu-\lambda_1$:

$$\frac{d}{dy}x^*(1)=\frac{\lambda_2}{2\lambda_1}\left[-1+\frac{\mu+\lambda_1}{\sqrt{(\mu+\lambda_1)^2-4\lambda_1\mu}}\right]=\frac{\lambda_2}{2\lambda_1}\left[-1+\frac{\mu+\lambda_1}{\sqrt{(\mu-\lambda_1)^2}}\right]=\frac{\lambda_2}{2\lambda_1}\left[\frac{\mu+\lambda_1-(\mu-\lambda_1)}{\mu-\lambda_1}\right]=\frac{\lambda_2}{\mu-\lambda_1}.$$

For future reference:

$$\frac{d}{dy}x^*(1)=\frac{\lambda_2}{\mu-\lambda_1}=\frac{\rho_2}{1-\rho_1}. \tag{114}$$

Returning to the fundamental equation (106) at $x=x^*(y)$, the RHS also equals zero, giving

$$\mu\cdot[(x^*(y)-y)P_y(y)-x^*(y)(1-y)\pi(0,0)]=0 \quad\Longrightarrow\quad P_y(y)=\frac{x^*(y)(1-y)}{x^*(y)-y}\pi(0,0). \tag{115}$$

By Corollary 2, $\pi(0,0)=\rho(1-\rho)$. Substituting this and Equation (115) into the fundamental equation (106):

$$\begin{aligned}\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)&=\mu(x-y)\frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho)-\mu x(1-y)\rho(1-\rho)\\\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)&=\mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y)-y}-x\right]\rho(1-\rho)\\\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)&=\mu(1-y)\left[\frac{y(x-x^*(y))}{x^*(y)-y}\right]\rho(1-\rho)\\\left[(1+\rho_1+\rho_2)xy-y-\rho_1x^2y-\rho_2xy^2\right]P(x,y)&=y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).\end{aligned}$$

Multiplying both sides by $\left[(1+\rho_1+\rho_2)xy-y-\rho_1x^2y-\rho_2xy^2\right]^{-1}$ isolates $P(x,y)$ and concludes the proof. □

6.1.2 Probabilistic approach for Model-A in state space S

Proof. The proof is similar to the analytical approach, but deriving the Class-2 marginal PGF $P_y(y)$ by means of probabilistic arguments. We begin by stating the fundamental blobal balance equation for Model-A (Equation (106)):

$$\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)=\mu\big[(x-y)P_y(y)-x(1-y)\pi(0,0)\big]. \tag{116}$$

Class-2 conditional distribution via $M/G/1$. Since class 1 has non-preemptive priority, class-2 customers are will not be taken into service whenever $N_1>0$; their *effective service time*, namely B, is therefore the duration of a complete Busy Period, of the single-class $M/M/1$ sub-queue with arrival rate λ_1 and service rate μ, so from class-2's viewpoint the system is an $M/G/1$ queue with arrivale rate λ_2 and generally distributed service times B with mean $\mathbb{E}[B]$. The LST and mean of B are presented in §3.1 (Equations (2)–(3)), now with λ_1 instead of λ:

$$\widetilde{B}(s)=\frac{(\mu+\lambda_1+s)-\sqrt{(\mu+\lambda_1+s)^2-4\mu\lambda_1}}{2\lambda_1}, \tag{117}$$

$$\mathbb{E}[B]=\frac{1}{\mu(1-\rho_1)}. \tag{118}$$

Applying the Pollaczek–Khinchine (PK) formula (§3.4, Equation (9)) to this $M/G/1$ system, with $\widetilde{B}(\lambda_2(1-y))=x^*(y)$ (cf. (113)) and $\lambda_2\mathbb{E}[B]=\rho_2/(1-\rho_1)$:

$$L_2(y)=\frac{(1-\lambda_2\mathbb{E}[B])(1-y)\widetilde{B}(\lambda_2(1-y))}{\widetilde{B}(\lambda_2(1-y))-y}=\frac{1-\rho}{1-\rho_1}\cdot\frac{x^*(y)(1-y)}{x^*(y)-y}. \tag{119}$$

By the PASTA property, the time-average distribution of N_2 conditional on (busy, $N_1=0$) coincides with the $M/G/1$ steady-state distribution, so $\mathbb{E}[y^{N_2}\mid \text{busy}, N_1=0]=L_2(y)$.

By the law of total expectation, $P_y(y)=\mathbb{P}(\text{busy},N_1=0)\cdot\mathbb{E}[y^{N_2}\mid\text{busy},N_1=0]$. So

$$P_y(y)=\mathbb{P}(\text{busy},N_1=0)\cdot\mathbb{E}[y^{N_2}\mid\text{busy},N_1=0]=\mathbb{P}(\text{busy},N_1=0)\cdot L_2(y).$$

We now can obtain $\mathbb{P}(\text{busy},N_1=0)$, from $\mathbb{P}(\text{busy})=\rho$ and $\mathbb{P}(N_1=0\mid\text{busy})=1-\rho_1$. So

$$\mathbb{P}(\text{busy},N_1=0)=\rho(1-\rho_1)$$

and therefore

$$P_y(y)=\rho(1-\rho_1)\cdot L_2(y)=\rho(1-\rho_1)\cdot\frac{1-\rho}{1-\rho_1}\cdot\frac{x^*(y)(1-y)}{x^*(y)-y}=\frac{x^*(y)(1-y)}{x^*(y)-y}\cdot(1-\rho)\rho \tag{120}$$

Substituting (120) and $\pi(0,0)=\rho(1-\rho)$ (Corollary 2) into (116):

$$\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)=\mu(x-y)\frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho)-\mu x(1-y)\rho(1-\rho)$$

$$\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)=\mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y)-y}-x\right]\rho(1-\rho)$$

$$\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\right]P(x,y)=\mu(1-y)\left[\frac{y(x-x^*(y))}{x^*(y)-y}\right]\rho(1-\rho)$$

$$\left[(1+\rho_1+\rho_2)xy-y-\rho_1x^2y-\rho_2xy^2\right]P(x,y)=y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).$$

Multiplying both sides by $\left[(1+\rho_1+\rho_2)xy-y-\rho_1x^2y-\rho_2xy^2\right]^{-1}$ isolates $P(x,y)$ and concludes the proof. $\square$

6.1.3 Partial-PGF recurrence approach for Model-A in state space S

This proof follows Cohen [Coh56]. It consists in keeping the class-1 index n_1 as a discrete parameter and forming a generating function only with respect to N_2, namely the PPGF $\widetilde{P}(n_1,y)=\sum_{n_2\geq 0}\pi(n_1,n_2)\,y^{n_2}$ introduced in (39). Multiplying the balance equations by y^{n_2} and summing over n_2 turns the two-dimensional balance equations into a second-order linear recurrence in n_1, whose characteristic roots fix the geometric structure of the solution in the priority class index.

Proof. We start by stating the balance equations for Model-A, with all jockeying and abandonment parameters set to zero (i.e. $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$) in the interior balance equation (Equation (25)) and our x-boundary equation (Equation (27)). We have

$$(\lambda_1 + \lambda_2 + \mu)\,\pi(n_1, n_2) = \mu\,\pi(n_1 + 1, n_2) + \lambda_1\,\pi(n_1 - 1, n_2) + \lambda_2\,\pi(n_1, n_2 - 1), \quad n_1 \geq 1,\ n_2 \geq 1, \tag{121}$$

$$(\lambda_1 + \lambda_2 + \mu)\,\pi(n_1, 0) = \mu\,\pi(n_1 + 1, 0) + \lambda_1\,\pi(n_1 - 1, 0), \quad n_1 \geq 1,\ n_2 = 0. \tag{122}$$

$$\begin{aligned}
&(\lambda_1 + \lambda_2 + \mu) \sum_{n_2=1}^{\infty} \pi(n_1, n_2) y^{n_2} \\
&\quad = \quad \mu \sum_{n_2=1}^{\infty} \pi(n_1 + 1, n_2) y^{n_2} + \lambda_1 \sum_{n_2=1}^{\infty} \pi(n_1 - 1, n_2) y^{n_2} + \lambda_2 \sum_{n_2=1}^{\infty} \pi(n_1, n_2 - 1) y^{n_2} \\
&LHS_1 \quad = \quad RHS_1 + RHS_2 + RHS_3 \\
&LHS_1 \\
&\quad = \quad (\lambda_1 + \lambda_2 + \mu) \sum_{n_2=1}^{\infty} \pi(n_1, n_2) y^{n_2} \\
&\quad = \quad (\lambda_1 + \lambda_2 + \mu) \left[\sum_{n_2=0}^{\infty} \pi(n_1, n_2) y^{n_2} - \pi(n_1, 0) y^0 \right] \\
&\quad = \quad (\lambda_1 + \lambda_2 + \mu) \left[\widetilde{P}(n_1, y) - \pi(n_1, 0) \right] \\
&RHS_1 \\
&\quad = \quad \mu \sum_{n_2=1}^{\infty} \pi(n_1 + 1, n_2) y^{n_2} \\
&\quad = \quad \mu \left[\sum_{n_2=0}^{\infty} \pi(n_1 + 1, n_2) y^{n_2} - \pi(n_1 + 1, 0) y^0 \right] \\
&\quad = \quad \mu \left[\widetilde{P}(n_1 + 1, y) - \pi(n_1 + 1, 0) \right] \\
&RHS_2 \\
&\quad = \quad \lambda_1 \sum_{n_2=1}^{\infty} \pi(n_1 - 1, n_2) y^{n_2} \\
&\quad = \quad \lambda_1 \left[\sum_{n_2=0}^{\infty} \pi(n_1 - 1, n_2) y^{n_2} - \pi(n_1 - 1, 0) y^0 \right] \\
&\quad = \quad \lambda_1 \left[\widetilde{P}(n_1 - 1, y) - \pi(n_1 - 1, 0) \right] \\
&RHS_3 \\
&\quad = \quad \lambda_2 \sum_{n_2=1}^{\infty} \pi(n_1, n_2 - 1) y^{n_2} \\
&\quad = \quad \lambda_2 y \sum_{n_2=1}^{\infty} \pi(n_1, n_2 - 1) y^{n_2 - 1} \\
&\quad = \quad \lambda_2 y \sum_{m=0}^{\infty} \pi(n_1, m) y^{m} \\
&\quad = \quad \lambda_2 y\, \widetilde{P}(n_1, y) \\
&(\lambda_1 + \lambda_2 + \mu) \big[\widetilde{P}(n_1, y) - \pi(n_1, 0) \big] \\
&\quad = \quad \mu \big[\widetilde{P}(n_1 + 1, y) - \pi(n_1 + 1, 0) \big] \\
&\qquad + \ \lambda_1 \big[\widetilde{P}(n_1 - 1, y) - \pi(n_1 - 1, 0) \big] \\
&\qquad + \ \lambda_2 y\, \widetilde{P}(n_1, y)
\end{aligned}$$

$$\begin{aligned}&[\mu+\lambda_1+\lambda_2(1-y)]\,\widetilde{P}(n_1,y)-\mu\widetilde{P}(n_1+1,y)-\lambda_1\widetilde{P}(n_1-1,y)\\&\quad=\quad(\lambda_1+\lambda_2+\mu)\pi(n_1,0)-\mu\pi(n_1+1,0)-\lambda_1\pi(n_1-1,0)\end{aligned}\tag{123}$$

Note that, using our x-boundary balance equation (Equation (122)), the right-hand side of the expression above (Equation (123)) vanishes completely. We obtain the homogeneous recurrence

$$[\mu+\lambda_1+\lambda_2(1-y)]\,\widetilde{P}(n_1,y)=\mu\widetilde{P}(n_1+1,y)+\lambda_1\widetilde{P}(n_1-1,y),\qquad n_1\geq 1.\tag{124}$$

Characteristic roots and mode selection. Equation (124) is a *second-order linear recurrence* in n_1 with constant coefficients in n_1. So its general solution is a linear combination of two geometric sequences $[\widetilde{x}^{\pm}(y)]^{n_1}$, where $\widetilde{x}^{\pm}(y)$ solve the characteristic equation

$$\mu\,\widetilde{x}^2-[\mu+\lambda_1+\lambda_2(1-y)]\,\widetilde{x}+\lambda_1=0,\tag{125}$$

that is,

$$\widetilde{x}^{\pm}(y)=\frac{\big(\mu+\lambda_1+\lambda_2(1-y)\big)\pm\sqrt{\big(\mu+\lambda_1+\lambda_2(1-y)\big)^2-4\lambda_1\mu}}{2\mu}.\tag{126}$$

The *characteristic polynomial* (Equation (125)) is the reciprocal to that of the bivariate PGF approaches (i.e. the reciprocal of $\lambda_1x^2-[\mu+\lambda_1+\lambda_2(1-y)]x+\mu$), thus its roots are the reciprocals of that kernel's roots: the positive root is the reciprocal of the negative one ($\widetilde{x}^{+}(y)=1/x^*(y)$), and conversely $\widetilde{x}^{-}(y)=1/x^{+}(y)$. By Vieta's formulas, $\widetilde{x}^{-}(y)\,\widetilde{x}^{+}(y)=\lambda_1/\mu=\rho_1<1$. Recall from §6.1.1 that $x^*(y)<1<x^{+}(y)$ for $y\in[0,1)$. The selection of the negative-sign root is motivated by the fact that we need the sequence $\{\widetilde{P}(n_1,y)\}_{n_1\geq 0}$ to be summable. In fact, we need

$$\sum_{n_1=0}^{\infty}\widetilde{P}(n_1,y)=P(1,y)\leq P(1,1)=1-\pi_0=\rho<\infty.$$

At $y=1$, we have

$$\widetilde{x}^{-}(1)=\rho_1\quad\text{and}\quad\widetilde{x}^{+}(1)=1,$$

the latter making the geometric sequence $[\widetilde{x}^{+}(y)]^{n_1}$ non-summable. We therefore define $\widetilde{x}^*(y):=\widetilde{x}^{-}(y)$,

$$\widetilde{x}^*(y):=\widetilde{x}^{-}(y)=\frac{\big(\mu+\lambda_1+\lambda_2(1-y)\big)-\sqrt{\big(\mu+\lambda_1+\lambda_2(1-y)\big)^2-4\lambda_1\mu}}{2\mu}=\rho_1\,x^*(y).\tag{127}$$

The last identity holds because $\widetilde{x}^*$ and x^* share the same numerator and differ only in the denominator (2μ versus $2\lambda_1$). Next,

$$\widetilde{P}(n_1,y)=\widetilde{P}(0,y)\,[\widetilde{x}^*(y)]^{n_1},\qquad n_1\geq 0,\tag{128}$$

where the relation also holds at $n_1=0$ trivially. Recall that at $y=1$, we have $\widetilde{x}^*(1)=\lambda_1/\mu=\rho_1$. Thus, $\widetilde{P}(n_1,1)=\widetilde{P}(0,1)\,\rho_1^{n_1}$ and $\widetilde{P}(0,1)=P_y(1)=\rho(1-\rho_1)$. The constant $\widetilde{P}(0,y)$ is resolved by its definition

$$\widetilde{P}(0,y)=\sum_{n_2=0}^{\infty}\pi(0,n_2)\,y^{n_2}=P(0,y)=P_y(y),$$

as determined in the bivariate PGF approaches. Summing the geometric law (Equation (128)) over $n_1\geq 0$ for $x\in(0,1]$ recovers

$$P(x,y)=\sum_{n_1=0}^{\infty}\widetilde{P}(n_1,y)\,x^{n_1}=P_y(y)\sum_{n_1=0}^{\infty}\big[\widetilde{x}^*(y)\,x\big]^{n_1}=\frac{P_y(y)}{1-\widetilde{x}^*(y)\,x}.\tag{129}$$

Now, in order to bring this to the form stated in Theorem 1, we define the kernel of the fundamental equation divided by μ as

$$K(x,y):=(1+\rho_1+\rho_2)xy-y-\rho_1x^2y-\rho_2xy^2=-y\rho_1\big(x-x^*(y)\big)\big(x-x^{+}(y)\big),\tag{130}$$

where $x^*(y)$ and $x^{+}(y)$ are the roots found in §6.1.1. By Vieta's product formula, $x^*(y)\,x^{+}(y)=\mu/\lambda_1=1/\rho_1$. This, together with $\widetilde{x}^*(y)$ (Equation (127)), gives $\widetilde{x}^*(y)=\rho_1x^*(y)=1/x^{+}(y)$. Now, looking at the denominator of our reconstructed PGF (Equation (129)), and using the factorisation of $K(x,y)$ (Equation (130)), we see that

$$1-\widetilde{x}^*(y)\,x = 1-\frac{x}{x^+(y)} = \frac{x^+(y)-x}{x^+(y)} = \frac{K(x,y)}{y\rho_1\,x^+(y)\big(x-x^*(y)\big)}.$$

Substituting this into our reconstructed $P(x,y)$ (Equation (129)), using $P_y(y)=\frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho)$ and also noting that $\rho_1\,x^*(y)\,x^+(y)=1$, we obtain

$$\begin{aligned}
P(x,y) &= \frac{x^*(y)(1-y)}{x^*(y)-y}\,\rho(1-\rho)\cdot\frac{y\rho_1\,x^+(y)\big(x-x^*(y)\big)}{K(x,y)}\\
&= \underbrace{\rho_1\,x^*(y)\,x^+(y)}_{=1}\;\frac{1}{K(x,y)}\,y(1-y)\,\frac{x-x^*(y)}{x^*(y)-y}\,\rho(1-\rho)\\
&= \big[(1+\rho_1+\rho_2)xy-y-\rho_1x^2y-\rho_2xy^2\big]^{-1}\cdot y(1-y)\,\frac{x-x^*(y)}{x^*(y)-y}\,\rho(1-\rho),
\end{aligned}$$

which recovers Equation (107) from Theorem 1 and concludes the proof. □

6.2 Analysis of Model-A: state space $\widetilde{S}$

Setting $\gamma_1=\gamma_2=\theta_1=\theta_2=0$ in the $\widetilde{S}$ balance equations (30)–(34) and forming the PPGF $\widetilde{P}(y,n)=\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)\,y^{n_2}$ yields the inhomogeneous recurrence:

$$\begin{aligned}
&[\lambda_1+\lambda_2+\mu]\,\widetilde{P}(y,n)\\
&\quad= \quad(\lambda_1+\lambda_2y)\widetilde{P}(y,n-1)+\mu\widetilde{P}(y,n+1)+\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}\tag{131}$$

Theorem 2 (Approximate PPGF on $\widetilde{S}$, heuristic)**.** (The following result rests on an unproved decay assumption; see the proof.) *Under the approximation that neglects the boundary-diagonal forcing term $\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1)$ in (131), the homogeneous recurrence admits the geometric solution*

$$\widetilde{P}(y,n)\;\approx\;\rho(1-\rho)\,[\widetilde{y}^*(y)]^n,\tag{132}$$

where

$$\widetilde{y}^*(y)=\frac{(\lambda+\mu)-\sqrt{(\lambda+\mu)^2-4\mu(\lambda_1+\lambda_2y)}}{2\mu},\qquad \lambda=\lambda_1+\lambda_2.\tag{133}$$

At $y=1$ this recovers $\widetilde{P}(1,n)=(1-\rho)\rho^{n+1}$, consistent with the $M/M/1(\lambda,\mu)$ steady-state marginal for n customers waiting.

Proof of Theorem 2. The inhomogeneous term $\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)$ prevents the direct application of a geometric-sequence ansatz. The diagonal probabilities $\widetilde{\pi}(n,n)=\pi(0,n)$ correspond to states where all n waiting customers are class 2; from the exact formula in Theorem 1 one can verify that $\pi(0,n)$ decays geometrically in n, but this is not derived here. *Assuming* this geometric decay, the forcing term is exponentially small and we obtain an approximation by neglecting it. The resulting homogeneous recurrence is

$$(\lambda_1+\lambda_2+\mu)\widetilde{P}(y,n)=(\lambda_1+\lambda_2y)\widetilde{P}(y,n-1)+\mu\widetilde{P}(y,n+1).\tag{134}$$

The characteristic polynomial of this recurrence is

$$\mu[\widetilde{y}(y)]^2-(\lambda_1+\lambda_2+\mu)\widetilde{y}(y)+(\lambda_1+\lambda_2y)=0,\tag{135}$$

with roots

$$\widetilde{y}^{\pm}(y)=\frac{(\lambda_1+\lambda_2+\mu)\pm\sqrt{(\lambda_1+\lambda_2+\mu)^2-4\mu(\lambda_1+\lambda_2y)}}{2\mu}.\tag{136}$$

At $y=1$ the roots are $\widetilde{y}_+(1)=1$ and $\widetilde{y}_-(1)=\rho$. The root $\widetilde{y}_+=1$ would produce a non-summable geometric series as $n\to\infty$, so only the negative-sign root is admissible:

$$\widetilde{y}^*(y)=\frac{(\lambda_1+\lambda_2+\mu)-\sqrt{(\lambda_1+\lambda_2+\mu)^2-4\mu(\lambda_1+\lambda_2y)}}{2\mu}.\tag{137}$$

The geometric ansatz gives $\widetilde{P}(y,n) = \widetilde{P}(y,0) \cdot [\widetilde{y}^*(y)]^n$. Using $\widetilde{P}(y,0) = \widetilde{\pi}(0,0) = \pi(0,0) = \rho(1-\rho)$ (Corollary 2):

$$\widetilde{P}(y,n) = \rho(1-\rho) \cdot [\widetilde{y}^*(y)]^n. \tag{138}$$

Since n counts customers *waiting* (the customer in service is not counted), $n+1$ is the total number in the system, so evaluating at $y = 1$:

$$\widetilde{P}(1,n) = \rho(1-\rho) \cdot \rho^n = (1-\rho)\rho^{n+1}, \tag{139}$$

□

which matches the $M/M/1(\lambda, \mu)$ stationary distribution for $n+1$ customers in the system [AR02].

6.3 Mean queue lengths and mean waiting times

6.3.1 Class-1 marginal PGF and mean queue length

Setting $y = 1$ in the fundamental equation (106) gives

$$\left[-\lambda_1(x-1)(x-1/\rho_1)\right] P(x,1) \;=\; \mu(x-1)\, P_y(1). \tag{140}$$

Cancelling the common factor $(x-1)$ for $x \neq 1$ and rearranging:

$$P(x,1) \;=\; \frac{\mu\, P_y(1)}{\mu - \lambda_1 x}. \tag{141}$$

The constant $P_y(1)$ is pinned by the normalisation $P(1,1) = 1 - \pi_0 = \rho$:

$$\rho \;=\; \frac{\mu\, P_y(1)}{\mu - \lambda_1} \quad\Longrightarrow\quad P_y(1) \;=\; \rho\,(1-\rho_1), \tag{142}$$

so that

$$P(x,1) \;=\; \frac{\rho(1-\rho_1)}{1-\rho_1 x}. \tag{143}$$

Differentiating by the quotient rule and evaluating at $x = 1$:

$$\mathbb{E}[N_1] \;=\; \frac{d}{dx}P(x,1)\bigg|_{x=1} \;=\; \frac{\rho(1-\rho_1)\cdot\rho_1}{(1-\rho_1 x)^2}\bigg|_{x=1} \;=\; \frac{\rho(1-\rho_1)\,\rho_1}{(1-\rho_1)^2} \;=\; \frac{\rho\rho_1}{1-\rho_1}. \tag{144}$$

6.3.2 Total queue length from the diagonal

Setting $x = y = z$ in the kernel polynomial $\lambda_1 x^2 - (\lambda_1 + \lambda_2 + \mu)x + \mu$ shows that $x^*(z) = z$ iff $z \in \{1,\, 1/\rho\}$; hence $x^*(z) \neq z$ for all $z \in (0,1)$ with $\rho < 1$. Therefore the ratio $(z - x^*(z))/(x^*(z) - z) = -1$ throughout $(0,1)$, and the formula of Theorem 1 reduces to:

$$P(z,z) \;=\; \frac{z(1-z)\,\rho(1-\rho)}{D(z,z)} \cdot (-1). \tag{145}$$

The denominator at $x = y = z$ is $D(z,z) = (1+\rho)z^2 - z - \rho z^3 = -\rho z(z-1)(z-1/\rho)$, so:

$$\begin{aligned} P(z,z) \;&=\; \frac{-z(1-z)\,\rho(1-\rho)}{-\rho z(z-1)(z-1/\rho)} \\ &=\; \frac{(1-z)(1-\rho)}{(z-1)(z-1/\rho)} \;=\; \frac{-(1-\rho)}{z-1/\rho} \;=\; \frac{\rho(1-\rho)}{1-\rho z}, \end{aligned} \tag{146}$$

confirming the diagonal identity of Corollary 2. Since $\frac{d}{dz}P(z,z)\big|_{z=1} = \frac{\partial P}{\partial x}(1,1) + \frac{\partial P}{\partial y}(1,1) = \mathbb{E}[N_1] + \mathbb{E}[N_2]$, differentiating by the quotient rule and evaluating at $z = 1$ gives:

$$\mathbb{E}[N_1] + \mathbb{E}[N_2] \;=\; \frac{d}{dz}\,\frac{\rho(1-\rho)}{1-\rho z}\bigg|_{z=1} \;=\; \frac{\rho(1-\rho)\cdot\rho}{(1-\rho z)^2}\bigg|_{z=1} \;=\; \frac{\rho^2(1-\rho)}{(1-\rho)^2} \;=\; \frac{\rho^2}{1-\rho}. \tag{147}$$

This is the $M/M/1(\lambda, \mu)$ queue-length formula for the combined system [AR02], as expected: the two classes together behave as a single $M/M/1$ queue.

6.3.3 Class-2 mean queue length by subtraction

$$
\begin{aligned}
\mathbb{E}[N_2] &= \big(\mathbb{E}[N_1] + \mathbb{E}[N_2]\big) - \mathbb{E}[N_1] = \frac{\rho^2}{1-\rho} - \frac{\rho\rho_1}{1-\rho_1} \\
&= \frac{\rho^2(1-\rho_1) - \rho\rho_1(1-\rho)}{(1-\rho)(1-\rho_1)} = \frac{\rho\big[\rho(1-\rho_1) - \rho_1(1-\rho)\big]}{(1-\rho)(1-\rho_1)} \\
&= \frac{\rho(\rho-\rho_1)}{(1-\rho)(1-\rho_1)} = \frac{\rho\rho_2}{(1-\rho_1)(1-\rho)}.
\end{aligned}
\tag{148}
$$

6.3.4 Mean waiting times via Little's Law

Little's Law applied to the *waiting sub-system* of each class: since N_i counts only customers *waiting* in queue (excluding the customer in service), Little's Law gives $\mathbb{E}[N_i] = \lambda_i\,\mathbb{E}[W_i]$, where $\mathbb{E}[W_i]$ is the mean waiting time in queue (time from arrival until service start, not including the service duration):

$$
\mathbb{E}[W_1] = \frac{\mathbb{E}[N_1]}{\lambda_1} = \frac{\rho\rho_1}{\lambda_1(1-\rho_1)} = \frac{\rho\cdot(\lambda_1/\mu)}{\lambda_1(1-\rho_1)} = \frac{\rho}{\mu(1-\rho_1)}, \tag{149}
$$

$$
\mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\lambda_2} = \frac{\rho\rho_2}{\lambda_2(1-\rho_1)(1-\rho)} = \frac{\rho\cdot(\lambda_2/\mu)}{\lambda_2(1-\rho_1)(1-\rho)} = \frac{\rho}{\mu(1-\rho_1)(1-\rho)}. \tag{150}
$$

Class-1 benefits from priority: $\mathbb{E}[W_2]$ carries the additional penalty factor $1/(1-\rho)$ relative to $\mathbb{E}[W_1]$, reflecting the delay that class-2 customers incur while class-1 customers in the queue are served first. Note that both $\mathbb{E}[W_1]$ and $\mathbb{E}[W_2]$ depend on ρ_2 through the total load $\rho = \rho_1 + \rho_2$; class-2 affects class-1 waiting times only via server utilisation, not through a direct queueing penalty.

7 Analysis of Model-B $(\gamma_1, \gamma_2 > 0,\ \theta_1 = \theta_2 = 0)$

Model-B introduces bidirectional jockeying ($\gamma_1, \gamma_2 > 0$) while retaining no abandonment ($\theta_1 = \theta_2 = 0$). Jockeying conserves the total number of customers in the system, so the stationary distribution still exists under the same condition as Model-A. However, jockeying breaks the per-class Poisson/renewal structure: a Class-2 customer can no longer identify its effective service time with a Class-1 busy period, so the Pollaczek–Khinchine probabilistic route is unavailable. Applying the method of characteristics to the fundamental equation yields a general solution, but closing it — i.e., determining the unknown boundary function $P_y(y)$ — requires solving a Volterra-type integral equation that does not admit a closed form. Model-B is therefore intentionally left open, in structural parallel with Model-X; its role is to motivate the tractable specialisation Model-B_2.

Stability. $\rho = (\lambda_1 + \lambda_2)/\mu < 1$. Jockeying conserves the total customer count and does not affect this condition.

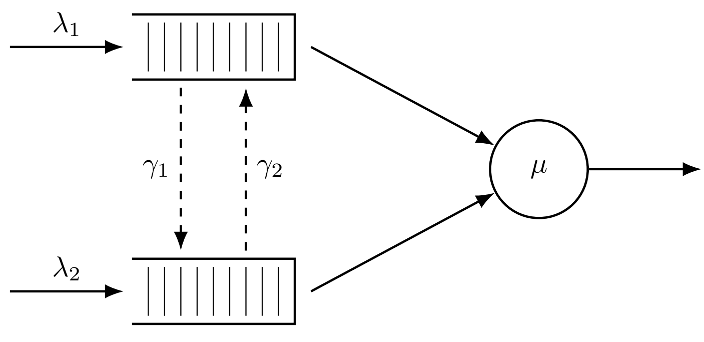

Figure 5: Queue layout for Model-B. Dashed arrows indicate bidirectional jockeying: γ_1 moves a Class-1 customer down to Queue 2 and γ_2 moves a Class-2 customer up to Queue 1. Both transfers conserve the total n.

Corollary 3. *The empty-system probability π_0 (server idle, no one waiting) and the both-queues-empty probability $\pi(0,0)$ (server busy, $N_1 = N_2 = 0$) of Model-B coincide with the Model-A baseline,*

$$
\pi_0 = 1-\rho, \qquad \pi(0,0) = \rho(1-\rho), \tag{151}
$$

and the diagonal of the PGF is $P(z,z) = \pi(0,0)/(1-\rho z)$.

Proof. Setting $x = y$ in the fundamental equation (42) with $\theta_1 = \theta_2 = 0$, both jockeying terms carry the factor $(x-y)$ and vanish, as does the boundary term $\mu(x-y)P_y(y)$; dividing the surviving terms by the common factor x leaves

$$\big[(\lambda_1+\lambda_2+\mu)x - \mu - (\lambda_1+\lambda_2)x^2\big]P(x,x) = -\mu(1-x)\,\pi(0,0). \tag{152}$$

Writing $\lambda = \lambda_1+\lambda_2$, the bracket factors as $(\mu-\lambda x)(x-1)$; cancelling $(x-1)$ against $-(1-x) = (x-1)$ on the right gives

$$P(x,x) = \frac{\mu\,\pi(0,0)}{\mu-\lambda x} = \frac{\pi(0,0)}{1-\rho x}. \tag{153}$$

The idle state (0) exchanges probability flux only with the both-queues-empty busy state, so the global balance across the idle–busy cut reads $\lambda\pi_0 = \mu\pi(0,0)$ (an arrival at total rate λ leaves (0); a service completion at rate μ returns to it), whence $\pi(0,0) = \rho\pi_0$. Evaluating (153) at $x = 1$ with $P(1,1) = 1-\pi_0$ gives $1-\pi_0 = \rho\pi_0/(1-\rho)$, so $(1-\pi_0)(1-\rho) = \rho\pi_0$, i.e. $\pi_0 = 1-\rho$ and $\pi(0,0) = \rho(1-\rho)$. This matches Model-A (109): jockeying conserves the total-customer process and cannot change the marginal idle probability. The same identities hold in Models B_2 and C_2 for the same conservation reason (24). ☐

Claim 2. *Under stability, $\rho < 1$, the bivariate PGF of Model-B satisfies, on each characteristic curve $\gamma_2 x + \gamma_1 y = C_1$ of the fundamental equation, the integral representation*

$$P(x,y(x)) \;=\; P_h(x,y(x)) \int_0^x \frac{h\big(t;\,C_1\big)}{P_h\big(t,\,y(t)\big)}\,dt, \tag{154}$$

where P_h is the homogeneous solution (166), $y(t) = (C_1-\gamma_2 t)/\gamma_1$ is the characteristic parametrisation, and $h(t;C_1)$ is an explicit forcing function depending on the unknown boundary trace $P_y(y) = P(0,y)$. Evaluating (154) along the diagonal $x = y = z$ and using the conservation identity $P(z,z) = \pi(0,0)/(1-\rho z)$ reduces the problem to a Volterra integral equation of the first kind *for P_y:*

$$\mathcal{F}(z) \;=\; \int_0^z \mathcal{K}(z,t)\,P_y\big(y(t;z)\big)\,dt, \tag{155}$$

where $\mathcal{K}$ and $\mathcal{F}$ are fully explicit (defined in (182)–(184) below). The kernel $\mathcal{K}$ does not factorise for general (γ_1,γ_2), so (155) does not reduce to a closed-form solution for P_y. Model-B is therefore left open.

Proof of Claim 2. The proof proceeds in five steps: (i) homogeneous solution via characteristics, (ii) rigorous sign analysis of the exponent E, (iii) particular solution via variation of parameters, (iv) pinning the free function from the boundary condition, and (v) diagonal evaluation yielding the Volterra equation.

Step 1: Homogeneous solution via the method of characteristics.
The fundamental equation (42) with $\theta_1 = \theta_2 = 0$ is a first-order linear PDE in $P(x,y)$. Setting the right-hand side to zero and dividing through by y yields the homogeneous PDE

$$\gamma_1 x(x-y)\,\frac{\partial P}{\partial x} - \gamma_2 x(x-y)\,\frac{\partial P}{\partial y} = -\big[(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big]P. \tag{156}$$

Following the method of characteristics (§3.6.3), we associate to this PDE the characteristic system

$$\frac{dx}{\gamma_1 x(x-y)} = \frac{dy}{-\gamma_2 x(x-y)} = \frac{dP}{-\big[(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big]P}. \tag{157}$$

Taking the ratio of the first two differentials, the $x(x-y)$ factors cancel, giving $dx/\gamma_1 = dy/(-\gamma_2)$. Integrating yields the first integral

$$C_1 = \gamma_2 x + \gamma_1 y \quad \text{(constant along each characteristic)}. \tag{158}$$

The characteristics are therefore the straight lines of slope $-\gamma_2/\gamma_1$ in the (x,y)-plane. Fix one such line and parametrise y by $y(x) = (C_1-\gamma_2 x)/\gamma_1$. To extract the equation for P along this curve we need the coefficient $A(x,y)$ and the denominator $x(x-y)$ after substitution. Setting

$$A(x,y) := (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy, \tag{159}$$

and substituting $y = (C_1 - \gamma_2 x)/\gamma_1$ so that $\lambda_2 xy = (\lambda_2 C_1/\gamma_1)x - (\lambda_2\gamma_2/\gamma_1)x^2$, we obtain

$$A\big(x, y(x)\big) = -\frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1}\, x^2 + \frac{\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1}{\gamma_1}\, x - \mu. \tag{160}$$

Similarly, $x - y(x) = [(\gamma_1+\gamma_2)x - C_1]/\gamma_1$, so $\gamma_1 x(x - y(x)) = x[(\gamma_1+\gamma_2)x - C_1]$. The dP/P equation from the characteristic system therefore becomes

$$\frac{dP}{P} = \frac{-A(x, y(x))}{\gamma_1 x(x - y(x))}\, dx = \frac{(\gamma_1\lambda_1 - \lambda_2\gamma_2)x^2 - \big[\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1\big]x + \gamma_1\mu}{\gamma_1 x\big[(\gamma_1+\gamma_2)x - C_1\big]}\, dx. \tag{161}$$

To integrate this, we decompose the integrand via partial fractions. Writing the integrand as $(1/\gamma_1)\, I(x)\, dx$ with $I(x) = K + M/x + N(C_1)/[(\gamma_1+\gamma_2)x - C_1]$, the partial fraction identity requires

$$K(\gamma_1+\gamma_2)x^2 + \big[M(\gamma_1+\gamma_2) + N - KC_1\big]x - MC_1 = (\gamma_1\lambda_1 - \lambda_2\gamma_2)x^2 - \big[\gamma_1(\lambda_1+\lambda_2+\mu) - \lambda_2 C_1\big]x + \gamma_1\mu.$$

Matching the x^2, x^0, and x^1 coefficients in turn:

$$K = \frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1+\gamma_2}, \qquad M = -\frac{\gamma_1\mu}{C_1}, \qquad N(C_1) = \gamma_1\left[\frac{\lambda_1+\lambda_2}{\gamma_1+\gamma_2}C_1 - (\lambda_1+\lambda_2+\mu) + \frac{\mu(\gamma_1+\gamma_2)}{C_1}\right]. \tag{162}$$

(For N: substituting K and M into the x^1 match gives $N = -\gamma_1(\lambda_1+\lambda_2+\mu) + \lambda_2 C_1 + KC_1 - M(\gamma_1+\gamma_2)$; expanding KC_1 and $-M(\gamma_1+\gamma_2)$ and collecting yields (162).) Integrating $dP/P = (1/\gamma_1)\, I(x)\, dx$ term by term gives

$$\ln P = \frac{K}{\gamma_1}x - \frac{\mu}{C_1}\ln x + \frac{N(C_1)}{\gamma_1(\gamma_1+\gamma_2)}\ln\big[(\gamma_1+\gamma_2)x - C_1\big] + \ln f(C_1), \tag{163}$$

where $f(C_1)$ is an arbitrary function of the characteristic constant. Now observe that

$$(\gamma_1+\gamma_2)x - C_1 = (\gamma_1+\gamma_2)x - (\gamma_2 x + \gamma_1 y) = \gamma_1(x - y), \tag{164}$$

so $\ln[(\gamma_1+\gamma_2)x - C_1] = \ln\gamma_1 + \ln(x-y)$. The constant $\ln\gamma_1$ is absorbed into a redefined arbitrary function $F(C_1)$. Reading off the exponent of $(x-y)$:

$$E(x,y) := \left.\frac{N(C_1)}{\gamma_1(\gamma_1+\gamma_2)}\right|_{C_1 = \gamma_2 x + \gamma_1 y} = \frac{\lambda_1+\lambda_2}{(\gamma_1+\gamma_2)^2}(\gamma_2 x + \gamma_1 y) - \frac{\lambda_1+\lambda_2+\mu}{\gamma_1+\gamma_2} + \frac{\mu}{\gamma_2 x + \gamma_1 y}. \tag{165}$$

Exponentiating (163) yields the general homogeneous solution

$$P_h(x,y) = F(\gamma_2 x + \gamma_1 y)\cdot\exp\!\left(\frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1(\gamma_1+\gamma_2)}x\right)\cdot x^{-\mu/(\gamma_2 x + \gamma_1 y)}\cdot(x-y)^{E(x,y)}, \tag{166}$$

where F is an arbitrary differentiable function to be determined.

Step 2: The exponent $E(x,y)$ is strictly positive on $(0,1)^2$.

Write $\lambda = \lambda_1 + \lambda_2$ and $\Gamma = \gamma_1 + \gamma_2$. Along any fixed characteristic, $C_1 = \gamma_2 x + \gamma_1 y$ is constant, so the exponent (165) depends only on $u := C_1 \in (0, \Gamma)$:

$$E(u) = \frac{\lambda}{\Gamma^2}\,u - \frac{\lambda+\mu}{\Gamma} + \frac{\mu}{u}. \tag{167}$$

(For $(x,y) \in (0,1)^2$ we have $u = \gamma_2 x + \gamma_1 y \in (0,\Gamma)$; $E(u)$ is continuous there, with $E(u) \to +\infty$ as $u \to 0^+$ and $E(u) \to 0^+$ as $u \to \Gamma^-$, so it is finite at every interior point although not uniformly bounded.) Multiplying through by $u\Gamma^2 > 0$ gives the equivalent quadratic

$$q(u) := \lambda u^2 - (\lambda+\mu)\Gamma u + \mu\Gamma^2. \tag{168}$$

The sign of E on $(0,\Gamma)$ equals the sign of $q(u)$ there. We factor q by finding its roots. The discriminant is $\Delta = (\lambda+\mu)^2\Gamma^2 - 4\lambda\mu\Gamma^2 = \Gamma^2(\mu-\lambda)^2$, so (using $\rho < 1 \Rightarrow \mu > \lambda \Rightarrow \mu - \lambda > 0$)

$$u_{1,2} = \frac{(\lambda+\mu)\Gamma \pm \Gamma(\mu-\lambda)}{2\lambda} \quad\Longrightarrow\quad u_1 = \frac{2\lambda\Gamma}{2\lambda} = \Gamma, \quad u_2 = \frac{2\mu\Gamma}{2\lambda} = \frac{\Gamma}{\rho} > \Gamma. \tag{169}$$

Hence

$$q(u) = \lambda(u-\Gamma)\Big(u - \tfrac{\Gamma}{\rho}\Big). \tag{170}$$

For every $u \in (0,\Gamma)$ both factors are strictly negative ($(u-\Gamma) < 0$ and $(u-\Gamma/\rho) < 0$ since $\Gamma/\rho > \Gamma$), so $q(u) = \lambda \cdot (\text{neg}) \cdot (\text{neg}) > 0$, and therefore

$$E(x,y) > 0 \quad \text{for all } (x,y) \in (0,1)^2. \tag{171}$$

Note that $E \to 0$ as $u \to \Gamma^-$ (i.e. as $(x,y) \to (1,1)$), while E is bounded and bounded away from zero in any compact subset of $(0,1)^2$.

Sign convention and diagonal limit. Because $E > 0$, the factor $(x-y)^E$ in P_h carries the complex phase $e^{i\pi E}$ whenever $x < y$. This phase is absorbed into the arbitrary function $F(C_1)$, and the ratio $P_h(x,y(x))/P_h(t,y(t))$ that appears in the integral representation is always real and positive: both numerator and denominator lie on the same characteristic (same C_1, same constant value of E), so the phases cancel exactly. Crucially, since $E > 0$ the factor $(y(t;z)-t)^E \to 0$ as $t \to z^-$, meaning P_h formally vanishes on the diagonal $x = y$. The diagonal equation that appears in Step 5 below is therefore understood as the $\lim_{x\to z^-}$ of (177); an asymptotic analysis near $t = z$ (expanding $y(t;z) - t \sim (\Gamma/\gamma_1)(z-t)$) confirms that the $0 \cdot \infty$ limit is finite and equals $\pi(0,0)/(1-\rho z)$.

Step 3: Particular solution via variation of parameters.

We now work with the full (inhomogeneous) fundamental equation restricted to a fixed characteristic. Along the curve $y(x) = (C_1 - \gamma_2 x)/\gamma_1$, the total derivative satisfies $\frac{d}{dx}P(x,y(x)) = \partial_x P + \frac{dy}{dx}\partial_y P = \partial_x P - \frac{\gamma_2}{\gamma_1}\partial_y P$, so the jockeying terms in (42) become

$$\gamma_1 xy(x-y)\,\partial_x P + \gamma_2 xy(y-x)\,\partial_y P = \gamma_1 xy(x-y)\Big(\partial_x P - \tfrac{\gamma_2}{\gamma_1}\partial_y P\Big) = \gamma_1 xy(x-y)\,\frac{d}{dx}P(x,y(x)).$$

Substituting into the fundamental equation and dividing by $\gamma_1 x(x-y(x))$ reduces the PDE to the first-order linear ODE (for fixed C_1)

$$\frac{dP}{dx} = f(x;C_1)\,P + h(x;C_1), \tag{172}$$

where the coefficient f and the forcing term h are

$$f(x;C_1) = -\frac{A(x,\,y(x))}{\gamma_1 x\big(x-y(x)\big)}, \tag{173}$$

$$h(x;C_1) = \frac{G(x,\,y(x))}{\gamma_1 x\,y(x)\big(x-y(x)\big)}, \tag{174}$$

with A as in (159) and

$$G(x,y) := \mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0). \tag{175}$$

Since $P_h(x,y(x))$ satisfies the homogeneous equation $\frac{d}{dx}P_h = f(x;C_1)\,P_h$, it is the fundamental solution of ODE (172). Applying variation of parameters (§3.6.2), the general solution of the inhomogeneous ODE is

$$P(x,y(x)) = P_h(x,y(x))\left[F(C_1) + \int_{x_0}^{x} \frac{h(t;C_1)}{P_h(t,y(t))}\,dt\right]. \tag{176}$$

Step 4: Pinning the free function from the boundary condition at $x = 0$.

Along any characteristic with constant $C_1 > 0$, the endpoint at $x = 0$ is $y(0) = C_1/\gamma_1$, giving

$$P\left(0,\,\frac{C_1}{\gamma_1}\right) = P_y\left(\frac{C_1}{\gamma_1}\right),$$

which is finite. However, P_h contains the factor $x^{-\mu/C_1} \to +\infty$ as $x \to 0^+$ (since $\mu/C_1 > 0$ for $C_1 > 0$), while the integral in (176) vanishes as $x_0 \to 0$ (the integration interval collapses). Therefore

$$\lim_{x\to 0^+} P(x,y(x)) = \lim_{x\to 0^+} P_h(x,y(x)) \cdot F(C_1) = +\infty \cdot F(C_1),$$

which can only be finite if $F(C_1) = 0$. Setting $x_0 = 0$ and $F(C_1) = 0$ in (176) gives the integral representation of Claim 2:

$$P(x,y(x)) = P_h(x,y(x)) \int_0^x \frac{h(t;C_1)}{P_h(t,y(t))}\,dt. \tag{177}$$

Step 5: Reduction to the Volterra equation.

By Corollary 3 the empty-state probabilities are $\pi_0 = 1-\rho$ and $\pi(0,0) = \rho(1-\rho)$, with diagonal value $P(z,z) = \pi(0,0)/(1-\rho z)$ by (153); these are the inputs to the diagonal evaluation below.

We simplify the forcing function h on the characteristic. Substituting $G(t, y(t))$ from (175) into (174):

$$\begin{aligned} h(t; C_1) &= \frac{\mu(t-y(t))\, P_y(y(t)) - \mu t(1-y(t))\, \pi(0,0)}{\gamma_1\, t\, y(t)\, (t-y(t))} \\ &= \frac{\mu\, P_y(y(t))}{\gamma_1\, t\, y(t)} - \frac{\mu(1-y(t))\, \pi(0,0)}{\gamma_1\, y(t)\, (t-y(t))}. \end{aligned} \tag{178}$$

We will evaluate (177) along the diagonal characteristic, which passes through (z,z) and has $C_1 = (\gamma_1+\gamma_2)z$. The parametrisation along that curve is $y(t;z) = ((\gamma_1+\gamma_2)z - \gamma_2 t)/\gamma_1$, so $y(0;z) = (\gamma_1+\gamma_2)z/\gamma_1 > z$ and $y(z;z) = z$. Therefore $y(t;z) > t$ for all $t \in [0,z)$, which means $t - y(t;z) < 0$ throughout the integration interval. Writing $t - y(t;z) = -(y(t;z) - t)$ in the denominator of (178):

$$h\big(t;\, (\gamma_1+\gamma_2)z\big) = \frac{\mu}{\gamma_1\, t\, y(t;z)}\, P_y\big(y(t;z)\big) + \frac{\mu\, (1-y(t;z))}{\gamma_1\, y(t;z)\, (y(t;z)-t)}\, \pi(0,0). \tag{179}$$

As established in Step 2, $E > 0$, so the homogeneous factor $(x-y)^E$ vanishes on the diagonal and $P_h\big(x, y(x;z)\big) \to 0$ as $x \to z^-$; simultaneously the integrand in (177) develops a non-integrable singularity $\sim (y(t;z)-t)^{-1-E}$ as $t \to z^-$, so the integral diverges. Their product is the finite $0 \cdot \infty$ limit

$$\frac{\pi(0,0)}{1-\rho z} = \lim_{x\to z^-} P_h\big(x, y(x;z)\big) \int_0^x \frac{h\big(t; (\gamma_1+\gamma_2)z\big)}{P_h\big(t, y(t;z)\big)}\, dt, \tag{180}$$

whose value is $P(z,z) = \pi(0,0)/(1-\rho z)$ by Corollary 3. Substituting (179) and separating the term containing the unknown P_y from the known remainder, this limit splits as

$$\begin{aligned} \frac{\pi(0,0)}{1-\rho z} &= \lim_{x\to z^-} \frac{\mu\, P_h\big(x, y(x;z)\big)}{\gamma_1} \int_0^x \frac{P_y\big(y(t;z)\big)}{t \cdot y(t;z) \cdot P_h\big(t, y(t;z)\big)}\, dt \\ &\quad + \lim_{x\to z^-} \frac{\mu\, P_h\big(x, y(x;z)\big)\, \pi(0,0)}{\gamma_1} \int_0^x \frac{1-y(t;z)}{y(t;z)\cdot (y(t;z)-t) \cdot P_h\big(t, y(t;z)\big)}\, dt. \end{aligned} \tag{181}$$

Each limit is a Hadamard finite part: the vanishing prefactor $P_h^{\text{diag}}(z) := \lim_{x\to z^-} P_h\big(x, y(x;z)\big)$ regularises the divergent integral, and neither factor is meaningful in isolation. Writing the kernel and known term *formally* as

$$\mathcal{K}(z,t) := \frac{\mu \cdot P_h^{\text{diag}}(z)}{\gamma_1 \cdot t \cdot y(t;z) \cdot P_h\big(t, y(t;z)\big)}, \tag{182}$$

$$\mathcal{B}(z) := \frac{\mu \cdot P_h^{\text{diag}}(z) \cdot \pi(0,0)}{\gamma_1} \int_0^z \frac{1-y(t;z)}{y(t;z)\cdot (y(t;z)-t)\cdot P_h\big(t, y(t;z)\big)}\, dt, \tag{183}$$

to be read through the regularised limit (181) *rather than pointwise*, and setting $\mathcal{F}(z) = \pi(0,0)/(1-\rho z) - \mathcal{B}(z)$, equation (181) becomes the Volterra integral equation of the first kind

$$\mathcal{F}(z) = \int_0^z \mathcal{K}(z,t)\, P_y\big(y(t;z)\big)\, dt, \tag{184}$$

confirming Claim 2. The kernel $\mathcal{K}$ depends on P_h through both arguments and does not simplify for general (γ_1, γ_2), so (184) does not yield a closed form. Resolving it would require Volterra inversion methods well beyond the scope of this thesis. □

Model-B is therefore left open at this point. Its role in the hierarchy is structural: it identifies the precise obstruction — the Volterra equation for P_y — that the tractable special cases B_2 and C_2 circumvent, each by a different mechanism. Setting $\gamma_2 = 0$ collapses the characteristic family from a two-parameter set of oblique lines to a one-parameter set of vertical lines, degenerating the PDE to an ODE in x alone; the boundary function $P_y(y)$ is then pinned by an analyticity argument at the interior singular point $x = y$. Model-C_2 removes jockeying entirely and pins P_y by a boundedness argument at the boundary of the domain. Both specialisations are analysed in the following sections.

8 Analysis of Model-B_2 ($\gamma_1 > 0$, $\gamma_2 = 0$, $\theta_1 = \theta_2 = 0$)

Model-B_2 restricts jockeying to the direction $1 \to 2$ only ($\gamma_1 > 0$, $\gamma_2 = 0$) and retains no abandonment ($\theta_1 = \theta_2 = 0$). Setting $\gamma_2 = 0$ eliminates the $\partial P/\partial y$ term from the fundamental equation, reducing it to a first-order linear ODE in x for each fixed $y \in [0, 1]$. An integrating factor then yields a formal integral expression for $P(x, y)$; the boundary function $P_y(y)$ is pinned by requiring $P(\cdot, y)$ to be holomorphic at the interior singular point $x = y$, where the integrating factor carries a branch singularity with negative exponent. This analyticity condition replaces the boundedness argument used for Model-C_2.

Stability. As in Models A and B, $\rho = (\lambda_1+\lambda_2)/\mu < 1$. Jockeying conserves the total customer count and does not affect the stability condition.

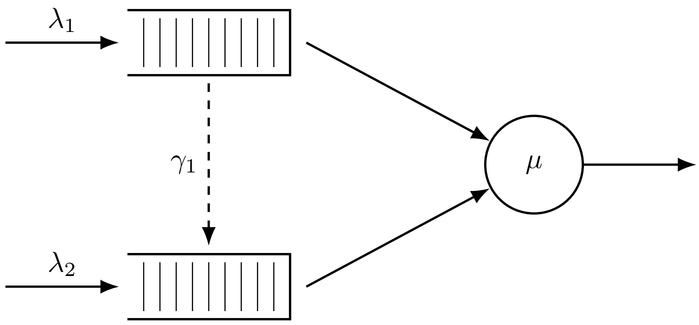

Figure 6: Queue layout for Model-B_2. Only the $1 \to 2$ jockeying direction is active (dashed, downward): Class-1 customers may defect to Queue 2, but no customer can move in the reverse direction ($\gamma_2 = 0$).

8.1 Reduced fundamental equation

Setting $\gamma_2 = 0$ and $\theta_1 = \theta_2 = 0$ in the general fundamental equation (42) (Lemma 1) annihilates the $\partial P/\partial y$ term, whose coefficient $xy\big[\gamma_2(y - x) + \theta_2(y - 1)\big]$ vanishes identically, and leaves only the γ_1 contribution to the $\partial P/\partial x$ term:

$$\begin{aligned}\big[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\big]P + \gamma_1 xy(x - y)\,\frac{\partial P}{\partial x}\\ = \mu(x - y)\,P_y(y) - \mu x(1 - y)\,\pi(0, 0).\end{aligned} \tag{185}$$

For each fixed $y \in (0, 1]$ this is a first-order linear ODE in x; the entire analysis of this section is built on it.

Theorem 3. *Consider the two-class non-preemptive priority queueing system of Model-B_2 with per-customer jockeying rate $\gamma_1 > 0$ from Queue 1 to Queue 2 ($\gamma_2 = 0$, $\theta_1 = \theta_2 = 0$). Under the stability condition $\rho = \lambda/\mu < 1$, $\lambda = \lambda_1 + \lambda_2$, the boundary generating function is*

$$P_y(y) = \frac{\mu\,\pi(0, 0)}{\mu - \lambda y}\cdot\frac{{}_1F_1\big(\alpha(y) + 1;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}, \tag{186}$$

and the bivariate PGF, for $0 \le x \le y$, is

$$P(x, y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}(y - x)^{-\beta}}{\gamma_1 y}\Big[P_y(y)\,\mathcal{H}_1(x, y) + (1 - y)\pi(0, 0)\,\mathcal{H}_2(x, y)\Big], \tag{187}$$

where ${}_1F_1$ is Kummer's confluent hypergeometric function, the exponents are

$$\alpha(y) = \frac{\mu}{\gamma_1 y}, \qquad \beta(y) = \frac{(1 - y)(\lambda y - \mu)}{\gamma_1 y}, \qquad b^\star(y) = \alpha + \beta + 1 = \frac{\mu + \lambda(1 - y) + \gamma_1}{\gamma_1}, \tag{188}$$

the auxiliary integrals are

$$\mathcal{H}_1(x, y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\alpha-1}(y - t)^{\beta}\, dt, \tag{189}$$

$$\mathcal{H}_2(x, y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\alpha}(y - t)^{\beta-1}\, dt, \tag{190}$$

and the empty-state probability $\pi(0, 0) = \rho(1 - \rho)$ is given by Corollary 4 below. On $y < x \le 1$ the same expression (187) holds with $\mathcal{H}_1, \mathcal{H}_2$ replaced by their finite-part continuations.

Corollary 4. *The empty-system probability π_0 (server idle, no one waiting) and the both-queues-empty probability $\pi(0,0)$ (server busy, $N_1 = N_2 = 0$) of Model-B_2 are*

$$\pi_0 = 1-\rho, \qquad \pi(0,0) = \rho(1-\rho), \tag{191}$$

identical to the Model-A baseline.

Proof. Because one-way jockeying conserves the total customer count and there is no abandonment, the total-count process L (equal to the number in queue N plus the single customer in service whenever the server is busy) is itself a birth–death chain: it increases at rate $\lambda = \lambda_1 + \lambda_2$ on any arrival and decreases at rate μ on any service completion, irrespective of class. Thus L is an $M/M/1$ queue, and by the memorylessness of the exponential clocks its stationary law is geometric, $\mathbb{P}(L=k) = (1-\rho)\rho^k$; in particular $\pi_0 = \mathbb{P}(L=0) = 1-\rho$.

For $\pi(0,0)$, set $x = y$ in the reduced fundamental equation (185). The convection term $\gamma_1 xy(x-y)\partial_x P$ and the boundary term $\mu(x-y)P_y(y)$ both carry the factor $(x-y)$ and vanish on the diagonal; dividing the surviving terms by the common factor x leaves

$$\big[(\lambda_1+\lambda_2+\mu)x - \mu - (\lambda_1+\lambda_2)x^2\big] P(x,x) = -\mu(1-x)\,\pi(0,0). \tag{192}$$

Factoring the bracket and cancelling $(x-1)$ against $-(1-x) = (x-1)$ on the right:

$$\begin{aligned}\big[-\lambda x^2 + (\lambda+\mu)x - \mu\big] P(x,x) &= (\mu - \lambda x)(x-1)\,P(x,x)\\ &= \mu(x-1)\,\pi(0,0),\end{aligned} \tag{193}$$

so that

$$P(x,x) = \frac{\mu\,\pi(0,0)}{\mu - \lambda x} = \frac{\pi(0,0)}{1-\rho x}. \tag{194}$$

Evaluating at $x=1$ with $P(1,1) = 1-\pi_0 = \rho$ (using $\pi_0 = 1-\rho$ just established) yields $\pi(0,0) = \rho(1-\rho)$. □

Proof of Theorem 3. Stage 1: Standard form and the integrating factor.

For any fixed $y \neq 0$, divide the reduced fundamental equation (185) through by the coefficient $\gamma_1 xy(x-y)$ of $\partial P/\partial x$. The factor y cancels against the overall factor y in the bracket multiplying P, putting the equation in standard first-order form $\frac{dP}{dx} + p(x;y)\,P = q(x;y)$ with

$$p(x;y) = \frac{(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy}{\gamma_1 x(x-y)}, \tag{195}$$

$$q(x;y) = \frac{\mu\,P_y(y)}{\gamma_1 xy} - \frac{\mu(1-y)\,\pi(0,0)}{\gamma_1 y(x-y)}. \tag{196}$$

Write the numerator of p as a polynomial in x,

$$N(x) = -\lambda_1 x^2 + \big[(\lambda_1+\lambda_2+\mu) - \lambda_2 y\big]x - \mu = -\lambda_1\big(x - x^*(y)\big)\big(x - x^+(y)\big),$$

which factors through the roots $x^*(y), x^+(y)$ of the Model-A kernel (Equation (113)), with $\lambda_1 x^* x^+ = \mu$. The partial-fraction coefficients of $p = N(x)/\big[\gamma_1 x(x-y)\big]$ follow by residues:

$$\begin{aligned} A &= -\lambda_1 && \text{(ratio of leading } x^2 \text{ terms)},\\ B &= \frac{N(0)}{0-y} = \frac{-\mu}{-y} = \frac{\mu}{y}, &&\\ C &= \frac{N(y)}{y} = \frac{-(\lambda y-\mu)(y-1)}{y} = \frac{(1-y)(\lambda y - \mu)}{y}, && \end{aligned} \tag{197}$$

where $N(y) = -\lambda y^2 + (\lambda+\mu)y - \mu = -(\lambda y - \mu)(y-1)$. Hence

$$p(x;y) = \frac{1}{\gamma_1}\left[-\lambda_1 + \frac{\mu/y}{x} + \frac{(1-y)(\lambda y-\mu)/y}{x-y}\right]. \tag{198}$$

We henceforth work on the principal triangle $0 \le x \le y$, on which $x - y \le 0$; the complementary region $y < x \le 1$ is recovered at the end by analytic continuation. There the real antiderivative of the simple-pole term is $\beta \ln|x-y| = \beta\ln(y-x)$, so integrating the partial-fraction form (198) term by term gives, up to an additive constant,

$$\int^x p(t;y)\,dt = \frac{1}{\gamma_1}\Big[-\lambda_1 x + \frac{\mu}{y}\ln x + \frac{(1-y)(\lambda y-\mu)}{y}\,\ln(y-x)\Big].$$

Exponentiating (an integrating factor is determined only up to a multiplicative constant, which we drop) yields the *real* integrating factor

$$\mu_I(x;y) = e^{-\lambda_1 x/\gamma_1}\, x^{\alpha(y)}\, (y-x)^{\beta(y)}, \tag{199}$$

with exponents $\alpha(y) = \mu/(\gamma_1 y) > 0$ and $\beta(y) = (1-y)(\lambda y - \mu)/(\gamma_1 y)$ as in (188); one verifies directly that $\mu_I'/\mu_I = -\lambda_1/\gamma_1 + \alpha/x + \beta/(x-y) = p(x;y)$, using $\frac{d}{dx}\ln(y-x)^\beta = \beta/(x-y)$. Working with $(y-x) > 0$ in place of the complex base $(x-y)$ keeps every quantity manifestly real throughout the principal triangle. Since $\alpha > 0$ we have $\mu_I \to 0$ as $x \to 0^+$, so the boundary term $P(0,y)\,\mu_I(0;y)$ vanishes when we integrate $\frac{d}{dx}[P\cdot\mu_I] = q\cdot\mu_I$ over $(0,x]$:

$$P(x,y)\,\mu_I(x;y) = \int_0^x q(t;y)\,\mu_I(t;y)\,dt. \tag{200}$$

It remains to evaluate the right-hand side. Multiplying q from (196) by μ_I and rewriting the second term with the elementary identity $(y-t)^\beta/(t-y) = -(y-t)^{\beta-1}$, the integrand separates into two manifestly real pieces,

$$q(t;y)\,\mu_I(t;y) = \frac{\mu}{\gamma_1 y}\Big[P_y(y)\,e^{-\lambda_1 t/\gamma_1}\,t^{\alpha-1}(y-t)^\beta + (1-y)\pi(0,0)\,e^{-\lambda_1 t/\gamma_1}\,t^{\alpha}(y-t)^{\beta-1}\Big], \tag{201}$$

the minus sign of (196) having been absorbed by that identity. With the auxiliary integrals $\mathcal{H}_1, \mathcal{H}_2$ of (189)–(190) and dividing through by $\mu_I(x;y) = e^{-\lambda_1 x/\gamma_1}x^\alpha(y-x)^\beta$, the solution on the principal triangle $0 \le x \le y$ is

$$P(x,y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Big[P_y(y)\,\mathcal{H}_1(x,y) + (1-y)\pi(0,0)\,\mathcal{H}_2(x,y)\Big], \tag{202}$$

which still contains the unknown boundary function $P_y(y)$; Stage 2 determines it.

Stage 2: Determination of $P_y(y)$ by analyticity at $x = y$.

Unlike Models A and C_2, no probabilistic (Pollaczek–Khinchine) route is available here. One-way jockeying makes the effective service experienced by a Class-2 customer depend on the Class-1 queue contents at the (random) jockeying instants, so the Class-2 input is no longer a Poisson stream feeding an $M/G/1$ subsystem and no busy-period reduction applies (cf. the discussion of Model-B in §7). We therefore determine $P_y(y)$ *purely analytically*, by imposing holomorphy of $P(\cdot,y)$ at the interior point $x = y$, where the solution (202) is *a priori* singular.

Under stability $\mu - \lambda y > 0$ for all $y \in (0,1)$, so the second exponent is negative,

$$\beta(y) = \frac{(1-y)(\lambda y - \mu)}{\gamma_1 y} < 0, \qquad 0 < y < 1. \tag{203}$$

The prefactor $(y-x)^{-\beta} = (y-x)^{|\beta|}$ in (202) therefore *vanishes* as $x \to y^-$, while the integrals $\mathcal{H}_1, \mathcal{H}_2$ may *diverge* there; the product is an indeterminate $0\cdot\infty$ form. Because $P(\cdot,y)$ is a probability generating function it must be holomorphic on the open unit disc, and in particular analytic at the interior point $x = y$. As we show next, the general solution carries a non-integer power $(y-x)^{|\beta|}$ — a branch term that vanishes at $x = y$ yet is not analytic there — and holomorphy forces its coefficient to zero; this single condition pins down $P_y(y)$.

Local analysis at $x = y$. To exhibit the two modes explicitly, read off the local behaviour of the ODE (195)–(196) at $x = y$. By the partial fraction (198), p has a simple pole at $x = y$ with residue β; by (196), q has the leading singularity $-D/(x-y)$ with $D = \mu(1-y)\pi(0,0)/(\gamma_1 y)$. The local model equation is therefore

$$P'(x,y) + \frac{\beta}{x-y}\,P(x,y) = \frac{-D}{x-y} + (\text{analytic near } x = y). \tag{204}$$

A constant particular solution $P \equiv P_*$ requires $\beta P_* = -D$, so, cancelling the common factor $(1-y)/(\gamma_1 y)$ shared by D and β,

$$P(y,y) = P_* = -\frac{D}{\beta} = -\frac{\mu\,\pi(0,0)}{\lambda y - \mu} = \frac{\mu\,\pi(0,0)}{\mu - \lambda y} = \frac{\rho(1-\rho)}{1-\rho y}, \tag{205}$$

consistent with the diagonal value (194) (using $\pi(0,0) = \rho(1-\rho)$). The general local solution adds a multiple of the homogeneous mode $(x-y)^{-\beta} = (x-y)^{|\beta|}$. Since $|\beta| \notin \mathbb{Z}$ generically this is a branch term: it vanishes at $x = y$ but is not analytic there, so holomorphy forces its coefficient to zero. The remaining task is to locate that coefficient inside the explicit solution (202).

No-branch condition and $P_y(y)$. Write the analytic, non-vanishing prefactor of (202) as $\Psi(x) = \mu\, e^{\lambda_1 x/\gamma_1} x^{-\alpha}/(\gamma_1 y)$, so that $P(x, y) = \Psi(x)\,(y - x)^{-\beta}\big[P_y(y)\,\mathcal{H}_1 + (1 - y)\pi(0, 0)\,\mathcal{H}_2\big]$. As $x \to y^-$ each integral admits a local expansion, obtained by Taylor-expanding its smooth integrand factor — $g_1(t) = e^{-\lambda_1 t/\gamma_1} t^{\alpha-1}$ for $\mathcal{H}_1$, $g_2(t) = e^{-\lambda_1 t/\gamma_1} t^{\alpha}$ for $\mathcal{H}_2$ — about $t = y$ and integrating the leading power of $(y - t)$:

$$\mathcal{H}_1(x, y) = \text{f.p.}\,\mathcal{H}_1(y, y) \;-\; \frac{g_1(y)}{\beta + 1}\,(y - x)^{\beta+1} + \cdots, \tag{206}$$

$$\mathcal{H}_2(x, y) = \text{f.p.}\,\mathcal{H}_2(y, y) \;-\; \frac{g_2(y)}{\beta}\,(y - x)^{\beta} + \cdots, \tag{207}$$

where f.p. denotes the Hadamard finite part (the $x \to y$ limit of the integral once its singular term has been subtracted) and the omitted terms are of higher order. Multiplying by $\Psi(x)(y - x)^{-\beta}$ sorts the contributions to $P(x, y)$ by their power of $(y - x)$:

- the singular part of $\mathcal{H}_2$ gives $-\Psi(y)(1 - y)\pi(0, 0)\,g_2(y)/\beta = -D/\beta = P_*$, exactly the regular value (205) (using $\Psi(y)\,g_2(y) = \mu/(\gamma_1 y)$);
- the singular part of $\mathcal{H}_1$ gives a term $\propto (y - x)^1$, analytic at $x = y$;
- the finite parts, riding on the prefactor, produce the branch term $\Psi(x)(y-x)^{-\beta}\big[P_y(y)\,\text{f.p.}\,\mathcal{H}_1 + (1 - y)\pi(0, 0)\,\text{f.p.}\,\mathcal{H}_2\big]$.

Only the last is non-analytic ($(y - x)^{|\beta|}$ with $|\beta| \notin \mathbb{Z}$), so holomorphy of $P(\cdot, y)$ at $x = y$ requires its bracket to vanish:

$$P_y(y)\,\text{f.p.}\,\mathcal{H}_1(y, y) + (1 - y)\,\pi(0, 0)\,\text{f.p.}\,\mathcal{H}_2(y, y) = 0. \tag{208}$$

To evaluate the finite parts, substitute $t = ys$ ($dt = y\,ds$) in (189)–(190) and pull out $y^{\alpha+\beta}$ from $t^{\alpha-1}(y - t)^{\beta}\,dt = y^{\alpha+\beta} s^{\alpha-1}(1 - s)^{\beta}\,ds$ (resp. $t^{\alpha}(y - t)^{\beta-1}\,dt = y^{\alpha+\beta} s^{\alpha}(1 - s)^{\beta-1}\,ds$); with $c := \lambda_1 y/\gamma_1$,

$$\mathcal{H}_1(y, y) = y^{\alpha+\beta}\int_0^1 s^{\alpha-1}(1 - s)^{\beta} e^{-cs}\,ds, \qquad \mathcal{H}_2(y, y) = y^{\alpha+\beta}\int_0^1 s^{\alpha}(1 - s)^{\beta-1} e^{-cs}\,ds.$$

Matching the Euler integral representation (12), written as $\int_0^1 t^{a-1}(1-t)^{b-1}e^{zt}\,dt = B(a, b)\,{}_1F_1(a; a+b; z)$, with $(a, b, z) = (\alpha, \beta + 1, -c)$ for $\mathcal{H}_1$ and $(a, b, z) = (\alpha + 1, \beta, -c)$ for $\mathcal{H}_2$ — read as its analytic continuation in the second parameter b beyond the convergence domain $b > 0$, since $\beta < 0$ by (203), which is precisely the finite part — and noting $a + b = b^\star(y)$ in both cases, gives

$$\text{f.p.}\,\mathcal{H}_1(y, y) = B(\alpha,\, \beta + 1)\, y^{\alpha+\beta}\, {}_1F_1\big(\alpha;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big), \tag{209}$$

$$\text{f.p.}\,\mathcal{H}_2(y, y) = B(\alpha + 1,\, \beta)\, y^{\alpha+\beta}\, {}_1F_1\big(\alpha + 1;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big). \tag{210}$$

Solving the no-branch condition (208) for $P_y(y)$ and inserting the finite parts (209)–(210), the common factor $y^{\alpha+\beta}$ cancels and

$$P_y(y) = -(1 - y)\,\pi(0, 0)\,\frac{\text{f.p.}\,\mathcal{H}_2(y, y)}{\text{f.p.}\,\mathcal{H}_1(y, y)} = -(1 - y)\,\pi(0, 0)\,\frac{B(\alpha + 1,\, \beta)}{B(\alpha,\, \beta + 1)}\,\frac{{}_1F_1\big(\alpha + 1;\, b^\star;\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha;\, b^\star;\, -\lambda_1 y/\gamma_1\big)}. \tag{211}$$

The Beta ratio reduces, via $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$ and $\Gamma(\beta + 1) = \beta\Gamma(\beta)$, to

$$\begin{aligned}\frac{B(\alpha + 1,\, \beta)}{B(\alpha,\, \beta + 1)} &= \frac{\Gamma(\alpha + 1)\Gamma(\beta)}{\Gamma(\alpha + \beta + 1)} \cdot \frac{\Gamma(\alpha + \beta + 1)}{\Gamma(\alpha)\Gamma(\beta + 1)} \\ &= \frac{\Gamma(\alpha + 1)}{\Gamma(\alpha)} \cdot \frac{\Gamma(\beta)}{\Gamma(\beta + 1)} = \frac{\alpha}{\beta},\end{aligned} \tag{212}$$

and the surviving prefactor simplifies, using $\alpha/\beta = \mu/\big[(1 - y)(\lambda y - \mu)\big]$, as

$$\begin{aligned}-(1 - y)\,\frac{\alpha}{\beta} &= -(1 - y) \cdot \frac{\mu}{(1 - y)(\lambda y - \mu)} \\ &= -\frac{\mu}{\lambda y - \mu} = \frac{\mu}{\mu - \lambda y}.\end{aligned} \tag{213}$$

Substituting (212)–(213) into (211) gives the boundary generating function

$$P_y(y) = \frac{\mu\,\pi(0,0)}{\mu - \lambda y} \cdot \frac{{}_1F_1\big(\alpha(y)+1;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}, \tag{214}$$

confirming (186). The prefactor equals the regular value $P(y,y)$ of (205), and the Kummer ratio is the boundary correction. As $y \to 0^+$ the argument $-\lambda_1 y/\gamma_1 \to 0$, so the ratio tends to 1 and $P_y(0) = \pi(0,0)$, consistent with $P_y(0) = \sum_{n_2} \pi(0,n_2)\cdot 0^{n_2} = \pi(0,0)$.

Stage 3: Recovery of $P(x,y)$.

Substituting (214) into the real-form expression (202) gives, for $0 \le x \le y$,

$$P(x,y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Bigg[\frac{\mu\,\pi(0,0)}{\mu-\lambda y}\cdot\frac{{}_1F_1\big(\alpha+1;\, b^\star;\, -\frac{\lambda_1 y}{\gamma_1}\big)}{{}_1F_1\big(\alpha;\, b^\star;\, -\frac{\lambda_1 y}{\gamma_1}\big)}\cdot\mathcal{H}_1(x,y) \\ + (1-y)\pi(0,0)\cdot\mathcal{H}_2(x,y)\Bigg], \tag{215}$$

which is the statement of the theorem (Equation (187)). The no-branch condition (208) guarantees that this extends analytically across $x = y$, with diagonal value $P(y,y) = \rho(1-\rho)/(1-\rho y)$ as computed in (205). On $y < x \le 1$ the same formula holds with $\mathcal{H}_1, \mathcal{H}_2$ read as their finite-part continuations; $\pi(0,0) = \rho(1-\rho)$ from Corollary 4 makes the solution fully explicit. *Reduction to Model-A as $\gamma_1 \to 0^+$.* The no-jockeying limit is cleanest at the level of the fundamental equation, but $\gamma_1 \to 0^+$ is a *singular* perturbation — the coefficient $\gamma_1 xy(x-y)$ of the highest derivative $\partial_x P$ vanishes — so one must check that the convection term itself, and not merely its coefficient, disappears; the derivative $\partial_x P$ could *a priori* blow up. The probabilistic bound rules this out. Each $P(\cdot,\cdot)$ is a probability generating function, so $|P| \le 1$ on the closed unit polydisc *uniformly* in γ_1; by Cauchy's estimate $\partial_x P$ is then bounded uniformly in γ_1 on every compact subset of the open polydisc, whence $\gamma_1 xy(x-y)\,\partial_x P \to 0$ there. The fundamental equation (185) therefore degenerates to the purely algebraic Model-A equation (106) (the coefficients of P and the right-hand side are γ_1-independent, and $\pi(0,0) = \rho(1-\rho)$ by Corollary 4). Since $\{P\}_{\gamma_1}$ is uniformly bounded it is a normal family, so it converges — along subsequences, hence in full by uniqueness of the polydisc-analytic generating-function solution of (106) — to the Model-A PGF (107). In particular the boundary function reduces to the Model-A value (115),

$$P_y(y) \xrightarrow[\gamma_1\to 0^+]{} \pi(0,0)\,\frac{x^*(y)\,(1-y)}{x^*(y)-y}, \tag{216}$$

with $x^*(y)$ the Model-A kernel root (113); this recovery needs no special-function asymptotics.

It is instructive to read the same limit off the closed form (214). Its prefactor satisfies $\mu\,\pi(0,0)/(\mu-\lambda y) = \pi(0,0)/(1-\rho y)$ and is γ_1-independent, so (216) is equivalent to the Kummer ratio tending to

$$\frac{{}_1F_1\big(\alpha+1;\, b^\star;\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha;\, b^\star;\, -\lambda_1 y/\gamma_1\big)} \xrightarrow[\gamma_1\to 0^+]{} (1-\rho y)\,\frac{x^*(y)\,(1-y)}{x^*(y)-y}. \tag{217}$$

Note the factor $(1-\rho y)$: the ratio does *not* tend to the bare rational expression $x^*(y)(1-y)/(x^*(y)-y)$, the discrepancy being precisely the prefactor. This is consistent with the three-term recurrence for ${}_1F_1$ in its first parameter [DLM, §13.3]: as $\gamma_1 \to 0^+$ the parameters $\alpha = \mu/(\gamma_1 y)$, $b^\star$ and $c = \lambda_1 y/\gamma_1$ all grow like $1/\gamma_1$, and to leading order that recurrence reduces to a quadratic in the ratio whose admissible root is the right-hand side of (217). □

8.2 Mean queue lengths and mean waiting times

Total queue length from the diagonal

Setting $x = y = z$ in the reduced fundamental equation (185), the jockeying convection term $\gamma_1 xy(x-y)\partial_x P$ vanishes at $x = y = z$ (factor $(x-y) = 0$), as does the boundary term $\mu(x-y)P_y(y)$. The remaining balance is identical to Model-A's, so the same diagonal cancellation gives:

$$P(z,z) = \frac{\rho(1-\rho)}{1-\rho z}, \tag{218}$$

the same formula as (146). Since $P(z,z) = \mathbb{E}[z^{N_1+N_2}] = \mathbb{E}[z^N]$ is the PGF of the total in-queue count N, the generating-function identity $\frac{d}{dz}P(z,z)\big|_{z=1} = \mathbb{E}[N_1] + \mathbb{E}[N_2]$ holds; differentiating by the quotient rule and evaluating at $z = 1$:

$$\mathbb{E}[N_1] + \mathbb{E}[N_2] \;=\; \frac{d}{dz}\,\frac{\rho(1-\rho)}{1-\rho z}\bigg|_{z=1} \;=\; \frac{\rho(1-\rho)\cdot\rho}{(1-\rho z)^2}\bigg|_{z=1} \;=\; \frac{\rho^2(1-\rho)}{(1-\rho)^2} \;=\; \frac{\rho^2}{1-\rho}. \tag{219}$$

One-way jockeying does not alter the total mean queue length relative to Model-A, since the diagonal PGF is unchanged.

Class-1 mean queue length

We extract $\mathbb{E}[N_1] = \frac{\partial P}{\partial x}\big|_{(1,1)}$ by first reducing the fundamental equation (185) at $y = 1$, and then differentiating the resulting marginal PGF at $x = 1$.

Step 1: marginal PGF $P(x,1)$. Evaluating (185) at $y = 1$, the bracket multiplying P becomes $(\lambda_1 + \mu)x - \mu - \lambda_1 x^2 = -\lambda_1(x-1)(x-1/\rho_1)$, the convection coefficient is $\gamma_1 x(x-1)$, and the right-hand side reduces to $\mu(x-1)P_y(1)$ (the $\pi(0,0)$ term carries the factor $(1-y) = 0$). Every term therefore shares the common factor $(x-1)$; cancelling it leaves the first-order linear ODE in x:

$$-\lambda_1(x - 1/\rho_1)\,P(x,1) \;+\; \gamma_1 x\,\frac{d}{dx}P(x,1) \;=\; \mu\,P_y(1). \tag{220}$$

The integrating factor for this ODE is $e^{-\lambda_1 x/\gamma_1}\,x^{\mu/\gamma_1}$ (noting $\lambda_1/\rho_1 = \mu$), and integrating from 0 to x (the boundary term vanishes since $0^{\mu/\gamma_1} = 0$):

$$P(x,1)\,e^{-\lambda_1 x/\gamma_1}\,x^{\mu/\gamma_1} \;=\; \frac{\mu\,P_y(1)}{\gamma_1}\int_0^x e^{-\lambda_1 t/\gamma_1}\,t^{\mu/\gamma_1 - 1}\,dt. \tag{221}$$

Rearranging and writing $\alpha = \mu/\gamma_1$ and $\mathcal{J}(x) = \int_0^x e^{-\lambda_1 t/\gamma_1}\,t^{\alpha-1}\,dt$:

$$P(x,1) \;=\; \frac{\mu\,P_y(1)}{\gamma_1}\cdot e^{\lambda_1 x/\gamma_1}\,x^{-\alpha}\,\mathcal{J}(x). \tag{222}$$

The constant $P_y(1)$ follows from $P(1,1) = 1 - \pi_0 = \rho$, which gives $P_y(1) = \rho\gamma_1/\big(\mu\,e^{\lambda_1/\gamma_1}\,\mathcal{J}(1)\big)$.

Step 2: differentiate $P(x,1)$ at $x = 1$. Writing $P(x,1) = C\cdot f(x)\cdot\mathcal{J}(x)$ with $C = \mu P_y(1)/\gamma_1$ and $f(x) = e^{\lambda_1 x/\gamma_1}\,x^{-\alpha}$, the product rule together with Leibniz's rule ($d\mathcal{J}/dx = e^{-\lambda_1 x/\gamma_1}\,x^{\alpha-1}$, the integrand at the upper limit) gives:

$$\frac{d}{dx}P(x,1) \;=\; C\Big[f'(x)\,\mathcal{J}(x) + f(x)\cdot e^{-\lambda_1 x/\gamma_1}\,x^{\alpha-1}\Big]. \tag{223}$$

The logarithmic derivative $f'(x)/f(x) = \lambda_1/\gamma_1 - \alpha/x = (\lambda_1 x - \mu)/(\gamma_1 x)$, and the second product term simplifies as $f(x)\,e^{-\lambda_1 x/\gamma_1}x^{\alpha-1} = x^{-1}$, so:

$$\frac{d}{dx}P(x,1) \;=\; \frac{\mu\,P_y(1)}{\gamma_1}\left[e^{\lambda_1 x/\gamma_1}\,x^{-\alpha}\cdot\frac{\lambda_1 x - \mu}{\gamma_1 x}\cdot\mathcal{J}(x) \;+\; x^{-1}\right]. \tag{224}$$

Step 3: evaluate at $x = 1$.

$$\begin{aligned}\mathbb{E}[N_1] \;&=\; \frac{d}{dx}P(x,1)\bigg|_{x=1}\\ &=\; \frac{\mu\,P_y(1)}{\gamma_1}\left[e^{\lambda_1/\gamma_1}\cdot\frac{\lambda_1 - \mu}{\gamma_1}\cdot\mathcal{J}(1) \;+\; 1\right],\end{aligned} \tag{225}$$

where $\mathcal{J}(1) = \int_0^1 e^{-\lambda_1 t/\gamma_1}\,t^{\mu/\gamma_1 - 1}\,dt = (\gamma_1/\lambda_1)^{\mu/\gamma_1}\,\gamma(\mu/\gamma_1,\,\lambda_1/\gamma_1)$ and $\gamma(\cdot,\cdot)$ is the lower incomplete gamma function. The factor $(\lambda_1 - \mu) < 0$ (since $\rho_1 = \lambda_1/\mu < 1$), confirming that $e^{\lambda_1/\gamma_1}(\lambda_1 - \mu)\mathcal{J}(1)/\gamma_1 + 1 > 0$. No elementary simplification of (225) is known; the expression is evaluated numerically. As $\gamma_1 \to 0^+$ the Model-A value (144) is recovered.

Class-2 mean queue length by subtraction

$$\mathbb{E}[N_2] \;=\; \big(\mathbb{E}[N_1] + \mathbb{E}[N_2]\big) - \mathbb{E}[N_1] \;=\; \frac{\rho^2}{1-\rho} - \mathbb{E}[N_1]. \tag{226}$$

Since jockeying defections reduce the class-1 queue, $\mathbb{E}[N_1]$ is smaller than the Model-A value $\rho\rho_1/(1-\rho_1)$, so $\mathbb{E}[N_2]$ correspondingly exceeds the Model-A value.

Mean waiting times via Little's Law

Little's Law applies here without qualification. It is a rate-conservation identity — for any subsystem in which customers are neither created nor destroyed internally, the time-average number present equals the long-run entry rate times the mean time an entrant spends inside — and it holds for *any* positive-recurrent system with finite mean queue length (here ensured by $\rho < 1$, which gives $\mathbb{E}[N_1] \leq \mathbb{E}[N] = \rho^2/(1-\rho) < \infty$ and so makes the jockeying rate $\gamma_1\mathbb{E}[N_1]$ below a genuine finite rate), requiring neither Poisson input nor independence. In particular it remains valid even though one-way jockeying has destroyed the per-class Poisson/renewal structure (the very obstruction that ruled out a probabilistic route in Stage 2). We apply it to each queue (the waiting line, excluding the customer in service), with $\mathbb{E}[W_i]$ understood as the mean time a customer spends in Queue i averaged over all customers that enter it — an arrival routed directly into service contributing a zero-length visit.

Class 1. Every class-1 arrival is counted as an entrant to Queue 1 at rate λ_1 (those finding the server idle enter service at once, contributing zero waiting time), and each leaves Queue 1 either by entering service or by jockeying to Queue 2. Since no other stream feeds Queue 1, its throughput is exactly λ_1, and Little's Law gives

$$\mathbb{E}[W_1] \;=\; \frac{\mathbb{E}[N_1]}{\lambda_1}. \tag{227}$$

Class 2. Queue 2 has *two* input streams: the class-2 Poisson arrivals at rate λ_2 and the jockeying transfers from Queue 1. Because each of the N_1 waiting class-1 customers jockeys independently at rate γ_1, the long-run transfer rate is $\gamma_1\,\mathbb{E}[N_1]$ (a genuine rate by the ergodic theorem, equal to the time average of the instantaneous jockeying intensity $\gamma_1 N_1$). The throughput of Queue 2 is therefore $\lambda_2 + \gamma_1\,\mathbb{E}[N_1]$, and Little's Law reads $\mathbb{E}[N_2] = \big(\lambda_2 + \gamma_1\,\mathbb{E}[N_1]\big)\,\mathbb{E}[W_2]$, so that

$$\mathbb{E}[W_2] \;=\; \frac{\mathbb{E}[N_2]}{\lambda_2 + \gamma_1\,\mathbb{E}[N_1]}. \tag{228}$$

Here $\mathbb{E}[W_2]$ is the mean time in the class-2 line averaged over *all* Queue-2 entrants of both streams; a jockeyed customer's Queue-2 clock starts at the transfer instant, so its total wait is an initial Queue-1 segment (of mean $\mathbb{E}[W_1]$) followed by this Queue-2 segment. Thus (228) is the per-Queue-2-entrant residence time, not the end-to-end delay of a tagged class-2 *arrival*.

The jockeying inflow is what distinguishes (228) from the single-stream Models A and C_2: the naive denominator λ_2 would undercount the entrants to Queue 2 and is correct only in the no-jockeying limit. Indeed, as $\gamma_1 \to 0^+$ the transfer rate $\gamma_1\,\mathbb{E}[N_1] \to 0$, so (228) reduces to $\mathbb{E}[N_2]/\lambda_2$ and both (227) and (228) recover the Model-A values (149)–(150).

9 Analysis of Model-C_2 ($\theta_1 > 0$, $\gamma_1 = \gamma_2 = \theta_2 = 0$)

In this section, we will derive $P(x, y)$ for this model where only true abandonments coming from Class-1 take place, while no abandonments in Class-2 or jockeying of any sort are taken into account.

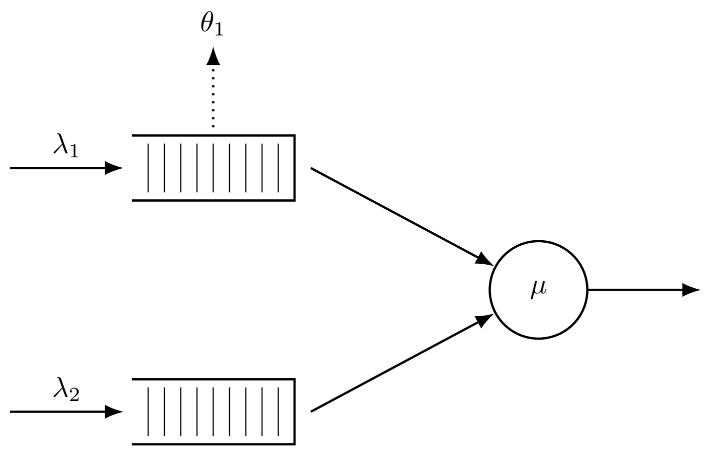

Figure 7: Queue layout for Model-C_2. Class-1 customers abandon the system at rate θ_1 per waiting customer (dotted arrow, upward from Queue 1). Class-2 never abandons ($\theta_2 = 0$) and there is no jockeying.

The rationale behind using this combination of parameters is that, thanks to the fact that Class-2 still sees Poisson arrivals, the same sort of probabilistic arguments we used for Model-A will hold. Moreover, we will see that the resulting equation when plugging this combination of parameters in our fundamental equation (Equation (233)) yields a *first-order linear ODE* in x and, $P_y(y)$ can be taken out of the integral, saving us from having to solve an equation of the Volterra family.

Theorem 4. *Consider a queueing system with 2 priority classes with Poisson arrival rates λ_1 and λ_2, where class-1 customers have non-preemptive priority over those of class-2, exponentially distributed service times of rate μ independently of the class, and per-customer abandonment of class-1 customers from the queue at rate θ_1 (with no class-2 abandonment and no jockeying). Under the stability condition*

$$\rho_2 \cdot {}_1F_1\Big(1;\, \tfrac{\mu}{\theta_1}+1;\, \tfrac{\lambda_1}{\theta_1}\Big) \;<\; 1 \quad\Longleftrightarrow\quad \lambda_2\,\mathbb{E}[B_C] \;<\; 1, \tag{229}$$

the probability generating function $P(x,y)$ of such a system is given by

$$\begin{aligned} P(x,y) \;=\; & \frac{e^{\lambda_1 x/\theta_1}\,\mu\,\pi(0,0)\,(1-y)}{\theta_1\, x^{\mu/\theta_1}(1-x)^{\lambda_2(1-y)/\theta_1}\big(\widetilde{B}_C(\lambda_2(1-y))-y\big)} \\ & \times\Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{H}_1(x,y)-\big(1-\widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{H}_2(x,y)\Big], \end{aligned} \tag{230}$$

where

$$\begin{aligned} \mathcal{H}_1(x,y) \;&=\; \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\,\mu/\theta_1-1}(1-t)^{\,\lambda_2(1-y)/\theta_1}\,dt, \\ \mathcal{H}_2(x,y) \;&=\; \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\,\mu/\theta_1}(1-t)^{\,\lambda_2(1-y)/\theta_1-1}\,dt, \end{aligned}$$

$\widetilde{B}_C(s)$ is the LST of the busy period of the $M/M/1{+}M$ subsystem of class-1 given by Equation (6), and $\pi(0,0)$ is given by Corollary 5 below.

Proof. The proof is structured in three stages.

Stage 1: Reduction to a first-order ODE in x and the integrating factor. Substituting $\theta_1>0$, $\gamma_1=\gamma_2=\theta_2=0$ into the general fundamental equation eliminates both convection terms in y, leaving a first-order linear ODE in x for each fixed $y\in[0,1]$ (Equation (233)). Partial-fraction decomposition of the ODE coefficient gives the integrating factor

$$\mu_I(x;y) \;=\; e^{-\lambda_1 x/\theta_1}\,x^{\alpha}\,(1-x)^{\beta(y)}, \qquad \alpha=\tfrac{\mu}{\theta_1}, \quad \beta(y)=\tfrac{\lambda_2(1-y)}{\theta_1},$$

which vanishes at $x=0$ and $x=1$ because $\alpha,\beta(y)>0$.

Stage 2: Determination of $P_y(y)$. Two independent routes are given in §§9.2–9.3:

- *Probabilistic (Pollaczek–Khinchine).* Because $\theta_2=\gamma_i=0$, the class-2 arrival stream remains Poisson; PASTA applies. Class-2 customers see an $M/G/1$ queue with effective service time B_C (the busy period of the class-1 $M/M/1{+}M$ subsystem), and the P-K formula yields $P_y(y)$ directly (Equation (240)).
- *Analytical (boundedness at $x=1$).* Since $\mu_I(1;y)=0$, the only way for $P(x,y)$ to remain bounded as $x\to 1$ is $\int_0^1 q(t;y)\mu_I(t;y)\,dt=0$ (Equation (250)). Evaluating this integral via the Kummer–Euler identity (257) recovers the same $P_y(y)$.

Stage 3: Recovery of $P(x,y)$. With $P_y(y)$ known, the ODE is integrated against the integrating factor (§9.4), yielding the closed form (264) stated in the theorem. □

Corollary 5. *The limiting probability $\pi(0,0)$ of the joint state $(N_1,N_2)=(0,0)$ with the server busy, and the empty-system probability π_0, are given by*

$$\pi(0,0) \;=\; \frac{(\lambda_1+\lambda_2)\big(1-\lambda_2\,\mathbb{E}[B_C]\big)}{\mu\,\big(1+\lambda_1\,\mathbb{E}[B_C]\big)} \;=\; (\rho_1+\rho_2)\,\frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{231}$$

$$\pi_0 \;=\; \frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{232}$$

where $\mathbb{E}[B_C]$ is the mean busy period of the $M/M/1{+}M$ subsystem of class-1, given by Equation (7). In the limit $\theta_1\to 0^+$, $\mathbb{E}[B_C]\to(\mu-\lambda_1)^{-1}$ and the expressions above recover the classical Model-A results $\pi(0,0)=\rho(1-\rho)$ and $\pi_0=1-\rho$.

9.1 Setup: reduction to a first-order linear ODE in x

Plugging $\theta_1 > 0$, $\gamma_1 = \gamma_2 = \theta_2 = 0$ into the general fundamental equation (42) (Lemma 1) annihilates the $\partial P/\partial y$ term entirely, since its coefficient $xy\big[\gamma_2(y-x)+\theta_2(y-1)\big]$ vanishes identically, and reduces the $\partial P/\partial x$ coefficient $xy\big[\gamma_1(x-y)+\theta_1(x-1)\big]$ to its single surviving abandonment term $\theta_1\, xy(x-1)$. What remains is

$$\big[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\big]\, P(x,y)+\theta_1\, xy(x-1)\,\frac{\partial}{\partial x}P(x,y) \;=\; \mu(x-y)\,P_y(y)-\mu x(1-y)\,\pi(0,0), \tag{233}$$

where the boundary function $P_y(y)=P(0,y)=\sum_{n_2=0}^{\infty}\pi(0,n_2)\,y^{n_2}$ is, as in Model-A, the partial PGF restricted to states where the server is busy and the priority queue is empty ($N_1=0$).

For any fixed y, Equation (233) is a *first-order linear ODE in x* whose unknowns are the function $P(x,y)$ on $[0,1]^2$ and the boundary trace $P_y(y)=P(0,y)$. The strategy in what follows is in two stages:

1. Identify $P_y(y)$ — first by a probabilistic argument exploiting the $M/G/1$ structure seen by Class-2 (§9.2), then re-derived analytically from boundedness of $P(x,y)$ at $x=1$ (§9.3).
2. Recover $P(x,y)$ by solving the ODE with $P_y(y)$ now known (§9.4).

We close the setup with the conditional-expectation decomposition that will drive the probabilistic derivation. By the law of total expectation, restricting the defining sum of $P_y(y)=\mathbb{E}[\mathbf{1}\{\text{busy}, N_1=0\}\,y^{N_2}]$ to the event $\{\text{busy}, N_1=0\}$ factors it into the mass of that event and the conditional PGF of N_2 given it:

$$P_y(y)\;=\;\mathbb{P}(\text{busy},\,N_1=0)\cdot\mathbb{E}\big[y^{N_2}\,\big|\,\text{busy},\,N_1=0\big]\,. \tag{234}$$

9.2 Probabilistic derivation of $P_y(y)$

The idea is, again, to adopt the perspective of a Class-2 customer waiting in line while the server is busy. Unlike in the case of jockeying, the customers from Class-1 truly abandon the system, leaving the Poisson arrival process of Class-2 undisturbed. Because Class-1 holds non-preemptive priority (§3.2), a waiting Class-2 customer is not taken into service while any Class-1 customer is present ($N_1>0$); its *effective service time* is therefore the duration of one complete busy period B_C of the Class-1 subsystem—now an $M/M/1+M$ queue with arrival rate λ_1, service rate μ and per-customer abandonment rate θ_1, rather than the pure $M/M/1$ of Model-A. By the *memorylessness* of the exponential service, arrival and abandonment clocks, these effective service times are i.i.d., so from Class-2's viewpoint the system is an $M/G/1$ queue with Poisson arrivals at rate λ_2 and general service times distributed as B_C.

The LST $\widetilde{B}_C(s)$ and mean $\mathbb{E}[B_C]$ of this busy period are derived via the first-step analysis in §3.3; see Equations (6)–(7).

The stability condition required for the Class-2 subsystem is that of the $M/G/1$ defined above, i.e. $\lambda_2\,\mathbb{E}[B_C]<1$:

$$\rho_2\cdot{}_1F_1\!\left(1;\,\frac{\mu}{\theta_1}+1;\,\frac{\lambda_1}{\theta_1}\right)\;<\;1. \tag{235}$$

Compared to Model-A, this stability condition is strictly weaker than the classical $\rho_1+\rho_2<1$, which aligns with the fact that abandonments lower the load of the system: for all $\theta_1>0$ the series is termwise dominated by the geometric series for $1/(1-\rho_1)$. Indeed, since $1^{(n)}=n!$ the Kummer series (10) reduces to

$${}_1F_1\!\left(1;\,\tfrac{\mu}{\theta_1}+1;\,\tfrac{\lambda_1}{\theta_1}\right)\;=\;\sum_{n=0}^{\infty}\frac{(\lambda_1/\theta_1)^n}{\left(\frac{\mu}{\theta_1}+1\right)^{(n)}}\;=\;\sum_{n=0}^{\infty}\frac{(\lambda_1/\theta_1)^n}{\prod_{k=1}^{n}\left(\frac{\mu}{\theta_1}+k\right)},$$

and since $\frac{\mu}{\theta_1}+k>\frac{\mu}{\theta_1}$ for every $k\ge 1$, the n-th term is bounded above by $(\lambda_1/\theta_1)^n/(\mu/\theta_1)^n=\rho_1^n$, with strict inequality for $n\ge 1$. Summing gives ${}_1F_1(1;\,\mu/\theta_1+1;\,\lambda_1/\theta_1)<\sum_{n\ge 0}\rho_1^n=1/(1-\rho_1)$. Abandonments in the priority queue therefore stabilise the non-priority queue.

The class-2 subsystem is therefore an $M/G/1$ queue with Poisson arrival rate λ_2, service-time LST $\widetilde{B}_C(s)$ and utilisation $\rho_B=\lambda_2\,\mathbb{E}[B_C]$. Applying the Pollaczek–Khinchine formula (9) (§3.4) with these identifications $\big(\lambda\mapsto\lambda_2,\ \widetilde{B}\mapsto\widetilde{B}_C,\ \rho_B=\lambda_2\mathbb{E}[B_C]\big)$ gives the PGF of the stationary number of class-2 customers in that $M/G/1$:

$$L_2(y)\;=\;\frac{(1-\lambda_2\mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y))-y}. \tag{236}$$

Because $\theta_2 = \gamma_i = 0$, the class-2 arrival stream is Poisson and satisfies the lack-of-anticipation assumption, so by the *PASTA* property (§3.2) the time-stationary distribution of the number of class-2 customers present coincides with the $M/G/1$ steady-state distribution. Conditioning on $\{\text{busy}, N_1 = 0\}$ — the event on which a class-2 customer is at the head of its queue — therefore identifies that count with N_2, giving

$$\mathbb{E}\left[y^{N_2} \,\middle|\, \text{busy}, N_1 = 0\right] \;=\; L_2(y). \tag{237}$$

Substituting the conditional identification (237) into the law-of-total-expectation decomposition (234) yields

$$P_y(y) \;=\; \mathbb{P}(\text{busy}, N_1 = 0)\left[\frac{(1-\lambda_2\mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}\right]. \tag{238}$$

By the definition of P_y, $P_y(0) = \pi(0,0)$. Evaluating (238) at $y = 0$,

$$\begin{aligned} P_y(0) \;=\; \pi(0,0) \;&=\; \mathbb{P}(\text{busy}, N_1 = 0)\left[\frac{(1-\lambda_2\mathbb{E}[B_C])\cdot 1\cdot \widetilde{B}_C(\lambda_2)}{\widetilde{B}_C(\lambda_2)}\right] \\ &=\; \mathbb{P}(\text{busy}, N_1 = 0)\,(1-\lambda_2\mathbb{E}[B_C]), \end{aligned} \tag{239}$$

so that $\mathbb{P}(\text{busy}, N_1 = 0) = \pi(0,0)/(1-\lambda_2\mathbb{E}[B_C])$. Substituting this back into (238), the $(1 - \lambda_2\mathbb{E}[B_C])$ factor cancels and we are left with

$$P_y(y) \;=\; \pi(0,0)\,\frac{(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}. \tag{240}$$

This is structurally the same expression as in the probabilistic argument for Model-A (Equation (115)), with the only difference being the effective service time: B_C is now the busy period of an $M/M/1+M$ subsystem with abandonment rate θ_1, rather than the busy period of a pure $M/M/1$.

9.3 Analytical re-derivation of $P_y(y)$ via boundedness at $x = 1$

For an analytic counterpart of the previous derivation, we put (233) in standard form:

$$\frac{\partial P}{\partial x} + \underbrace{\frac{(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy}{\theta_1\, x(x-1)}}_{p(x;\,y)} P \;=\; \underbrace{\frac{\mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0)}{\theta_1\, xy(x-1)}}_{q(x;\,y)}. \tag{241}$$

The numerator of $p(x;y)$ is the same polynomial we have seen in every model so far. With $x^*(y)$ and $x^+(y)$ as its roots (Equation (113)), we can write

$$(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy \;=\; -\lambda_1\big(x - x^*(y)\big)\big(x - x^+(y)\big), \tag{242}$$

with $x^*(y)\,x^+(y) = \mu/\lambda_1$. Partial fraction decomposition gives

$$\frac{-\lambda_1\big(x - x^*(y)\big)\big(x - x^+(y)\big)}{x(x-1)} \;=\; A + \frac{B}{x} + \frac{C}{x-1}, \tag{243}$$

where $A = -\lambda_1$ is read off from the ratio of leading coefficients of the quadratic terms in numerator and denominator, $B = \mu$ from setting $x = 0$ and using $\lambda_1\, x^* x^+ = \mu$, and $C = \lambda_2(1-y)$ by the cover-up rule at $x = 1$. Writing the numerator as $N(x) = (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy$, the coefficient of $1/(x-1)$ is its residue $C = N(1) = (\lambda_1+\lambda_2+\mu) - \mu - \lambda_1 - \lambda_2 y = \lambda_2(1-y)$; equivalently, since $N(x) = -\lambda_1\big(x - x^*(y)\big)\big(x - x^+(y)\big)$, this is the identity $\big(1 - x^*(y)\big)\big(1 - x^+(y)\big) = -\lambda_2(1-y)/\lambda_1$ evaluated at $x = 1$. Thus

$$p(x;y) \;=\; \frac{1}{\theta_1}\left[-\lambda_1 + \frac{\mu}{x} + \frac{\lambda_2(1-y)}{x-1}\right]. \tag{244}$$

Integrating term by term,

$$\int p(x;y)\,dx \;=\; \frac{1}{\theta_1}\left[-\lambda_1 x + \mu\ln x + \lambda_2(1-y)\ln(1-x)\right], \tag{245}$$

which yields the integrating factor

$$\mu_I(x;y) \;=\; e^{-\lambda_1 x/\theta_1}\cdot x^{\alpha}\cdot(1-x)^{\beta(y)}, \tag{246}$$

where, in parallel with Model-B_2, we set

$$\alpha \;=\; \frac{\mu}{\theta_1}, \qquad \beta(y) \;=\; \frac{\lambda_2(1-y)}{\theta_1}. \tag{247}$$

Under our stability assumptions both exponents are positive ($\alpha > 0$ always; $\beta(y) \geq 0$ for $y \in [0,1]$), so μ_I vanishes at both $x = 0$ and $x = 1$. From $\frac{d}{dx}[P\cdot\mu_I] = q\cdot\mu_I$, integration over $(0,x]$ reads

$$P(x,y)\,\mu_I(x;y) - P(0,y)\,\mu_I(0;y) \;=\; \int_0^x q(t;y)\,\mu_I(t;y)\,dt, \tag{248}$$

and the boundary term vanishes, yielding

$$P(x,y) \;=\; \frac{1}{\mu_I(x;y)}\int_0^x q(t;y)\,\mu_I(t;y)\,dt. \tag{249}$$

Now, in the same fashion as the root-cancellation argument used for Model-A, we exploit boundedness of PGFs to determine $P_y(y)$ by observing the behaviour of $P(x,y)$ as $x \to 1$. Since $\mu_I(x;y) \to 0$, the only way to keep the right-hand side of (249) bounded is

$$\int_0^1 q(t;y)\,\mu_I(t;y)\,dt \;=\; 0. \tag{250}$$

Substituting the explicit forms of q and μ_I into (250), the integrand is

$$q(t;y)\,\mu_I(t;y) \;=\; \frac{\mu\big[(t-y)\,P_y(y) - t(1-y)\,\pi(0,0)\big]}{\theta_1\,y}\cdot\frac{e^{-\lambda_1 t/\theta_1}\,t^{\alpha}(1-t)^{\beta(y)}}{t\,(t-1)}, \tag{251}$$

where $t^{\alpha}/t = t^{\alpha-1}$ and, with $b = \beta(y)$, the elementary identity $(1-t)^{\beta(y)}/(t-1) = -(1-t)^{\beta(y)-1}$ flips the sign and lowers the exponent. The constant prefactor $-\mu/(\theta_1 y)$, nonzero for $y \in (0,1]$, divides out of the vanishing condition (250), leaving

$$\int_0^1 e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha-1}(1-t)^{\,\beta(y)-1}\big[(t-y)\,P_y(y) - t(1-y)\,\pi(0,0)\big]\,dt \;=\; 0, \tag{252}$$

with α and $\beta(y)$ as in (247). Introducing the auxiliary integrals

$$\mathcal{J}_0(y) \;=\; \int_0^1 e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha-1}(1-t)^{\,\beta(y)-1}\,dt, \tag{253}$$

$$\mathcal{J}_1(y) \;=\; \int_0^1 e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha}(1-t)^{\,\beta(y)-1}\,dt, \tag{254}$$

and distributing the bracket over $t^{\alpha-1}$, note that $t\cdot t^{\alpha-1} = t^{\alpha}$ feeds $\mathcal{J}_1$ while $t^{\alpha-1}$ feeds $\mathcal{J}_0$. The two terms regroup as

$$P_y(y)\int_0^1 e^{-\lambda_1 t/\theta_1}\big(t^{\alpha} - y\,t^{\alpha-1}\big)(1-t)^{\beta(y)-1}\,dt \;-\; (1-y)\,\pi(0,0)\int_0^1 e^{-\lambda_1 t/\theta_1}\,t^{\alpha}(1-t)^{\beta(y)-1}\,dt \;=\; 0,$$

so the boundedness condition (250) becomes

$$P_y(y)\big[\mathcal{J}_1(y) - y\,\mathcal{J}_0(y)\big] \;=\; (1-y)\,\pi(0,0)\,\mathcal{J}_1(y), \tag{255}$$

which gives

$$P_y(y) \;=\; \pi(0,0)\,(1-y)\,\frac{\mathcal{J}_1(y)}{\mathcal{J}_1(y) - y\,\mathcal{J}_0(y)}. \tag{256}$$

The continued-fraction representation (6) admits an equivalent integral form. Setting $a = \alpha = \mu/\theta_1$, $b = s/\theta_1$, $c = \lambda_1/\theta_1$, the Euler integral representation (12) identifies each of the integrals below as a Beta-function multiple of a ${}_1F_1$ value:

$$\widetilde{B}_C(s) \;=\; \frac{\displaystyle\int_0^1 t^{a}(1-t)^{b-1}e^{-ct}\,dt}{\displaystyle\int_0^1 t^{a-1}(1-t)^{b-1}e^{-ct}\,dt} \;=\; \frac{B(a+1,b)\,{}_1F_1(a+1;\,a+1+b;\,-c)}{B(a,b)\,{}_1F_1(a;\,a+b;\,-c)}. \tag{257}$$

That this ratio equals the continued fraction (6) follows from the first-passage recursion (8) derived in §3.3, equivalently from the contiguous-function relations for ${}_1F_1$ [DLM].

Setting $s = \lambda_2(1-y)$ — so that $b = \beta(y)$ matches the exponent in (247) — yields exactly the ratio of our $\mathcal{J}_i$:

$$\widetilde{B}_C\big(\lambda_2(1-y)\big) \;=\; \frac{\mathcal{J}_1(y)}{\mathcal{J}_0(y)}. \tag{258}$$

Dividing numerator and denominator of (256) by $\mathcal{J}_0(y)$ then recovers exactly Equation (240). The boundedness route, which uses no probabilistic structure, reaches the same Pollaczek–Khinchine form only through the Beta-integral identity (257), which is the analytic shadow of the $M/G/1$/busy-period decomposition used in §9.2.

9.4 Recovery of $P(x, y)$ from $P_y(y)$

With $P_y(y)$ now in hand, we return to the ODE (233) and substitute (240) into its right-hand side. Writing $\zeta(y) \equiv \widetilde{B}_C(\lambda_2(1-y))$ for brevity,

$$\begin{aligned}
\text{RHS} &= \mu(x-y)\,\pi(0,0)\,\frac{(1-y)\,\zeta(y)}{\zeta(y)-y} \;-\; \mu x(1-y)\,\pi(0,0)\\
&= \mu\,\pi(0,0)\,(1-y)\left[\frac{(x-y)\,\zeta(y)}{\zeta(y)-y} - x\right]\\
&= \mu\,\pi(0,0)\,(1-y)\,\frac{(x-y)\,\zeta(y) - x\big[\zeta(y)-y\big]}{\zeta(y)-y}\\
&= \mu\,\pi(0,0)\,y(1-y)\,\frac{x-\zeta(y)}{\zeta(y)-y},
\end{aligned}$$

which is structurally identical to the RHS found for Model-A, though now with the more involved effective service time B_C. With $p(x;y)$ and $\mu_I(x;y)$ as in (244) and (246), the source $q(x;y)$ of the standard-form ODE now reads

$$q(x;y) \;=\; \frac{\mu\,\pi(0,0)\,(1-y)\,\big[x-\zeta(y)\big]}{\theta_1\,x(x-1)\,\big[\zeta(y)-y\big]}, \tag{259}$$

and the implicit solution (249) becomes

$$P(x,y) \;=\; \frac{1}{\mu_I(x;y)}\int_0^x q(t;y)\,\mu_I(t;y)\,dt. \tag{260}$$

To evaluate the integral, we use the partial-fraction identity

$$\frac{t-\zeta(y)}{t(t-1)} \;=\; \frac{\zeta(y)(t-1)+t\big(1-\zeta(y)\big)}{t(t-1)} \;=\; \frac{\zeta(y)}{t} + \frac{1-\zeta(y)}{t-1},$$

and define the auxiliary integrals

$$\mathcal{H}_1(x,y) \;=\; \int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha-1}(1-t)^{\,\beta(y)}\,dt, \tag{261}$$

$$\mathcal{H}_2(x,y) \;=\; \int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha}(1-t)^{\,\beta(y)-1}\,dt, \tag{262}$$

with α and $\beta(y)$ as in (247).

Note on the $\mathcal{J}$–$\mathcal{H}$ correspondence. The integrals $\mathcal{H}_1, \mathcal{H}_2$ are the partial (upper limit x) counterparts of the full integrals $\mathcal{J}_0, \mathcal{J}_1$ from (250). Evaluating at $x = 1$ gives $\mathcal{H}_2(1,y) = \mathcal{J}_1(y)$ exactly. However, $\mathcal{H}_1(1,y)$ is *not* $\mathcal{J}_0(y)$: the former has the exponent $\beta(y)$ on $(1-t)$ while the latter has $\beta(y)-1$, which under the Euler representation (12) corresponds to incrementing the second Kummer parameter by one. Both families are evaluated by the same representation; the exponent shift is harmless because $\mathcal{H}_1$ and $\mathcal{H}_2$ always appear together in the combination $\zeta(y)\,\mathcal{H}_1 - (1-\zeta(y))\,\mathcal{H}_2$, not individually.

Multiplying the source $q(t;y)$ by the integrating factor $\mu_I(t;y) = e^{-\lambda_1 t/\theta_1}t^{\alpha}(1-t)^{\beta(y)}$ and inserting the partial fraction, the integrand splits into the two pieces defining $\mathcal{H}_1, \mathcal{H}_2$:

$$q(t;y)\,\mu_I(t;y) \;=\; \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\left[\frac{\zeta(y)}{t} + \frac{1-\zeta(y)}{t-1}\right] e^{-\lambda_1 t/\theta_1}\,t^{\alpha}(1-t)^{\beta(y)}$$

$$= \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\Big[\zeta(y)\,e^{-\lambda_1 t/\theta_1}\,t^{\alpha-1}(1-t)^{\beta(y)} - \big(1-\zeta(y)\big)\,e^{-\lambda_1 t/\theta_1}\,t^{\alpha}(1-t)^{\beta(y)-1}\Big],$$

where the second term used $(1-t)^{\beta(y)}/(t-1) = -(1-t)^{\beta(y)-1}$, the same identity used to simplify the boundedness condition (250); this is the source of both the minus sign and the exponent shift $\beta(y) \to \beta(y)-1$ on $\mathcal{H}_2$. Since $\pi(0,0)$ and $\zeta(y)$ come out of the integral,

$$\int_0^x q(t;y)\,\mu_I(t;y)\,dt \;=\; \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\Big[\zeta(y)\,\mathcal{H}_1(x,y) \;-\; \big(1-\zeta(y)\big)\,\mathcal{H}_2(x,y)\Big]. \tag{263}$$

Dividing by $\mu_I(x;y)$ and restoring $\zeta(y) = \widetilde{B}_C(\lambda_2(1-y))$, we arrive at the closed form

$$\begin{aligned} P(x,y) \;&=\; \frac{e^{\lambda_1 x/\theta_1}\,\mu\,\pi(0,0)\,(1-y)}{\theta_1\,x^{\alpha}(1-x)^{\beta(y)}\big(\widetilde{B}_C(\lambda_2(1-y))-y\big)} \\ &\times\Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{H}_1(x,y) - \big(1-\widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{H}_2(x,y)\Big], \end{aligned} \tag{264}$$

which is the statement of the theorem. This closed form is the product of a probabilistic derivation of $P_y(y)$ via Pollaczek–Khinchine, combined with an analytic integration of the ODE in x; the same result is reached from the purely analytic route of §9.3.

Proof of Corollary 5. The fully idle state—the server empty with no customer present, of probability π_0—communicates with the rest of the chain through a single neighbour: the busy-and-empty state $(N_1,N_2)=(0,0)$ of probability $\pi(0,0)$, in which exactly one customer is in service and both queues are empty (recall the standing convention $\pi_0 \neq \pi(0,0)$). The system leaves the idle state only upon an arrival of either class, at combined Poisson rate $\lambda_1+\lambda_2$ (Poisson superposition), and re-enters it only upon the service completion that empties the busy-and-empty state, at rate μ; no abandonment can fire from $\pi(0,0)$, since both queues are empty. Equating the probability flow across the cut separating the idle state from its single neighbour (global balance for the idle state, as in (29)) therefore gives

$$(\lambda_1+\lambda_2)\,\pi_0 \;=\; \mu\,\pi(0,0), \tag{265}$$

so $\pi(0,0) = (\rho_1+\rho_2)\,\pi_0$.

To fix π_0, we exploit the $M/G/1$ structure used in §9.2. Class-2 customers form an $M/G/1$ queue with arrival rate λ_2 and effective service time B_C. By the standard $M/G/1$ work-conservation (utilisation) identity, the long-run fraction of time this $M/G/1$ is busy is $\rho_B = \lambda_2\,\mathbb{E}[B_C]$, so its idle probability is $1-\rho_B$. The event "this $M/G/1$ is idle" coincides with the absence of any Class-2 customer from the system, so

$$\mathbb{P}(\text{no Class-2 present}) \;=\; 1-\lambda_2\,\mathbb{E}[B_C]. \tag{266}$$

Conditional on no Class-2 being present, the priority subsystem evolves freely as the same $M/M/1+M$ queue used to define B_C, with arrival rate λ_1, service rate μ and abandonment rate θ_1. This subsystem alternates between idle periods (no Class-1 customer present) and busy periods B_C, the successive idle+busy spans forming an alternating renewal process whose regeneration epochs are the instants at which the subsystem empties. An idle period lasts until the next Class-1 arrival; by the memorylessness of the exponential interarrival clock this duration is $\mathrm{Exp}(\lambda_1)$, with mean $1/\lambda_1$. The mean regeneration cycle is therefore $1/\lambda_1 + \mathbb{E}[B_C]$, and by the renewal–reward theorem the stationary fraction of time the subsystem is idle equals the ratio of the mean idle period to the mean cycle:

$$\mathbb{P}(\text{Class-1 absent} \mid \text{no Class-2 present}) \;=\; \frac{1/\lambda_1}{1/\lambda_1+\mathbb{E}[B_C]} \;=\; \frac{1}{1+\lambda_1\,\mathbb{E}[B_C]}. \tag{267}$$

The fully idle event $\{\text{server idle}, N_1=N_2=0\}$ is exactly the intersection $A\cap B$ of the events $B = \{\text{no Class-2 present}\}$ and $A = \{\text{Class-1 absent}\}$, since the server is idle precisely when no customer of either class is in the system. By the multiplication (chain) rule $\mathbb{P}(A\cap B) = \mathbb{P}(B)\,\mathbb{P}(A \mid B)$—using the conditional idle fraction just computed for the factor $\mathbb{P}(A\mid B)$, and *not* independence of A and B—we obtain

$$\pi_0 \;=\; \big(1-\lambda_2\,\mathbb{E}[B_C]\big)\cdot\frac{1}{1+\lambda_1\,\mathbb{E}[B_C]} \;=\; \frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]}, \tag{268}$$

which is (232). Substituting into (265) yields (231).

For the Model-A limit, $\theta_1 \to 0^+$ gives $\mathbb{E}[B_C] \to (\mu - \lambda_1)^{-1}$ (the busy period of the $M/M/1+M$ subsystem collapses to that of the pure $M/M/1$ priority subsystem). Then

$$\lambda_1\,\mathbb{E}[B_C] \to \frac{\lambda_1}{\mu-\lambda_1} = \frac{\rho_1}{1-\rho_1}, \qquad \lambda_2\,\mathbb{E}[B_C] \to \frac{\lambda_2}{\mu-\lambda_1} = \frac{\rho_2}{1-\rho_1},$$

so that the denominator $1 + \lambda_1\,\mathbb{E}[B_C] \to 1/(1-\rho_1)$ and the numerator

$$1 - \lambda_2\,\mathbb{E}[B_C] \to 1 - \frac{\rho_2}{1-\rho_1} = \frac{1-\rho_1-\rho_2}{1-\rho_1} = \frac{1-\rho}{1-\rho_1}.$$

Substituting into (232),

$$\pi_0 \to \frac{(1-\rho)/(1-\rho_1)}{1/(1-\rho_1)} = 1-\rho,$$

and then $\pi(0,0) = (\rho_1+\rho_2)\,\pi_0 \to \rho(1-\rho)$ via (265), recovering the classical Model-A baseline values. ☐

9.5 Mean queue lengths and mean waiting times

Why the diagonal no longer simplifies

Setting $x = y = z$ in the fundamental equation (233), the abandonment convection $\theta_1 xy(x-1)\partial_x P$ does *not* vanish (since $(x-1)|_{x=z} \neq 0$ for $z \in (0,1)$). The diagonal equation reads:

$$\begin{aligned}&\big[(\lambda_1+\lambda_2+\mu)z^2 - \mu z - \lambda z^3\big]P(z,z) + \theta_1 z^2(z-1)\,\frac{\partial P}{\partial x}(z,z)\\ &\quad= -\mu z(1-z)\,\pi(0,0),\end{aligned} \tag{269}$$

Comparison with Model-A is instructive. There the same diagonal substitution $x = y = z$ also annihilates the boundary term $\mu(x-y)\,P_y(y)$, but with $\theta_1 = \gamma_1 = \gamma_2 = 0$ there is *no* convection term at all, so the diagonal equation reduced to a purely algebraic relation in the single unknown $P(z,z)$, with closed form $P(z,z) = \pi(0,0)/(1-\rho z)$ (Equation (146)), from which $\mathbb{E}[N_1]+\mathbb{E}[N_2]$ followed by a single differentiation. Here the surviving convection $\theta_1 z^2(z-1)\,\partial_x P(z,z)$ injects the off-diagonal slope $\partial_x P(z,z)$ as a *second* unknown into the lone diagonal relation, and no companion equation closes the system; consequently no closed form for $P(z,z)$ —and hence for the total $\mathbb{E}[N_1]+\mathbb{E}[N_2]$—is available. Abandonment removes customers and reduces both means relative to Model-A.

Class-2 mean queue length via $P(1,y)$

Setting $x = 1$ in the fundamental equation (233), the abandonment term $\theta_1 xy(x-1)\partial_x P$ vanishes identically, and the coefficient of $P(1,y)$ simplifies to $\lambda_2 y(1-y)$:

$$\lambda_2 y(1-y)\,P(1,y) \;=\; \mu(1-y)\big[P_y(y) - \pi(0,0)\big]. \tag{270}$$

Cancelling $(1-y)$ for $y \neq 1$ and substituting the known $P_y(y)$ (240):

$$P(1,y) \;=\; \frac{\mu\big[P_y(y)-\pi(0,0)\big]}{\lambda_2 y} \;=\; \frac{\mu\,\pi(0,0)}{\lambda_2}\cdot\frac{1-\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y))-y}. \tag{271}$$

This is the marginal PGF of N_2. To differentiate at $y = 1$ (where the expression is $0/0$), set $\varepsilon = 1-y \to 0$ and use the LST expansion $\widetilde{B}_C(\lambda_2\varepsilon) = 1 - \lambda_2\mathbb{E}[B_C]\,\varepsilon + \frac{1}{2}\lambda_2^2\mathbb{E}[B_C^2]\,\varepsilon^2 + O(\varepsilon^3)$:

$$\begin{aligned}1-\widetilde{B}_C(\lambda_2\varepsilon) &= \lambda_2\mathbb{E}[B_C]\,\varepsilon - \tfrac{1}{2}\lambda_2^2\mathbb{E}[B_C^2]\,\varepsilon^2 + O(\varepsilon^3),\\ \widetilde{B}_C(\lambda_2\varepsilon) - y &= \big(1-\lambda_2\mathbb{E}[B_C]\,\varepsilon + \tfrac{1}{2}\lambda_2^2\mathbb{E}[B_C^2]\,\varepsilon^2\big) - (1-\varepsilon)\\ &= (1-\lambda_2\mathbb{E}[B_C])\,\varepsilon + \tfrac{1}{2}\lambda_2^2\mathbb{E}[B_C^2]\,\varepsilon^2 + O(\varepsilon^3),\end{aligned} \tag{272}$$

so that (writing $a = \lambda_2\mathbb{E}[B_C]$ and $c = \frac{1}{2}\lambda_2^2\mathbb{E}[B_C^2]$):

$$P(1,1-\varepsilon) \;=\; \frac{\mu\,\pi(0,0)}{\lambda_2}\cdot\frac{a\varepsilon - c\varepsilon^2 + O(\varepsilon^3)}{(1-a)\varepsilon + c\varepsilon^2 + O(\varepsilon^3)} \;=\; \frac{\mu\,\pi(0,0)}{\lambda_2}\cdot\frac{a-c\varepsilon}{(1-a)+c\varepsilon} + O(\varepsilon^2). \tag{273}$$

Differentiating $f(\varepsilon) = (a - c\varepsilon)/((1-a) + c\varepsilon)$ by the quotient rule:

$$f'(\varepsilon) \;=\; \frac{-c\big((1-a)+c\varepsilon\big) - (a-c\varepsilon)\cdot c}{\big((1-a)+c\varepsilon\big)^2} \;=\; \frac{-c}{(1-a+c\varepsilon)^2}, \tag{274}$$

and since $d/dy\, P(1,y)\big|_{y=1} = -d/d\varepsilon\, P(1,1-\varepsilon)\big|_{\varepsilon=0}$:

$$\mathbb{E}[N_2] \;=\; \frac{d}{dy}P(1,y)\bigg|_{y=1} \;=\; \frac{\mu\,\pi(0,0)}{\lambda_2}\cdot\frac{c}{(1-a)^2} \;=\; \frac{\mu\,\pi(0,0)\,\lambda_2\,\mathbb{E}[B_C^2]}{2\,(1-\lambda_2\mathbb{E}[B_C])^2}. \tag{275}$$

As $\theta_1 \to 0^+$, $\mathbb{E}[B_C] \to (\mu-\lambda_1)^{-1}$ and $\mathbb{E}[B_C^2] \to 2\mu/(\mu-\lambda_1)^3$; substituting recovers the Model-A value $\mathbb{E}[N_2] = \rho\rho_2/((1-\rho_1)(1-\rho))$.

Class-1 mean queue length

We extract $\mathbb{E}[N_1] = \frac{\partial P}{\partial x}\big|_{(1,1)}$ by evaluating the theorem formula at $y = 1$ (limit) and then differentiating in x.

Step 1: marginal PGF $P(x,1)$. As $y \to 1$, the exponent $\beta(y) = \lambda_2(1-y)/\theta_1 \to 0$ so $(1-x)^{\beta(y)} \to 1$ for fixed $x \in (0,1)$, and consequently $\mathcal{H}_1(x,y) \to \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\mu/\theta_1 - 1}\, dt$ and $\mathcal{H}_2(x,y)$ tends to a finite limit (its integrand $e^{-\lambda_1 t/\theta_1} t^{\mu/\theta_1}(1-t)^{-1}$ is integrable on $[0,x]$ for $x<1$). Inside the theorem's bracket the two integrals carry different weights: by the LST expansion of §9.2, $\widetilde{B}_C(\lambda_2(1-y)) \to \widetilde{B}_C(0) = 1$ while $1 - \widetilde{B}_C(\lambda_2(1-y)) = \lambda_2\mathbb{E}[B_C]\,(1-y) + O((1-y)^2) \to 0$, so the $\mathcal{H}_2$ term vanishes in the limit and the bracket collapses to $\mathcal{H}_1(x,1)$ alone. Combining this with the scalar prefactor, whose y-dependent part satisfies $(1-y)/(\widetilde{B}_C(\lambda_2(1-y)) - y) \to 1/(1-\lambda_2\mathbb{E}[B_C])$ (again from the expansion $\widetilde{B}_C(\lambda_2(1-y)) - y = (1-\lambda_2\mathbb{E}[B_C])(1-y) + O((1-y)^2)$), the theorem formula gives:

$$P(x,1) \;=\; \frac{\mu\,\pi(0,0)}{\theta_1\,(1-\lambda_2\mathbb{E}[B_C])}\cdot e^{\lambda_1 x/\theta_1}\, x^{-\mu/\theta_1}\int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\mu/\theta_1-1}\, dt. \tag{276}$$

Step 2: differentiate $P(x,1)$ at $x=1$. Write $P(x,1) = C\cdot f(x)\cdot\mathcal{K}(x)$ with $C = \mu\pi(0,0)/(\theta_1(1-\lambda_2\mathbb{E}[B_C]))$, $f(x) = e^{\lambda_1 x/\theta_1}\, x^{-\mu/\theta_1}$, and $\mathcal{K}(x) = \int_0^x e^{-\lambda_1 t/\theta_1}\, t^{\mu/\theta_1-1}\, dt$. By the product rule and Leibniz's rule ($d\mathcal{K}/dx = e^{-\lambda_1 x/\theta_1}\, x^{\mu/\theta_1-1}$):

$$\frac{d}{dx}P(x,1) \;=\; C\Big[f'(x)\,\mathcal{K}(x) + f(x)\cdot e^{-\lambda_1 x/\theta_1}\, x^{\mu/\theta_1-1}\Big]. \tag{277}$$

The logarithmic derivative $f'(x)/f(x) = \lambda_1/\theta_1 - (\mu/\theta_1)/x = (\lambda_1 x - \mu)/(\theta_1 x)$, so:

$$\frac{d}{dx}P(x,1) \;=\; C\left[e^{\lambda_1 x/\theta_1}\, x^{-\mu/\theta_1}\cdot\frac{\lambda_1 x-\mu}{\theta_1 x}\cdot\mathcal{K}(x) \;+\; x^{-1}\right]. \tag{278}$$

Step 3: evaluate at $x=1$.

$$\begin{aligned}\mathbb{E}[N_1] \;&=\; \frac{d}{dx}P(x,1)\bigg|_{x=1}\\ &=\; \frac{\mu\,\pi(0,0)}{\theta_1\,(1-\lambda_2\mathbb{E}[B_C])}\left[e^{\lambda_1/\theta_1}\cdot\frac{\lambda_1-\mu}{\theta_1}\cdot\mathcal{K}(1) \;+\; 1\right],\end{aligned} \tag{279}$$

where $\mathcal{K}(1) = \int_0^1 e^{-\lambda_1 t/\theta_1}\, t^{\mu/\theta_1-1}\, dt = (\theta_1/\lambda_1)^{\mu/\theta_1}\,\gamma(\mu/\theta_1,\,\lambda_1/\theta_1)$, with $\gamma(\cdot,\cdot)$ the lower incomplete gamma function. The factor $(\lambda_1-\mu) < 0$ (since $\rho_1 < 1$), so the bracket is positive. No elementary simplification is known; $\mathbb{E}[N_1]$ is evaluated numerically. As $\theta_1 \to 0^+$, $\mathbb{E}[N_1]$ recovers the Model-A value (144).

Little's Law with abandonment

The universal relation $\mathbb{E}[N_i] = \lambda_i\,\mathbb{E}[W_i]$ holds for any stable system. Here λ_i is the *arrival* rate of all class-i customers, including those that will eventually abandon, so $\mathbb{E}[W_i]$ is the mean time in queue averaged over both served and abandoned customers:

$$\mathbb{E}[W_1] \;=\; \frac{\mathbb{E}[N_1]}{\lambda_1}, \qquad \mathbb{E}[W_2] \;=\; \frac{\mathbb{E}[N_2]}{\lambda_2}. \tag{280}$$

Using the throughput $\mu(1-\pi_0)$ instead would give the conditional waiting time of customers who complete service only — a different object. As $\theta_1 \to 0^+$ both means recover the Model-A values (144)–(150).

10 Analysis of Model-B_2^1 ($\gamma_1 > 0$ head-of-line, $\gamma_2 = \theta_1 = \theta_2 = 0$)

Model-B_2^1 is a deliberately simplified variant of the one-way jockeying model Model-B_2. In Model-B_2 *every* waiting class-1 customer may jockey to the class-2 queue, so the aggregate jockeying rate out of a state grows linearly with the queue length, $\gamma_1 n_1$. Here, by contrast, only the *customer at the head of Queue 1* — the one that would be the next class-1 customer to enter service — is allowed to jockey, at the fixed rate γ_1 whenever $n_1 \geq 1$. The aggregate jockeying rate is therefore $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ rather than $\gamma_1 n_1$.

This single change is decisive for tractability. In Model-B_2 the linear coefficient $\gamma_1 n_1$ produces, upon forming the joint PGF, a term proportional to $xy(x-y)\,\partial P/\partial x$, turning the fundamental equation into a first-order linear ODE in x that requires the integrating-factor machinery of Section 8. The flat rate $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$, on the other hand, contributes only the *algebraic* factor $\gamma_1 y(x-y)$ to the coefficient of $P(x,y)$; *no derivative of P appears.* The fundamental equation is then purely algebraic and can be solved by the Kernel Method exactly as in Model-A (Section 6). The analysis below mirrors the analytical treatment of Model-A step by step, and the resulting PGF turns out to be *structurally identical* to that of Model-A, the only difference being a shift in the characteristic roots.

Stability. Jockeying relocates a waiting customer between the two queues but does not remove anyone from the system; the total number of customers is therefore conserved by the jockeying mechanism, exactly as in Model-A. Since there is no abandonment, the chain is positive recurrent if and only if the total traffic intensity satisfies

$$\rho = \frac{\lambda_1 + \lambda_2}{\mu} < 1, \tag{281}$$

with $\rho_1 = \lambda_1/\mu$ governing the priority class.

10.1 Balance equations

Because the jockeying rate is now constant in n_1 rather than linear, the balance equations are *not* obtained by zeroing parameters in the general Lemma 1 (whose jockeying coefficient is $\gamma_1 n_1$); they must be stated directly. The jockeying transition $(n_1, n_2) \to (n_1 - 1, n_2 + 1)$ occurs at the fixed rate γ_1 whenever $n_1 \geq 1$. On the state space S the global balance equations read

$$\begin{aligned} &[\lambda_1 + \lambda_2 + \mu + \gamma_1]\,\pi(n_1, n_2) \\ &\quad = \quad \mu\,\pi(n_1+1, n_2) + \lambda_1 \pi(n_1 - 1, n_2) + \lambda_2 \pi(n_1, n_2 - 1) + \gamma_1 \pi(n_1 + 1, n_2 - 1) \qquad n_1 \geq 1,\ n_2 \geq 1, \end{aligned} \tag{282}$$

$$[\lambda_1 + \lambda_2 + \mu + \gamma_1]\,\pi(n_1, 0) \ = \ \mu\,\pi(n_1+1, 0) + \lambda_1 \pi(n_1 - 1, 0) \qquad n_1 \geq 1,\ n_2 = 0, \tag{283}$$

$$\begin{aligned} &[\lambda_1 + \lambda_2 + \mu]\,\pi(0, n_2) \\ &\quad = \quad \mu\,\pi(1, n_2) + \gamma_1 \pi(1, n_2 - 1) + \lambda_2 \pi(0, n_2 - 1) + \mu\,\pi(0, n_2 + 1) \qquad n_1 = 0,\ n_2 \geq 1, \end{aligned} \tag{284}$$

$$[\lambda_1 + \lambda_2 + \mu]\,\pi(0,0) = (\lambda_1 + \lambda_2)\pi_0 + \mu\,\pi(1,0) + \mu\,\pi(0,1), \tag{285}$$

$$(\lambda_1 + \lambda_2)\pi_0 = \mu\,\pi(0,0). \tag{286}$$

Two structural remarks are in order. First, jockeying is an *out*-transition from any state with $n_1 \geq 1$ (hence the $+\gamma_1$ in the left-hand sides of (282) and (283)) and an *in*-transition into any state with $n_2 \geq 1$ from the state $(n_1 + 1, n_2 - 1)$ (hence the $\gamma_1 \pi(n_1 + 1, n_2 - 1)$ in (282) and the $\gamma_1 \pi(1, n_2 - 1)$ in (284)). Second, when $n_2 = 0$ there is no jockeying in-flow, since it would have to originate from the non-existent state $(n_1 + 1, -1)$; and when $n_1 = 0$ there is no jockeying out-flow. The remaining transitions — Poisson arrivals at λ_1, λ_2 and service completions at μ — are exactly those of Model-A.

10.2 Fundamental PGF equation

Lemma 3 (Fundamental equation for Model-B_2^1)**.** *Assume positive recurrence, so that the stationary distribution exists and $P(x,y) = \sum_{n_1, n_2 \geq 0} \pi(n_1, n_2) x^{n_1} y^{n_2}$ is analytic on the closed unit* *polydisc. Then $P(x,y)$ and the boundary function $P_y(y) = P(0,y)$ satisfy*

$$\begin{aligned}&\big[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 xy^2 + \gamma_1 y(x-y)\big]\,P(x,y)\\ =\quad &\big[\mu(x-y) + \gamma_1 y(x-y)\big]\,P_y(y) - \mu x(1-y)\,\pi(0,0).\end{aligned} \tag{287}$$

Proof. Apart from the jockeying terms, equations (282)–(286) coincide with the balance equations of Model-A. Multiplying each balance equation by $x^{n_1}y^{n_2}$, summing over its domain and multiplying through by xy, the Model-A part reproduces exactly the left- and right-hand sides of the Model-A fundamental equation (106),

$$\big[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 xy^2\big]\,P(x,y) = \mu(x-y)P_y(y) - \mu x(1-y)\,\pi(0,0),$$

so it remains only to collect the jockeying contribution. Its *out*-flow appears in every state with $n_1 \geq 1$:

$$\gamma_1 \sum_{n_1\geq 1}\sum_{n_2\geq 0} \pi(n_1,n_2)\,x^{n_1}y^{n_2} = \gamma_1\big[P(x,y) - P_y(y)\big],$$

while its *in*-flow enters every state with $n_2 \geq 1$ from (n_1+1, n_2-1):

$$\gamma_1 \sum_{n_1\geq 0}\sum_{n_2\geq 1} \pi(n_1+1,n_2-1)\,x^{n_1}y^{n_2} = \gamma_1\,\frac{y}{x}\sum_{m_1\geq 1}\sum_{m_2\geq 0}\pi(m_1,m_2)\,x^{m_1}y^{m_2} = \gamma_1\,\frac{y}{x}\big[P(x,y)-P_y(y)\big],$$

where we substituted $m_1 = n_1+1,\ m_2 = n_2 - 1$. The net jockeying contribution (out minus in) is therefore $\gamma_1\big(1-\frac{y}{x}\big)\big[P(x,y)-P_y(y)\big]$; multiplying by xy gives $\gamma_1 y(x-y)\big[P(x,y)-P_y(y)\big]$. Adding this to the Model-A equation and moving the P_y part to the right-hand side yields (287). ☐

Remark. Equation (287) contains *no derivative* of P. The flat jockeying rate has contributed the algebraic factor $\gamma_1 y(x-y)$ to the coefficient of $P(x,y)$ where Model-B_2, with its linear rate $\gamma_1 n_1$, contributes the convective term $\gamma_1 xy(x-y)\,\partial P/\partial x$. Setting $\gamma_1 = 0$ recovers (106) verbatim.

10.3 Closed-form solution

The following theorem is the main result of this section; its proof follows the analytical (Kernel Method) treatment of Model-A in §6.1.1.

Theorem 5 (Closed-form PGF for Model-B_2^1)**.** *Consider the two-class non-preemptive priority queue with Poisson arrival rates λ_1, λ_2, common exponential service rate μ, and head-of-line class-1 jockeying at rate $\gamma_1 > 0$ (with $\gamma_2 = \theta_1 = \theta_2 = 0$). Under the stability condition (281), the joint PGF $P(x,y) = \mathbb{E}[x^{N_1}y^{N_2}]$ is the rational function*

$$P(x,y) \;=\; \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,\big(x^*(y)-y\big)\,\big(x^+(y)-x\big)}, \tag{288}$$

where $x^*(y)$ and $x^+(y)$ are the two roots of the kernel quadratic

$$\lambda_1 x^2 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \gamma_1\big]x + (\mu + \gamma_1 y) = 0, \tag{289}$$

with $x^*(y)$ the root inside the unit disk. This is the same rational form *as Model-A (Theorem 1); the two differ only in the kernel, where μ has been replaced by $\mu + \gamma_1 y$.*

Corollary 6. *Under (281), the empty-state probabilities of Model-B_2^1 coincide with those of Model-A,*

$$\pi_0 = 1-\rho, \qquad \pi(0,0) = \rho(1-\rho). \tag{290}$$

In particular $P(1,1) = 1 - \pi_0 = \rho$ and the diagonal generating function satisfies $P(z,z) = \pi(0,0)/(1-\rho z)$.

Proof of Corollary 6. Set $x = y = z$ in the fundamental equation (287). The jockeying factor $\gamma_1 y(x-y)$ vanishes on the diagonal — a direct consequence of the fact that jockeying conserves the total number of customers — and so does the term $\mu(x-y)P_y(y)$, leaving

$$\big[(\lambda_1+\lambda_2+\mu)z^2 - \mu z - (\lambda_1+\lambda_2)z^3\big]P(z,z) = -\mu z(1-z)\,\pi(0,0).$$

Dividing by z and factoring the bracket as $-(\lambda z - \mu)(z-1)$, with $\lambda = \lambda_1 + \lambda_2$,

$$(\lambda z - \mu)(1-z)\,P(z,z) = -\mu(1-z)\,\pi(0,0).$$

Cancelling $(1-z) \neq 0$ for $z \in [0,1)$ gives the diagonal identity

$$P(z,z) = \frac{\mu\, \pi(0,0)}{\mu - \lambda z} = \frac{\pi(0,0)}{1-\rho z}. \tag{291}$$

Setting $z = 1$ and using the normalisation $P(1,1) = 1-\pi_0$ yields $1-\pi_0 = \pi(0,0)/(1-\rho)$. The idle balance (286) gives $\lambda\pi_0 = \mu\pi(0,0)$, i.e. $\pi(0,0) = \rho\pi_0$; substituting, $(1-\pi_0)(1-\rho) = \rho\pi_0$, hence $\pi_0 = 1-\rho$ and $\pi(0,0) = \rho(1-\rho)$. □

10.3.1 Analytical approach for Model-B_2^1

Proof of Theorem 5. We apply the *Kernel Method* to the fundamental equation (287). Fix $y \in (0,1]$ and seek the values of x at which the coefficient of $P(x,y)$ — the *kernel* — vanishes. Since $P(x,y)$ is a PGF it is bounded on the closed unit disk, so at any such root the right-hand side of (287) must vanish as well. Dividing the kernel by $y \neq 0$,

$$0 = (\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy + \gamma_1(x-y),$$

and rearranging into a quadratic in x gives exactly (289),

$$\lambda_1 x^2 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \gamma_1\big]x + (\mu + \gamma_1 y) = 0,$$

whose two roots are

$$x^{\pm}(y) = \frac{A(y) \pm \sqrt{A(y)^2 - 4\lambda_1(\mu+\gamma_1 y)}}{2\lambda_1}, \qquad A(y) := \lambda_1 + \lambda_2(1-y) + \mu + \gamma_1. \tag{292}$$

This is precisely the Model-A kernel quadratic (113) with μ replaced by $\mu + \gamma_1 y$. We label the root carrying the negative sign $x^*(y)$ and the other $x^+(y)$. By Vieta's formulas the product of the roots is $x^*(y)\, x^+(y) = (\mu + \gamma_1 y)/\lambda_1$.

To locate the roots relative to the unit disk we evaluate the kernel quadratic, written as $f(x) := \lambda_1 x^2 - A(y)x + (\mu + \gamma_1 y)$, at $x = 1$:

$$f(1) = \lambda_1 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \gamma_1\big] + (\mu + \gamma_1 y) = -(1-y)\,(\lambda_2 + \gamma_1) < 0 \qquad \text{for } y \in [0,1).$$

Since f is an upward parabola ($\lambda_1 > 0$) with $f(1) < 0$, the point $x = 1$ lies strictly between its two roots, so $x^*(y) < 1 < x^+(y)$ for all $y \in [0,1)$. Hence $x^*(y)$ is the unique root inside the unit disk, and at $y = 1$ one checks $f(1) = 0$ with $x^*(1) = 1$ and $x^+(1) = (\mu+\gamma_1)/\lambda_1 > 1$ (the latter exceeding 1 because $\lambda_1 < \mu < \mu + \gamma_1$ under stability). For later use, we record $\frac{d}{dy}x^*(1)$. The cleanest route is implicit differentiation of $f(x,y) := \lambda_1 x^2 - A(y)x + (\mu + \gamma_1 y) = 0$ at the point $(x,y) = (1,1)$, where $f_y = \lambda_2 x + \gamma_1$ and $f_x = 2\lambda_1 x - A(y)$ evaluate to $f_y(1,1) = \lambda_2 + \gamma_1$ and $f_x(1,1) = \lambda_1 - \mu - \gamma_1$, so $\frac{d}{dy}x^*(1) = -f_y/f_x$:

$$\frac{d}{dy}x^*(1) = \frac{\lambda_2 + \gamma_1}{\mu + \gamma_1 - \lambda_1}. \tag{293}$$

Unlike Model-A's (114), the numerator carries an extra γ_1: the constant term $\mu + \gamma_1 y$ of the kernel (289) is itself y-dependent here, whereas in Model-A (and in Model-C_2^1 below) the constant term is independent of y.

Setting $x = x^*(y)$ in (287) and equating the right-hand side to zero,

$$\big[\mu(x^*(y) - y) + \gamma_1 y(x^*(y) - y)\big]P_y(y) - \mu x^*(y)(1-y)\,\pi(0,0) = 0,$$

that is, $(x^*(y) - y)(\mu + \gamma_1 y)P_y(y) = \mu x^*(y)(1-y)\pi(0,0)$, whence the boundary function is

$$P_y(y) = \frac{\mu\, x^*(y)\,(1-y)\,\pi(0,0)}{(\mu + \gamma_1 y)\,\big(x^*(y) - y\big)}. \tag{294}$$

As a consistency check, set $y = 0$ in (294): the factor $\mu + \gamma_1 y$ becomes μ and $1-y$ becomes 1, so $P_y(0) = \mu\, x^*(0)\,\pi(0,0)/\big(\mu\, x^*(0)\big) = \pi(0,0)$ (using $x^*(0) > 0$, the smaller of the two positive roots), recovering the boundary value $P_y(0) = \pi(0,0)$.

Finally, substitute (294) and $\pi(0,0) = \rho(1-\rho)$ (Corollary 6) back into (287). Writing the kernel as $-\lambda_1 y\,(x - x^*(y))(x - x^+(y))$ and simplifying the right-hand side,

$$\big[\mu(x-y) + \gamma_1 y(x-y)\big]P_y(y) - \mu x(1-y)\,\pi(0,0) = \mu(1-y)\,\pi(0,0)\left[\frac{(x-y)\,x^*(y)}{x^*(y) - y} - x\right]$$

$$= \mu(1-y)\,\pi(0,0)\,\frac{y\,\big(x - x^*(y)\big)}{x^*(y) - y},$$

where the bracket collapses because $(x-y)x^* - x(x^*-y) = y(x-x^*)$. Dividing by the factored kernel, the common factor $\big(x - x^*(y)\big)$ cancels and we obtain

$$P(x,y) = \frac{\mu(1-y)\,\pi(0,0)\,y\,(x - x^*(y))/(x^*(y)-y)}{-\lambda_1 y\,(x - x^*(y))(x - x^+(y))} = \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,\big(x^*(y)-y\big)\,\big(x^+(y)-x\big)},$$

which is (288) and concludes the proof. □

10.3.2 Probabilistic approach for Model-B_2^1

For Model-A, the boundary function $P_y(y)$ admits an independent derivation by the *Pollaczek–Khinchine* (PK) formula (§6.1.2): conditional on $N_1 = 0$, class 2 experiences an $M/G/1$ queue whose service times are i.i.d. copies of a class-1 busy period, and the PK formula then delivers $P_y(y)$. *This route does not transfer to Model-B_2^1.* Head-of-line jockeying injects class-1 customers into Queue 2 at the instants when $N_1 \geq 1$; the resulting arrival stream into Queue 2 is the superposition of the exogenous Poisson stream λ_2 and a state-dependent jockeying stream. This merged stream is *neither* Poisson *nor* independent of the system state — it is correlated with N_1. Since the PK formula requires PASTA, which in turn demands both a Poisson arrival process *and* the lack-of-anticipation property, and the merged input violates both, class 2 does not see a renewal $M/G/1$ input and no PK representation of $P_y(y)$ is available. The analytical derivation of §10.3.1 is consequently the only route, and we retain this subsection solely to record why the probabilistic one is unavailable — in the same spirit as the discussion of Models B and B_2.

10.4 Limits and sanity checks

As $\gamma_1 \to 0^+$ the kernel quadratic (289) reduces to Model-A's quadratic (113), so $x^*(y) \to x_A^*(y)$, and formula (288) reduces to the factored form of Theorem 1; the empty-state probabilities already equal those of Model-A by Corollary 6. Setting $x = y = 1$ in (288) through the diagonal identity (291) gives $P(1,1) = \rho = 1 - \pi_0$, confirming normalisation, and the diagonal $P(z,z) = \pi(0,0)/(1-\rho z)$ is exactly that of the combined $M/M/1(\lambda,\mu)$ queue — as it must be, since jockeying conserves the total customer count.

10.5 Mean queue lengths and mean waiting times

10.5.1 Class-1 marginal PGF and mean queue length

Setting $y = 1$ in the fundamental equation (287), the term $\mu x(1-y)\pi(0,0)$ vanishes and the coefficient of $P(x,1)$ factors as $-(x-1)\big(\lambda_1 x - (\mu+\gamma_1)\big)$, while the right-hand side becomes $(x-1)(\mu+\gamma_1)P_y(1)$. Cancelling $(x-1)$ for $x \neq 1$,

$$P(x,1) \;=\; \frac{(\mu+\gamma_1)\,P_y(1)}{\mu+\gamma_1-\lambda_1 x} \;=\; \frac{P_y(1)}{1-\rho_B\,x}, \qquad \rho_B := \frac{\lambda_1}{\mu+\gamma_1}. \tag{295}$$

This is a geometric class-1 marginal: the priority queue behaves as an $M/M/1$ queue whose effective *clearance rate* is $\mu+\gamma_1$, since a waiting class-1 customer at the head of Queue 1 departs that queue either by entering service (rate μ) or by jockeying (rate γ_1). The constant $P_y(1)$ is pinned by the normalisation $P(1,1) = 1-\pi_0 = \rho$, giving $P_y(1) = \rho(1-\rho_B)$, so that $P(x,1) = \rho(1-\rho_B)/(1-\rho_B x)$. Differentiating by the quotient rule and evaluating at $x=1$,

$$\mathbb{E}[N_1] \;=\; \frac{d}{dx}P(x,1)\Big|_{x=1} \;=\; \frac{\rho(1-\rho_B)\,\rho_B}{(1-\rho_B)^2} \;=\; \frac{\rho\,\rho_B}{1-\rho_B}. \tag{296}$$

This is Model-A's formula (144) with the priority load ρ_1 replaced by the *jockeying-reduced* load $\rho_B = \lambda_1/(\mu+\gamma_1) \leq \rho_1$.

10.5.2 Total queue length from the diagonal

Because jockeying conserves the total number of customers, the combined process N_1+N_2 (together with the in-service customer) is the queue-length process of an ordinary $M/M/1(\lambda,\mu)$ queue, and the diagonal generating function (291) is $P(z,z) = \rho(1-\rho)/(1-\rho z)$. Since $\frac{d}{dz}P(z,z)\big|_{z=1} = \mathbb{E}[N_1]+\mathbb{E}[N_2]$, differentiating gives

$$\mathbb{E}[N_1]+\mathbb{E}[N_2] \;=\; \frac{d}{dz}\,\frac{\rho(1-\rho)}{1-\rho z}\bigg|_{z=1} \;=\; \frac{\rho^2(1-\rho)}{(1-\rho)^2} \;=\; \frac{\rho^2}{1-\rho}, \tag{297}$$

identical to the combined $M/M/1$ result of Model-A (148).

10.5.3 Class-2 mean queue length by subtraction

Subtracting (296) from (297),

$$\begin{aligned}\mathbb{E}[N_2] \;&=\; \frac{\rho^2}{1-\rho} - \frac{\rho\,\rho_B}{1-\rho_B} \;=\; \frac{\rho\big[\rho(1-\rho_B)-\rho_B(1-\rho)\big]}{(1-\rho)(1-\rho_B)} \\ &=\; \frac{\rho\,(\rho-\rho_B)}{(1-\rho)(1-\rho_B)}.\end{aligned} \tag{298}$$

The excess load admits a transparent decomposition,

$$\rho-\rho_B = \frac{\lambda_1+\lambda_2}{\mu} - \frac{\lambda_1}{\mu+\gamma_1} = \frac{(\lambda_1+\lambda_2)(\mu+\gamma_1)-\lambda_1\mu}{\mu(\mu+\gamma_1)} = \frac{\lambda_2\mu+\lambda_1\gamma_1+\lambda_2\gamma_1}{\mu(\mu+\gamma_1)} = \rho_2 + \frac{\lambda_1\gamma_1}{\mu(\mu+\gamma_1)},$$

so $\rho-\rho_B$ exceeds the bare class-2 load ρ_2 precisely by the load that jockeying transfers from Queue 1 into Queue 2. At $\gamma_1 = 0$ it collapses to ρ_2 and (298) recovers Model-A's $\mathbb{E}[N_2] = \rho\rho_2/[(1-\rho_1)(1-\rho)]$.

10.5.4 Mean waiting times via Little's Law

We apply **Little's Law** to each queue separately, recalling that N_i counts only the customers *waiting*, so $\mathbb{E}[N_i] = \Lambda_i\,\mathbb{E}[W_i]$ with Λ_i the long-run rate at which customers *enter* Queue i and $\mathbb{E}[W_i]$ their mean residence time in that queue (from entry until they leave it, by service entry or by jockeying).

Every class-1 customer enters Queue 1 exactly once, by exogenous arrival, so $\Lambda_1 = \lambda_1$ and

$$\mathbb{E}[W_1] \;=\; \frac{\mathbb{E}[N_1]}{\lambda_1} \;=\; \frac{\rho\,\rho_B}{\lambda_1(1-\rho_B)} \;=\; \frac{\rho}{\mu+\gamma_1-\lambda_1}. \tag{299}$$

Compared with Model-A's $\mathbb{E}[W_1] = \rho/(\mu-\lambda_1)$ (149), the denominator is enlarged by γ_1: head-of-line jockeying drains the priority queue faster and shortens class-1 waits.

Queue 2 receives a *mixed* input: the exogenous Poisson stream at rate λ_2 together with the jockeying stream. The latter fires at rate γ_1 whenever $N_1 \ge 1$, so its long-run rate is $\gamma_1\,\mathbb{P}(N_1 \ge 1)$. The probability that a class-1 customer is waiting follows from the geometric marginal (295) by subtracting the $N_1 = 0$ mass from the busy mass,

$$\mathbb{P}(N_1 \ge 1) = P(1,1) - P(0,1) = \rho - \rho(1-\rho_B) = \rho\rho_B,$$

since $P(1,1) = 1-\pi_0 = \rho$ and $P(0,1) = P_y(1) = \rho(1-\rho_B)$. The total entrance rate into Queue 2 is therefore $\Lambda_2 = \lambda_2 + \gamma_1\rho\rho_B$, and Little's Law gives the mean residence time per Queue-2 entrant,

$$\mathbb{E}[W_2] \;=\; \frac{\mathbb{E}[N_2]}{\Lambda_2} \;=\; \frac{\rho\,(\rho-\rho_B)}{(1-\rho)(1-\rho_B)\,\big(\lambda_2+\gamma_1\rho\rho_B\big)}. \tag{300}$$

As $\gamma_1 \to 0^+$ the jockeying stream disappears, $\Lambda_2 \to \lambda_2$ and $\rho_B \to \rho_1$, and (300) recovers Model-A's $\mathbb{E}[W_2] = \rho/[\mu(1-\rho_1)(1-\rho)]$ (150).

11 Analysis of Model-C_2^1 ($\theta_1 > 0$ head-of-line, $\gamma_1 = \gamma_2 = \theta_2 = 0$)

Model-C_2^1 is the abandonment counterpart of Model-B_2^1, and a simplified variant of the class-1 abandonment model Model-C_2. In Model-C_2 *every* waiting class-1 customer is impatient and reneges at the linear rate $\theta_1 n_1$, turning the class-1 subsystem into an $M/M/1{+}M$ (Erlang-A) queue and the fundamental equation into a first-order ODE. Here only the *customer at the head of Queue 1* may abandon, at the fixed rate θ_1 whenever $n_1 \ge 1$; the aggregate abandonment rate is $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$ rather than $\theta_1 n_1$. As with Model-B_2^1, the flat rate contributes an *algebraic* term to the fundamental equation rather than a derivative, and the Kernel Method applies directly. Unlike jockeying, abandonment is a *true departure*: it removes a customer from the system and so breaks the conservation of the total customer count that Model-B_2^1 enjoyed. The empty-state probabilities and the class-2 mean queue length consequently require a separate argument.

Stability. Because only the head of Queue 1 abandons, the aggregate abandonment rate is bounded by θ_1 regardless of queue length, so — in contrast to the Erlang-A class-1 subsystem of Model-C_2, which is positive recurrent for all loads — a genuine stability condition is required. As shown in Corollary 7 below, the chain is positive recurrent if and only if $\pi_0 > 0$, that is

$$(\mu + \theta_1)(\mu - \lambda) + \lambda_1\theta_1 \;>\; 0, \qquad \lambda = \lambda_1 + \lambda_2. \tag{301}$$

This is *weaker* than $\rho < 1$: if $\rho < 1$ then $\mu - \lambda > 0$ and the condition holds trivially, but it can also hold with $\rho \geq 1$ provided θ_1 is large enough — head abandonment accelerates the clearance of the priority queue and can stabilise the system at offered loads exceeding the service capacity. Moreover (301) implies $\rho_C := \lambda_1/(\mu + \theta_1) < 1$, so the priority subqueue is itself stable whenever the condition holds.

11.1 Balance equations

The abandonment transition $(n_1, n_2) \to (n_1 - 1, n_2)$ occurs at the fixed rate θ_1 whenever $n_1 \geq 1$. Since both a service completion and a head abandonment from $(n_1 + 1, n_2)$ land in (n_1, n_2), the two mechanisms combine into the coefficient $\mu + \theta_1$ on the in-flows. On S,

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + \theta_1]\,\pi(n_1, n_2) \\
&\quad= (\mu + \theta_1)\pi(n_1 + 1, n_2) + \lambda_1\pi(n_1 - 1, n_2) + \lambda_2\pi(n_1, n_2 - 1) && n_1 \geq 1,\ n_2 \geq 1, && (302)\\
&[\lambda_1 + \lambda_2 + \mu + \theta_1]\,\pi(n_1, 0) = (\mu + \theta_1)\pi(n_1 + 1, 0) + \lambda_1\pi(n_1 - 1, 0) && n_1 \geq 1,\ n_2 = 0, && (303)\\
&[\lambda_1 + \lambda_2 + \mu]\,\pi(0, n_2) \\
&\quad= (\mu + \theta_1)\pi(1, n_2) + \lambda_2\pi(0, n_2 - 1) + \mu\,\pi(0, n_2 + 1) && n_1 = 0,\ n_2 \geq 1, && (304)\\
&[\lambda_1 + \lambda_2 + \mu]\,\pi(0, 0) = (\lambda_1 + \lambda_2)\pi_0 + (\mu + \theta_1)\pi(1, 0) + \mu\,\pi(0, 1), && && (305)\\
&(\lambda_1 + \lambda_2)\pi_0 = \mu\,\pi(0, 0). && && (306)
\end{aligned}$$

The contrast with Model-C_2 is the same as before: there the in-flow coefficients $\mu + \theta_1(n_1 + 1)$ and the out-flow coefficient $\theta_1 n_1$ grow with the queue length, whereas here they are the constants $\mu + \theta_1$ and θ_1 throughout.

11.2 Fundamental PGF equation

Lemma 4 (Fundamental equation for Model-C_2^1)**.** *Assume positive recurrence. Then the joint PGF $P(x, y)$ and the boundary function $P_y(y) = P(0, y)$ satisfy*

$$\begin{aligned}
&\big[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2 + \theta_1 y(x - 1)\big]\,P(x, y) \\
&\quad= \big[\mu(x - y) + \theta_1 y(x - 1)\big]\,P_y(y) - \mu x(1 - y)\,\pi(0, 0).
\end{aligned} \tag{307}$$

Proof. As in Lemma 3, the non-abandonment terms of (302)–(306) reproduce the Model-A fundamental equation (106) after multiplication by $x^{n_1}y^{n_2}$, summation, and multiplication by xy. The abandonment *out*-flow appears in every state with $n_1 \geq 1$,

$$\theta_1 \sum_{n_1 \geq 1}\sum_{n_2 \geq 0} \pi(n_1, n_2)\,x^{n_1}y^{n_2} = \theta_1\big[P(x, y) - P_y(y)\big],$$

and its *in*-flow enters every state (n_1, n_2) from $(n_1 + 1, n_2)$,

$$\theta_1 \sum_{n_1 \geq 0}\sum_{n_2 \geq 0} \pi(n_1 + 1, n_2)\,x^{n_1}y^{n_2} = \theta_1\,x^{-1} \sum_{m_1 \geq 1}\sum_{n_2 \geq 0} \pi(m_1, n_2)\,x^{m_1}y^{n_2} = \theta_1\,x^{-1}\big[P(x, y) - P_y(y)\big],$$

with $m_1 = n_1 + 1$. The net contribution is $\theta_1\big(1 - x^{-1}\big)\big[P(x, y) - P_y(y)\big]$; multiplying by xy gives $\theta_1 y(x - 1)\big[P(x, y) - P_y(y)\big]$. Adding to the Model-$A$ equation and rearranging yields (307). □

Remark. Again no derivative of P appears: the algebraic factor $\theta_1 y(x - 1)$ replaces the convective term $\theta_1 xy(x - 1)\,\partial P/\partial x$ of Model-C_2. Setting $\theta_1 = 0$ recovers (106). Unlike the jockeying factor of Model-B_2^1, the abandonment factor $\theta_1 y(x - 1)$ does *not* vanish on the diagonal $x = y$, which is the analytic signature of the broken conservation.

11.3 Closed-form solution

Theorem 6 (Closed-form PGF for Model-C_2^1)**.** *Consider the two-class non-preemptive priority queue with Poisson arrival rates λ_1, λ_2, common exponential service rate μ, and head-of-line class-1 abandonment at rate $\theta_1 > 0$ (with $\gamma_1 = \gamma_2 = \theta_2 = 0$). Under the stability condition (301), the joint PGF $P(x,y) = \mathbb{E}[x^{N_1} y^{N_2}]$ is the rational function*

$$P(x,y) \;=\; \frac{\mu(\mu+\theta_1)\,(1-y)\,\pi(0,0)}{\lambda_1\left(x^+(y)-x\right)\left[(\mu+\theta_1 y)\,x^*(y) - y(\mu+\theta_1)\right]}, \tag{308}$$

where $x^*(y)$ and $x^+(y)$ are the two roots of the kernel quadratic

$$\lambda_1 x^2 - \left[\lambda_1 + \lambda_2(1-y) + \mu + \theta_1\right]x + (\mu+\theta_1) = 0, \tag{309}$$

with $x^*(y)$ the root inside the unit disk, and $\pi(0,0)$ is given by Corollary 7. As $\theta_1 \to 0^+$ the formula recovers Model-A (Theorem 1) exactly.

Corollary 7. *Under (301), the empty-state probabilities of Model-C_2^1 are*

$$\pi_0 = \frac{(\mu+\theta_1)(\mu-\lambda) + \lambda_1\theta_1}{\mu(\mu+\theta_1) + \lambda_1\theta_1}, \qquad \pi(0,0) = \frac{\lambda\left[(\mu+\theta_1)(\mu-\lambda) + \lambda_1\theta_1\right]}{\mu\left[\mu(\mu+\theta_1) + \lambda_1\theta_1\right]}, \tag{310}$$

with $\lambda = \lambda_1 + \lambda_2$. At $\theta_1 \to 0^+$ these reduce to $\pi_0 \to 1-\rho$ and $\pi(0,0) \to \rho(1-\rho)$, recovering Model-A.

Proof of Corollary 7. Two evaluations of the fundamental equation (307) suffice. First set $y=1$: the term $\mu x(1-y)\pi(0,0)$ vanishes, the coefficient of $P(x,1)$ factors as $-(x-1)\big(\lambda_1 x - (\mu+\theta_1)\big)$, and the right-hand side becomes $(x-1)(\mu+\theta_1)P_y(1)$. Cancelling $(x-1)$,

$$P(x,1) \;=\; \frac{(\mu+\theta_1)\,P_y(1)}{\mu+\theta_1-\lambda_1 x} \;=\; \frac{P_y(1)}{1-\rho_C\,x}, \qquad \rho_C := \frac{\lambda_1}{\mu+\theta_1}. \tag{311}$$

This geometric marginal shows that the priority queue clears at the effective rate $\mu+\theta_1$ (service completions plus head abandonments). Evaluating at $x=1$ and using $P(1,1) = 1-\pi_0$ gives

$$1-\pi_0 \;=\; \frac{P_y(1)}{1-\rho_C} \quad\Longrightarrow\quad P_y(1) \;=\; (1-\pi_0)(1-\rho_C). \tag{312}$$

Second, set $x=1$ in (307). The coefficient of $P(1,y)$ collapses to $\lambda_2 y(1-y)$ and the right-hand side to $\mu(1-y)\left[P_y(y) - \pi(0,0)\right]$, yielding the class-2 marginal relation

$$\lambda_2\, y\, P(1,y) \;=\; \mu\left[P_y(y) - \pi(0,0)\right]. \tag{313}$$

Letting $y \to 1$ and using (312) and the idle balance (306), $\pi(0,0) = \lambda\pi_0/\mu$,

$$\lambda_2(1-\pi_0) \;=\; \mu\left[P_y(1) - \pi(0,0)\right] \;=\; \mu(1-\pi_0)(1-\rho_C) - \lambda\pi_0.$$

This is a linear equation in π_0. Expanding and collecting the π_0 terms on the left,

$$\left[\mu(1-\rho_C) + \lambda - \lambda_2\right]\pi_0 = \mu(1-\rho_C) - \lambda_2, \qquad \text{i.e.} \qquad \pi_0 = \frac{\mu(1-\rho_C) - \lambda_2}{\mu(1-\rho_C) + \lambda_1},$$

where we used $\lambda - \lambda_2 = \lambda_1$. Substituting $\mu(1-\rho_C) = \mu(\mu+\theta_1-\lambda_1)/(\mu+\theta_1)$ and multiplying numerator and denominator by $\mu+\theta_1$, the numerator becomes $\mu(\mu+\theta_1-\lambda_1) - \lambda_2(\mu+\theta_1) = (\mu+\theta_1)(\mu-\lambda) + \lambda_1\theta_1$ and the denominator $\mu(\mu+\theta_1-\lambda_1) + \lambda_1(\mu+\theta_1) = \mu(\mu+\theta_1) + \lambda_1\theta_1$, yielding the first identity in (310); the second follows from $\pi(0,0) = \lambda\pi_0/\mu$. The denominator $\mu(\mu+\theta_1) + \lambda_1\theta_1$ is strictly positive, so $\pi_0 > 0$ is equivalent to the stability condition (301). □

11.3.1 Analytical approach for Model-C_2^1

Proof of Theorem 6. We again apply the Kernel Method to (307). Dividing the coefficient of $P(x,y)$ by $y \neq 0$ and rearranging gives the kernel quadratic (309),

$$\lambda_1 x^2 - \left[\lambda_1 + \lambda_2(1-y) + \mu + \theta_1\right]x + (\mu+\theta_1) = 0,$$

which is the Model-A kernel quadratic with μ replaced by $\mu+\theta_1$ *both* in the linear coefficient and in the constant term. Its roots are

$$x^{\pm}(y)=\frac{A(y)\pm\sqrt{A(y)^2-4\lambda_1(\mu+\theta_1)}}{2\lambda_1},\qquad A(y):=\lambda_1+\lambda_2(1-y)+\mu+\theta_1, \tag{314}$$

with product $x^*(y)\,x^+(y)=(\mu+\theta_1)/\lambda_1$ by Vieta's formulas. Evaluating the quadratic $f(x):=\lambda_1x^2-A(y)x+(\mu+\theta_1)$ at $x=1$,

$$f(1)=\lambda_1-A(y)+(\mu+\theta_1)=-\lambda_2(1-y)<0\qquad\text{for } y\in[0,1),$$

so $x=1$ lies between the two roots and $x^*(y)<1<x^+(y)$ for $y\in[0,1)$; at $y=1$, $x^*(1)=1$ and $x^+(1)=(\mu+\theta_1)/\lambda_1>1$ (the latter by $\rho_C<1$). Hence $x^*(y)$ is the unique root inside the unit disk. Differentiating (314) at $y=1$,

$$\frac{d}{dy}x^*(1)=\frac{\lambda_2}{\mu+\theta_1-\lambda_1}, \tag{315}$$

the analogue of (114) and (293).

Setting $x=x^*(y)$ in (307) forces the right-hand side to vanish,

$$\big[\mu(x^*(y)-y)+\theta_1y(x^*(y)-1)\big]P_y(y)=\mu x^*(y)(1-y)\,\pi(0,0),$$

and, regrouping the bracket as $\mu(x^*-y)+\theta_1y(x^*-1)=(\mu+\theta_1y)x^*-y(\mu+\theta_1)$, the boundary function is

$$P_y(y)=\frac{\mu\,x^*(y)\,(1-y)\,\pi(0,0)}{(\mu+\theta_1y)\,x^*(y)-y(\mu+\theta_1)}. \tag{316}$$

At $y=0$ the denominator is $\Delta(0)=\mu\,x^*(0)$, so $P_y(0)=\mu x^*(0)\pi(0,0)/(\mu\,x^*(0))=\pi(0,0)$ (using $x^*(0)>0$, the smaller positive root), as required.

Substituting (316) into (307), writing the kernel as $-\lambda_1y\,(x-x^*(y))(x-x^+(y))$, and abbreviating the denominator of (316) by $\Delta(y):=(\mu+\theta_1y)x^*(y)-y(\mu+\theta_1)$, the right-hand side becomes

$$\begin{aligned}\big[\mu(x-y)+\theta_1y(x-1)\big]P_y(y)-\mu x(1-y)\pi(0,0)&=\frac{\mu(1-y)\pi(0,0)}{\Delta(y)}\Big\{\big[\mu(x-y)+\theta_1y(x-1)\big]x^*(y)-x\,\Delta(y)\Big\}\\&=\frac{\mu(1-y)\pi(0,0)}{\Delta(y)}\,(\mu+\theta_1)\,y\,\big(x-x^*(y)\big),\end{aligned}$$

where, after substituting $\Delta(y)=(\mu+\theta_1y)x^*(y)-y(\mu+\theta_1)$, the curly bracket is *linear* in $x^*(y)$ and collapses term-by-term to $(\mu+\theta_1)y\big(x-x^*(y)\big)$ (no use of the kernel relation is required). Dividing by the factored kernel, the factor $\big(x-x^*(y)\big)$ cancels and

$$P(x,y)=\frac{\mu(\mu+\theta_1)(1-y)\,\pi(0,0)\,y\,(x-x^*(y))/\Delta(y)}{-\lambda_1y\,(x-x^*(y))(x-x^+(y))}=\frac{\mu(\mu+\theta_1)(1-y)\,\pi(0,0)}{\lambda_1\,(x^+(y)-x)\,\Delta(y)},$$

which is (308) and concludes the proof. □

11.3.2 Probabilistic approach for Model-C_2^1

As for Model-B_2^1, the Pollaczek–Khinchine route used for Models A and C_2 does not transfer. In Model-C_2 the class-1 subsystem seen by class 2 is a clean $M/M/1{+}M$ (Erlang-A) queue whose busy period B_C is an i.i.d. renewal quantity, and class 2 sees an $M/G/1$ queue with service B_C. Under head-only abandonment this clean renewal structure is lost: abandonment acts at the fixed rate θ_1 only while a class-1 customer waits ($N_1\geq 1$) but is inactive when the single class-1 present is in service, so the departure rate of the class-1 subsystem is state-dependent at its boundary and the class-1 busy period is not the busy period of any single $M/M/1$ or $M/M/1{+}M$ queue. Consequently class 2 does not see a renewal $M/G/1$ input and no Pollaczek–Khinchine representation of $P_y(y)$ is available. We retain this subsection, in parallel with the other models, only to record why the analytical route of §11.3.1 is the operative one.

11.4 Limits and sanity checks

As $\theta_1\to 0^+$ the kernel quadratic (309) reduces to Model-A's, so $x^*(y)\to x_A^*(y)$ and $\Delta(y)\to\mu(x_A^*(y)-y)$; together with $\pi(0,0)\to\rho(1-\rho)$ from Corollary 7, formula (308) reduces to the factored Model-A PGF of Theorem 1. Setting $x=1$ in (308) and $y\to1$ reproduces $P(1,1)=1-\pi_0$ via (312), confirming normalisation.

11.5 Mean queue lengths and mean waiting times

11.5.1 Class-1 marginal PGF and mean queue length

The class-1 marginal is the geometric law (311); pinning the constant by $P(1,1) = 1 - \pi_0$ as in (312) gives $P(x,1) = (1-\pi_0)(1-\rho_C)/(1-\rho_C x)$. Differentiating and evaluating at $x = 1$,

$$\mathbb{E}[N_1] \;=\; \frac{d}{dx}P(x,1)\bigg|_{x=1} \;=\; \frac{(1-\pi_0)\,\rho_C}{1-\rho_C}, \qquad \rho_C = \frac{\lambda_1}{\mu+\theta_1}, \tag{317}$$

with $1-\pi_0 = \lambda(\mu+\theta_1)/\big[\mu(\mu+\theta_1)+\lambda_1\theta_1\big]$ from Corollary 7. As in Model-B_2^1, the priority queue behaves as an $M/M/1$ queue with effective clearance rate $\mu+\theta_1$, the only difference from Model-B_2^1's (296) being the prefactor $1-\pi_0$, which here is no longer equal to ρ because abandonment alters the idle probability.

11.5.2 Class-2 mean queue length

Abandonment is a true departure and so breaks the *conservation of the total customer count* that underlies the diagonal argument: in Model-B_2^1 the jockeying factor $\gamma_1 y(x-y)$ vanishes on the diagonal $x = y$, whereas here the abandonment factor $\theta_1 y(x-1)$ does not. Concretely, setting $x = y = z$ in (307) leaves the term $\theta_1 z(z-1)P_y(z)$ on the right-hand side, so the diagonal equation no longer closes and the route that furnished $\mathbb{E}[N_1]+\mathbb{E}[N_2]$ for Model-B_2^1 is unavailable. We instead differentiate the class-2 marginal. From the marginal relation (313), $P(1,y) = \mu\big[P_y(y) - \pi(0,0)\big]/(\lambda_2 y)$, and since $\mathbb{E}[N_2] = \frac{d}{dy}P(1,y)\big|_{y=1}$,

$$\mathbb{E}[N_2] \;=\; \frac{\mu}{\lambda_2}\,P_y'(1) - (1-\pi_0), \tag{318}$$

where we used $\frac{\mu}{\lambda_2}\big[P_y(1) - \pi(0,0)\big] = P(1,1) = 1-\pi_0$. It remains to compute $P_y'(1)$ from (316). Both the numerator $\mu x^*(y)(1-y)\pi(0,0)$ and the denominator $\Delta(y) = (\mu+\theta_1 y)x^*(y) - y(\mu+\theta_1)$ vanish at $y = 1$ (since $x^*(1) = 1$ gives $\Delta(1) = (\mu+\theta_1)-(\mu+\theta_1) = 0$), so $P_y(y)$ is a $0/0$ form there. Writing $N(y)$ and $\Delta(y)$ for numerator and denominator, a second-order Taylor expansion about $y = 1$ gives the value and the slope

$$P_y(1) = \frac{N'(1)}{\Delta'(1)} = (1-\pi_0)(1-\rho_C), \qquad P_y'(1) = \frac{N''(1)\,\Delta'(1) - N'(1)\,\Delta''(1)}{2\,\Delta'(1)^2},$$

which is why both the first and the second root sensitivities of x^* at $y = 1$ are needed, namely (315) together with

$$\frac{d^2}{dy^2}x^*(1) = \frac{2(\mu+\theta_1)\,\lambda_2^2}{(\mu+\theta_1-\lambda_1)^3}$$

(the latter obtained by differentiating $x^*(y)$ in (314) twice). Substituting these into $P_y'(1)$, then into (318), and simplifying yields the closed form

$$\mathbb{E}[N_2] \;=\; \frac{\lambda_2\,\lambda\,(\mu+\theta_1)\big[(\mu+\theta_1)^2 - \lambda_1\theta_1\big]}{(\mu+\theta_1-\lambda_1)\,\big[\mu(\mu+\theta_1)+\lambda_1\theta_1\big]\,\big[(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1\big]}, \tag{319}$$

with $\lambda = \lambda_1+\lambda_2$. The last bracket in the denominator is exactly the stability numerator of π_0 from (310), so $\mathbb{E}[N_2]$ is finite precisely under the stability condition (301) and diverges as $\pi_0 \to 0^+$. As $\theta_1 \to 0^+$ the right-hand side reduces to $\lambda_2\lambda/[(\mu-\lambda_1)(\mu-\lambda)] = \rho\rho_2/[(1-\rho_1)(1-\rho)]$, recovering Model-$A$'s (148).

11.5.3 Mean waiting times via Little's Law

We apply **Little's Law** to each queue. Every class-1 customer enters Queue 1 once by exogenous arrival and later leaves it either by entering service or by abandoning from the head; in both cases it leaves Queue 1, so the entrance rate is $\Lambda_1 = \lambda_1$ and

$$\mathbb{E}[W_1] \;=\; \frac{\mathbb{E}[N_1]}{\lambda_1} \;=\; \frac{(1-\pi_0)\,\rho_C}{\lambda_1(1-\rho_C)}. \tag{320}$$

Here $\mathbb{E}[W_1]$ is the mean time a class-1 customer spends *waiting* in Queue 1, averaged over all class-1 arrivals, whether they are eventually served or abandon. Queue 2, by contrast, receives no jockeying input and its customers never abandon ($\theta_2 = 0$), so every class-2 arrival is eventually served and the entrance rate is simply $\Lambda_2 = \lambda_2$:

$$\mathbb{E}[W_2] = \frac{\mathbb{E}[N_2]}{\lambda_2} = \frac{\lambda\,(\mu+\theta_1)\big[(\mu+\theta_1)^2 - \lambda_1\theta_1\big]}{(\mu+\theta_1-\lambda_1)\,\big[\mu(\mu+\theta_1)+\lambda_1\theta_1\big]\,\big[(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1\big]}. \tag{321}$$

Both reduce to the corresponding Model-A expressions as $\theta_1 \to 0^+$.

11.6 Comparison of the head-of-line models

Table 4 summarises the structural differences between the two head-of-line variants and their full-rate counterparts. The common thread is that replacing a length-proportional mechanism rate ($\gamma_1 n_1$ or $\theta_1 n_1$) by a head-of-line rate ($\gamma_1\,\mathbf{1}_{\{n_1\geq 1\}}$ or $\theta_1\,\mathbf{1}_{\{n_1\geq 1\}}$) converts the governing ODE into an algebraic equation, so that $P(x,y)$ becomes rational. For Model-B_2^1 the result is structurally identical to Model-A, the sole change being $\mu \to \mu + \gamma_1 y$ in the kernel; for Model-C_2^1 the broken conservation introduces the extra factor $(\mu+\theta_1)$ and the modified boundary denominator $\Delta(y)$, and shifts the stability boundary to (301).

Table 4: Structural comparison across the priority-queue models.

Property	**A**	$\mathbf{B}_2$	$\mathbf{B}_2^1$	$\mathbf{C}_2$	$\mathbf{C}_2^1$
Mechanism rate	—	$\gamma_1 n_1$	$\gamma_1\,\mathbf{1}_{\{n_1\geq 1\}}$	$\theta_1 n_1$	$\theta_1\,\mathbf{1}_{\{n_1\geq 1\}}$
PGF equation type	algebraic	ODE	**algebraic**	ODE	**algebraic**
$P(x,y)$ form	rational	integral+${}_1F_1$	**rational**	integral+${}_1F_1$	**rational**
Same form as Model-A?	✓	×	✓	×	(similar)
$\pi(0,0)$	$\rho(1-\rho)$	$\rho(1-\rho)$	$\rho(1-\rho)$	(231)	(310)
π_0	$1-\rho$	$1-\rho$	$1-\rho$	(232)	(310)
Conserves total N?	✓	✓	✓	×	×
Stability condition	$\rho<1$	$\rho<1$	$\rho<1$	$\lambda_2\mathbb{E}[B_C]<1$	(301)

12 Comparison of Models A, B_2, and C_2

The three solved models represent the three tractable specializations of Model-X: a pure baseline (A), a jockeying extension (B_2), and an abandonment extension (C_2). Each is obtained from Model-X by zeroing all but one of the extra mechanisms, and each produces a closed-form or integral-form expression for $P(x,y)$. Table 5 collects the key structural facts; the remainder of this section traces the consequences.

Table 5: Structural comparison of Models A, B_2, and C_2.

	Model-A	**Model-B_2**	**Model-C_2**
Extra mechanism	none	jockeying $1 \to 2$ ($\gamma_1 > 0$)	class-1 abandonment ($\theta_1 > 0$)
Stability condition	$\rho < 1$	$\rho < 1$	$\rho_2 \cdot {}_1F_1 < 1$
Fundamental equation	algebraic PGF equation	1st-order PDE in x, y	1st-order ODE in x
Pinning of $P_y(y)$	zero of kernel: $x = x^*(y)$	analyticity at $x = y$	boundedness at $x = 1$
Probabilistic route	yes (P-K with B)	no (jockeying breaks Poisson)	yes (P-K with B_C)
$P_y(y)$	rational in $x^*(y)$	Kummer ratio (214)	P-K / Kummer (240)
$P(x, y)$	closed form (106)	integral form (187)	integral form (264)
$\pi(0, 0)$	$\rho(1 - \rho)$	$\rho(1 - \rho)$	$(\rho_1 + \rho_2)\pi_0$ via $\mathbb{E}[B_C]$
π_0	$1 - \rho$	$1 - \rho$	$> 1 - \rho$ (abandonment increases idle time)
$P(z, z)$	$\rho(1 - \rho)/(1 - \rho z)$	$\rho(1 - \rho)/(1 - \rho z)$	no closed form
$\mathbb{E}[N_1] + \mathbb{E}[N_2]$	$\rho^2/(1 - \rho)$	$\rho^2/(1 - \rho)$	$< \rho^2/(1 - \rho)$
$\mathbb{E}[N_1]$ closed form	$\rho\rho_1/(1 - \rho_1)$	no	no

Model-A: the rational baseline

Model-A imposes no extra departure mechanism. Class-1 operates as an independent $M/M/1(\lambda_1, \mu)$ queue; class-2 sees an $M/G/1$ with service time equal to a class-1 busy period (by PASTA and the priority decomposition of §3.2). The Pollaczek–Khinchine formula (9) delivers $P_y(y)$ as a rational function of $x^*(y)$, and $P(x, y)$ is a ratio of polynomials in x, y, and $x^*(y)$. All performance measures — π_0, $\pi(0, 0)$, $\mathbb{E}[N_i]$, $\mathbb{E}[W_i]$ — are in closed form. Model-A also furnishes the structural identities that persist across the hierarchy: $P(z, z) = \rho(1 - \rho)/(1 - \rho z)$ (total queue is $M/M/1$), $\pi_0 = 1 - \rho$, and $\pi(0, 0) = \rho(1 - \rho)$.

Effect of jockeying: Model-A $\to$ Model-B_2

Adding one-way jockeying ($\gamma_1 > 0$, $\gamma_2 = 0$) couples the two queues: a class-1 customer waiting in Queue 1 may defect to Queue 2 at rate γ_1. The consequences are asymmetric.

What is preserved. Jockeying transfers customers but does not remove them, so the total-customer process remains a standard $M/M/1(\lambda, \mu)$ birth–death chain. Consequently $P(z, z)$, π_0, $\pi(0, 0)$, and $\mathbb{E}[N]$ are identical to Model-A for all $\gamma_1 > 0$.

What is destroyed. Jockeying breaks the per-class Poisson/renewal structure: a class-2 customer can no longer identify its effective service time with a class-1 busy period, because class-1 customers now arrive at Queue 2 from inside the system as well as from outside. The P-K route is therefore unavailable.

What is new. The fundamental PGF equation gains the convection term $\gamma_1 xy(x-y)\partial_x P$, which reduces it to a first-order ODE in x for each fixed y (since $\gamma_2 = 0$ kills the ∂_y term). The solution closes via an integrating factor with a branch singularity at the *interior* point $x = y$; the branch coefficient must vanish for P to be holomorphic, and this analyticity condition pins $P_y(y)$ as a Kummer ratio. The individual means $\mathbb{E}[N_1]$ and $\mathbb{E}[N_2]$ change with γ_1 (class-1 becomes shorter, class-2 longer), with no closed form for the split.

Effect of abandonment: Model-A $\to$ Model-C_2

Adding class-1 abandonment ($\theta_1 > 0$, $\theta_2 = 0$, $\gamma_i = 0$) allows each waiting class-1 customer to leave the system at rate θ_1.

What is preserved. Class-2 customers still see Poisson arrivals (abandonment only affects class-1; the class-2 arrival process is undisturbed). PASTA therefore still applies, and the $M/G/1$ subsystem for class-2 is intact — now with effective service time B_C (the busy period of the $M/M/1+M$ queue of §3.3) replacing B. The P-K formula (9) applies with B_C and delivers $P_y(y)$.

What is destroyed. Abandonment is a true departure: it reduces the total number of customers and breaks the conservation property. Consequently $P(z,z)$ no longer has the simple $M/M/1$ form (146), $\pi_0 > 1-\rho$, $\pi(0,0)$ requires $\mathbb{E}[B_C]$, and $\mathbb{E}[N] < \rho^2/(1-\rho)$. The stability condition is also generalized: $\rho < 1$ is no longer necessary; the relevant condition is $\lambda_2\,\mathbb{E}[B_C] < 1$, which is strictly weaker for all $\theta_1 > 0$.

What is new. The fundamental equation gains the convection term $\theta_1 xy(x-1)\partial_x P$, which already reduces it to a first-order ODE in x for each fixed y (with $\theta_2 = 0$ killing the ∂_y term). The integrating factor now has its singularity at the *boundary* $x = 1$ with a positive exponent, so μ_I vanishes there and mere boundedness of P forces $\int_0^1 q\,\mu_I\,dt = 0$. This determines $P_y(y)$ without any reference to holomorphy. Both $\mathbb{E}[N_1]$ and $\mathbb{E}[N_2]$ decrease relative to Model-A as θ_1 increases: class-2 benefits indirectly because class-1 abandonments free the server sooner.

Structural comparison of Models B_2 and C_2

Both models reduce the fundamental equation to a first-order linear ODE in x and both close in confluent hypergeometric functions; yet the mechanism that pins $P_y(y)$ is structurally different. The contrast is as follows.

	Model-C_2 $(\theta_1 > 0)$	**Model-B_2** $(\gamma_1 > 0)$
Convection coefficient	$\theta_1\,x(x-1)$	$\gamma_1\,x(x-y)$
Operative singular point	boundary $x = 1$	interior $x = y$
Exponent at singularity	$\beta(y) = \lambda_2(1-y)/\theta_1 > 0$ $(\mu_I \to 0)$	$\beta(y) = (1-y)(\lambda y-\mu)/(\gamma_1 y) < 0$ $(\mu_I \to \infty)$
Pinning principle	*boundedness*: $\int_0^1 q\,\mu_I\,dt = 0$	*analyticity*: branch coefficient $= 0$
$\pi(0,0)$	$M/G/1$+renewal (conservation broken)	$\rho(1-\rho)$ exactly (conservation holds)
Probabilistic backbone	class-2 is $M/G/1$; P-K with B_C	class-1 is $M/M/1{+}M$ (reneging $= \gamma_1$); no P-K

In words: in Model-C_2 the abandonment convection $\theta_1 x(x-1)$ places the singularity at the boundary $x = 1$ with a positive integrating-factor exponent, so μ_I vanishes there and mere boundedness forces the vanishing integral that determines $P_y(y)$. The Poisson character of the class-2 stream survives abandonment, so P-K delivers $P_y(y)$ probabilistically and an $M/G/1$ renewal argument delivers $\pi(0,0)$. In Model-B_2, the jockeying convection $\gamma_1 x(x-y)$ places the singularity at the interior point $x = y$ with a negative exponent, so μ_I blows up, the homogeneous mode vanishes, and only the stronger requirement of holomorphy — the absence of the non-integer branch $(x-y)^{|\beta|}$ — pins $P_y(y)$. Jockeying destroys the per-class Poisson/renewal structure, so no P-K identity is available; what remains probabilistic is the total-customer $M/M/1$ behaviour and the $M/M/1{+}M$ reading of the class-1 marginal, which is the origin of the ${}_1F_1$ factors.

Common Erlang-A structure

A common object appears in both B_2 and C_2:

$$ {}_1F_1\!\left(1;\ \frac{\mu}{r}+1;\ \frac{\lambda_1}{r}\right), \tag{322} $$

with $r = \gamma_1$ (Model-B_2) or $r = \theta_1$ (Model-C_2). This is the Erlang-A utilisation factor: in C_2 it equals $\mu\,\mathbb{E}[B_C]$ (the mean class-1 busy period scaled by μ), while in B_2 it controls the Kummer ratio in $P_y(y)$ through the exponent $\alpha(y) = \mu/(\gamma_1 y)$. In both cases the function evaluates at a negative argument $(-\lambda_1/r)$ in C_2 and $(-\lambda_1 y/\gamma_1)$ in B_2, and the ratio ${}_1F_1(\alpha+1;\cdot;\cdot)/{}_1F_1(\alpha;\cdot;\cdot)$ acts as a boundary correction to the Model-A rational formula.

Model-B_2 degenerates to Model-A only in the singular limit $\gamma_1 \to 0^+$, in which $\alpha, \beta \to \infty$ and the Kummer ratio collapses to the rational expression $x^*(y)(1-y)/(x^*(y)-y)$ of Model-A; this limit is not uniform, consistent with γ_1 multiplying the highest-order term of the ODE. Model-C_2 degenerates to Model-A uniformly as $\theta_1 \to 0^+$: $\mathbb{E}[B_C] \to (\mu - \lambda_1)^{-1}$, and all C_2 formulas recover their Model-A counterparts.

12.1 Traffic-intensity sweep

The structural comparison above is now made quantitative. Holding the service rate at $\mu = 1$ and the arrival split fixed at $\lambda_1{:}\lambda_2 = 3{:}4$, the offered load $\rho = \lambda_1 + \lambda_2$ is swept across $[0.10, 0.95]$ and the stationary performance of the three models is solved exactly on the truncated CTMC. Figure 8 collects the mean queue length $\mathbb{E}[N]$, the priority-class queue $\mathbb{E}[N_1]$, the priority waiting time $\mathbb{E}[W_1] = \mathbb{E}[N_1]/\lambda_1$ (Little's Law), and the idle probability π_0; Figure 9 isolates $\pi(0,0)$ against the Model-A parabola $\rho(1-\rho)$.

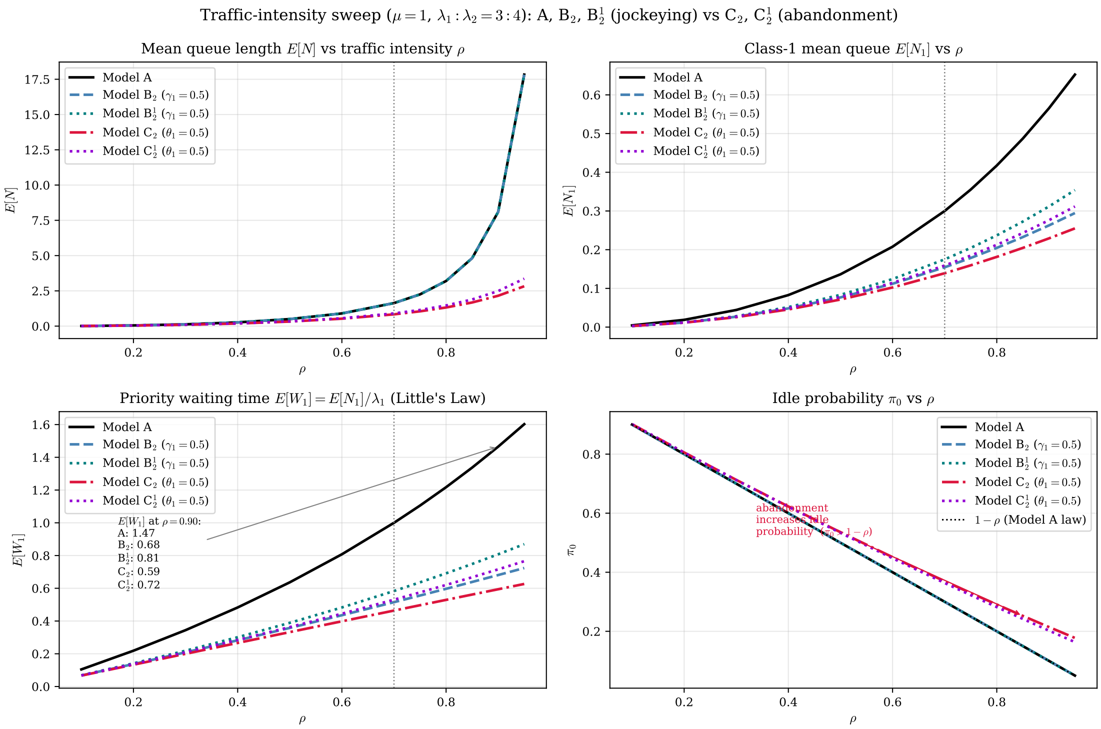

Figure 8: Performance versus traffic intensity ρ for the five solved models — the baseline A, the jockeying pair B_2, B_2^1 ($\gamma_1 = 0.5$), and the abandonment pair C_2, C_2^1 ($\theta_1 = 0.5$) — at $\mu = 1$ and $\lambda_1{:}\lambda_2 = 3{:}4$; the vertical line marks the canonical $\rho = 0.70$. *Top-left:* total queue $\mathbb{E}[N]$ — A, B_2 and B_2^1 coincide because jockeying conserves N (Corollaries 2, 6), while C_2 and C_2^1 lie strictly below because abandonment removes customers (C_2^1 by less). *Top-right:* priority queue $\mathbb{E}[N_1]$ — the head-of-line variants drain it less than their full-rate cousins. *Bottom-left:* priority waiting time $\mathbb{E}[W_1]$. *Bottom-right:* idle probability π_0, with the $1-\rho$ law (Models A, B_2, B_2^1) shown dotted; C_2 and C_2^1 sit above it for every ρ, as predicted by Corollaries 5, 7.

Two mechanisms produce two qualitatively different responses. Jockeying is internal: the $1 \to 2$ transition relabels a waiting customer without changing the total count, so the B_2 and A curves for $\mathbb{E}[N]$ and π_0 are indistinguishable across the entire sweep — the server sees the same workload. Abandonment is a true departure at rate $\theta_1 n_1$, so C_2 carries a strictly lighter queue and a strictly larger idle fraction: the priority backlog is bled off before it can be served. At $\rho = 0.70$ (Table 6) this gives $\pi_0 = 0.3696$ for C_2 against 0.3000 for A and B_2, and $\mathbb{E}[N] = 0.8358$ against 1.6333.

The divergence widens with load. At $\rho = 0.90$ the baseline class-2 queue reaches $\mathbb{E}[N_2] = 7.5346$, whereas C_2 holds it to 1.9245 — a factor of nearly four — because the self-regulating valve $\theta_1 n_1$ acts most strongly precisely when the queue is long. The priority queue $\mathbb{E}[N_1]$ is bounded for C_2 for the same reason and never exceeds 0.2292 over the sweep, while the A value grows monotonically toward the $\rho \to 1$ singularity. We note for completeness that near criticality the dense CTMC truncation leaves a boundary tail mass of order 6×10^{-4} for A and B_2 at $\rho = 0.95$ (against 1.5×10^{-7} for C_2, whose tail is light); the $\rho \geq 0.92$ points for the non-abandoning models therefore carry a small downward truncation bias and should be read as lower bounds.

Table 6: Comparison of Models A, B_2 and B_2^1 ($\gamma_1 = 0.5$), and C_2 and C_2^1 ($\theta_1 = 0.5$) at three traffic intensities with $\lambda_1{:}\lambda_2 = 3{:}4$, $\mu = 1$. The primed models use the head-of-line rate $\mathbf{1}_{\{n_1 \geq 1\}}$; the unprimed use the full rate n_1.

ρ	Model	$E[N_1]$	$E[N_2]$	$E[N]$	$E[W_1]$	$E[W_2]$	π_0	$\pi(0,0)$
	A	0.1364	0.3636	0.5000	0.6364	1.2727	0.5000	0.2500
	B_2	0.0767	0.4233	0.5000	0.3578	1.4817	0.5000	0.2500
0.50	B_2^1	0.0833	0.4167	0.5000	0.3889	1.4583	0.5000	0.2500
	C_2	0.0712	0.2521	0.3233	0.3323	0.8823	0.5356	0.2678
	C_2^1	0.0778	0.2593	0.3370	0.3630	0.9074	0.5333	0.2667
	A	0.3000	1.3333	1.6333	1.0000	3.3333	0.3000	0.2100
	B_2	0.1545	1.4788	1.6333	0.5151	3.6970	0.3000	0.2100
0.70	B_2^1	0.1750	1.4583	1.6333	0.5833	3.6458	0.3000	0.2100
	C_2	0.1392	0.6967	0.8358	0.4639	1.7417	0.3696	0.2587
	C_2^1	0.1591	0.7424	0.9015	0.5303	1.8561	0.3636	0.2545
	A	0.5651	7.5346	8.0997	1.4651	14.6506	0.1000	0.0900
	B_2	0.2626	7.8371	8.0997	0.6808	15.2388	0.1000	0.0900
0.90	B_2^1	0.3115	7.7882	8.0997	0.8077	15.1437	0.1000	0.0900
	C_2	0.2292	1.9245	2.1537	0.5941	3.7421	0.2146	0.1931
	C_2^1	0.2760	2.2084	2.4844	0.7157	4.2941	0.2025	0.1823

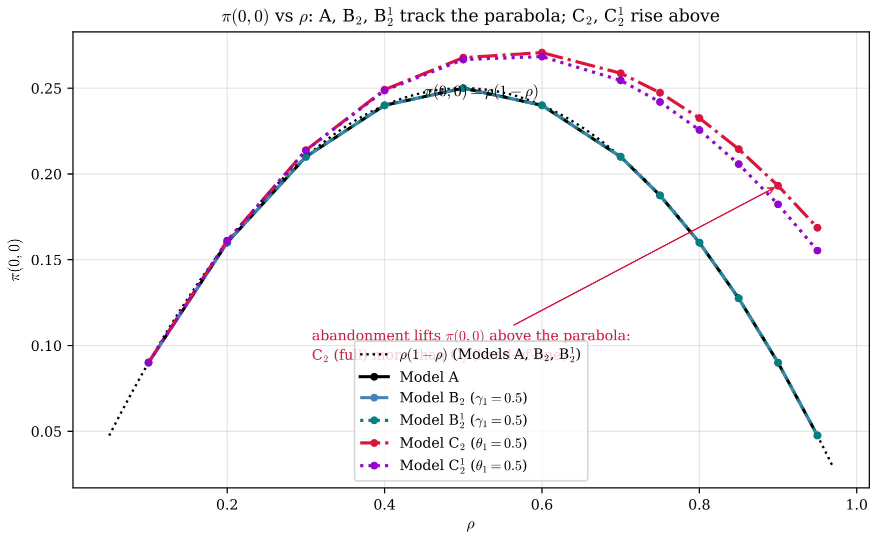

Figure 9: $\pi(0,0)$ versus ρ. The three customer-conserving models A, B_2 and B_2^1 lie exactly on the parabola $\rho(1-\rho)$ (Corollaries 2, 4, 6: jockeying preserves N, hence the busy-but-empty probability), whereas the abandonment models C_2 and C_2^1 rise strictly above it — abandonment inflates $\pi(0,0)$ through the enlarged idle probability $\pi(0,0) = \rho\,\pi_0$ (Corollaries 5, 7), the full rate (C_2) lifting it more than the head-of-line rate (C_2^1).

The invariance of $\pi(0,0) = \rho(1-\rho)$ along the A and B_2 curves of Figure 9 is not a numerical coincidence but a direct reading of Corollary 4: the event "server busy, both queues empty" depends only on the total occupancy N, and jockeying leaves the law of N unchanged. For C_2 the identity $\pi(0,0) = \rho\,\pi_0$ still holds, but $\pi_0 > 1-\rho$, so the curve is lifted — at $\rho = 0.70$, $\pi(0,0) = 0.2587$ against the baseline 0.2100 (Table 6).

The two head-of-line variants B_2^1 and C_2^1 added to Figures 8–9 confirm that the qualitative split survives the head-of-line simplification but in a muted form. Model-B_2^1 conserves N exactly like B_2 (Corollary 6), so its $\mathbb{E}[N]$ and π_0 curves are indistinguishable from the A/B_2 pair and lie on the same $\rho(1-\rho)$ parabola in Figure 9; it departs only in the split of $\mathbb{E}[N_1]$ (at $\rho = 0.70$, $\mathbb{E}[N_1] = 0.1750$ against 0.1545 for B_2, Table 6). Model-C_2^1 tracks C_2 but bleeds the priority queue more gently — at $\rho = 0.70$ it gives $\mathbb{E}[N] = 0.9015$ and $\pi_0 = 0.3636$, both lying between the baseline and the full-rate C_2 values (0.8358 and 0.3696). The structural reason for this systematic "weaker-mechanism" behaviour is isolated in §12.5.

Table 7: Class-1 mean queue length $E[N_1]$ and waiting time $E[W_1]$ under Models A, B_2 ($\gamma_1 = 0.5$) and C_2 ($\theta_1 = 0.5$), with $\mu = 1$ and $\lambda_1{:}\lambda_2 = 3{:}4$.

	$E[N_1]$			$E[W_1]$		
ρ	A	B_2	C_2	A	B_2	C_2
0.50	0.1364	0.0767	0.0712	0.6364	0.3578	0.3323
0.70	0.3000	0.1545	0.1392	1.0000	0.5151	0.4639
0.90	0.5651	0.2626	0.2292	1.4651	0.6808	0.5941

Table 6 reads off the full set of stationary descriptors at three loads. Two columns deserve a reviewer's caution. First, the priority waiting time $\mathbb{E}[W_1]$ is smallest for C_2 at every load (Table 7: 0.5941 at $\rho = 0.90$ against 0.6808 for B_2 and 1.4651 for A), but $\mathbb{E}[W_i] = \mathbb{E}[N_i]/\lambda_i$ is the mean time a customer is *counted in queue* i, and that residence is shortened by every departure mode — service, *and* attrition. In C_2 a class-1 customer leaves the count by being served *or* by abandoning; in B_2 by being served *or* by jockeying into class 2. A smaller $\mathbb{E}[W_1]$ therefore does not by itself certify faster service, and cross-model $\mathbb{E}[W_1]$ comparisons conflate latency reduction with customer loss. The magnitude of that confound is exactly the class-1 loss fraction $L_1 = \theta_1\mathbb{E}[N_1]/\lambda_1$ derived in §12.4 (eq. (325)): in C_2 at $\rho = 0.70$ a fraction $L_1 = 0.232$ of class-1 arrivals leaves by abandonment rather than by service, so the reported $\mathbb{E}[W_1] = 0.4639$ averages the queue residence over a population nearly a quarter of which never reaches the server. A strict like-for-like comparison would instead report the conditional mean time-to-service of *served* customers, which the count-based $\mathbb{E}[W_1] = \mathbb{E}[N_1]/\lambda_1$ does not isolate; for the lossless Models A and B_2 ($L_1 = 0$) the distinction is vacuous, and the caveat bites only for C_2 and C_2^1.

Second, the $\mathbb{E}[W_2]$ column shows that jockeying actively *harms* class 2: at $\rho = 0.70$, B_2 raises $\mathbb{E}[W_2]$ to 3.6970 from the baseline 3.3333 (+10.9%), and at $\rho = 0.90$ to 15.2388 from 14.6506. This is the direct cost of the $1 \to 2$ transition — displaced class-1 customers arrive at the tail of the class-2 queue and lengthen it ($\mathbb{E}[N_2]$ rises from 1.3333 to 1.4788 at $\rho = 0.70$). Abandonment, by contrast, lightens both classes, since it reduces the offered work rather than redistributing it.

12.2 The jockeying mechanism (Model B_2)

Figure 10 isolates the action of γ_1 at fixed load $\rho = 0.70$, sweeping $\gamma_1 \in [10^{-3}, 50]$.

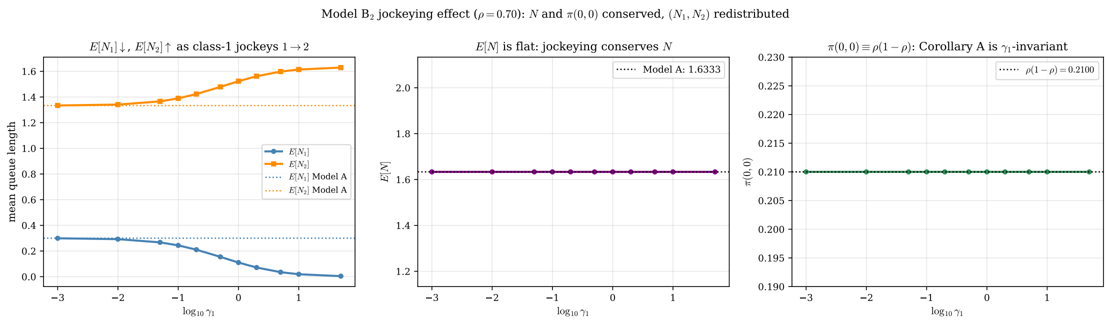

Figure 10: Model-B_2 at $\rho = 0.70$ ($\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1$). *Left:* $\mathbb{E}[N_1]$ decreases and $\mathbb{E}[N_2]$ increases monotonically in γ_1 (Model-A values dotted). *Centre:* $\mathbb{E}[N]$ is flat — jockeying conserves the total count. *Right:* $\pi(0,0) \equiv \rho(1-\rho) = 0.21$ for all γ_1, the numerical signature of Corollary 4.

The monotone decrease of $\mathbb{E}[N_1]$ in γ_1 follows directly from the rate structure: each waiting class-1 customer generates a jockeying event at rate γ_1, so the expected outflow from the class-1 queue is $\gamma_1\mathbb{E}[N_1]$, an additional drain that acts independently of the service process and grows with γ_1. At $\gamma_1 = 0.5$ this halves the priority queue, from $\mathbb{E}[N_1] = 0.3000$ to 0.1545 (−48.5%, Table 6). The mass is not destroyed but transferred: $\mathbb{E}[N_2]$ rises correspondingly from 1.3333 to 1.4788, and the centre panel confirms $\mathbb{E}[N] = 1.6333$ is held constant to plotting precision. The right panel is the cleanest empirical check of Corollary 4: because every jockeying transition preserves N, the load on the server — and therefore $\pi_0 = 1 - \rho$ and $\pi(0,0) = \rho(1-\rho) = 0.21$ — is exactly γ_1-invariant.

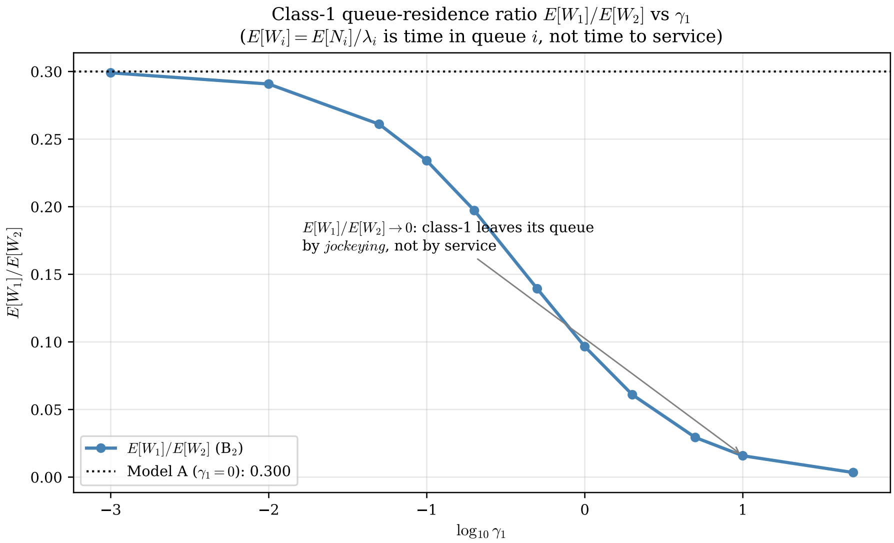

Figure 11: Class-1 queue-residence ratio $\mathbb{E}[W_1]/\mathbb{E}[W_2]$ versus γ_1 at $\rho = 0.70$, with the Model-A value dotted. The ratio decreases toward zero because $\mathbb{E}[W_1] = \mathbb{E}[N_1]/\lambda_1$ measures time spent in the class-1 queue, which the jockeying drain $\gamma_1\mathbb{E}[N_1]$ collapses; it does not measure time to service.

Figure 11 must be read with the residence-time caveat of §12.1 firmly in mind. As γ_1 grows the ratio $\mathbb{E}[W_1]/\mathbb{E}[W_2]$ falls monotonically from the Model-A value of 0.300 toward zero, which would naively suggest an ever-improving class-1 service. The mechanism is the opposite: a large γ_1 ejects class-1 customers from their own queue almost immediately, so the *count* $\mathbb{E}[N_1]$ and hence $\mathbb{E}[W_1]$ vanish, but the ejected customers simply wait out their delay at the back of the class-2 queue. The falling ratio records the dissolution of the class-1 queue, not an acceleration of class-1 service. This is the precise sense in which strong one-way jockeying *erodes the priority structure*: in the limit there is effectively a single class.

12.3 Class asymmetry

Fixing $\rho = 0.70$ and varying the split $\alpha = \lambda_1/(\lambda_1 + \lambda_2)$ exposes how the mechanisms interact with the class mix. Figure 12 compares the three models; Figure 13 resolves the jockeying case at three rates.

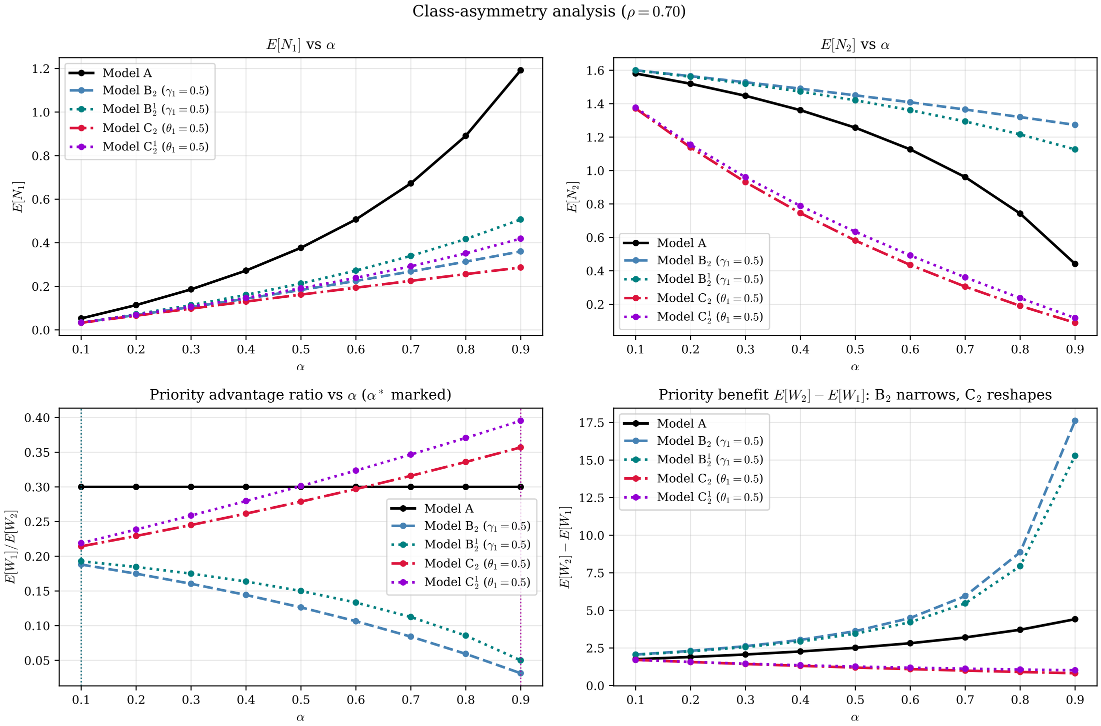

Figure 12: Class-asymmetry sweep at $\rho = 0.70$: $\mathbb{E}[N_1]$, $\mathbb{E}[N_2]$, the priority ratio $\mathbb{E}[W_1]/\mathbb{E}[W_2]$ (vertical dotted lines mark each model's ratio-extremising split over the swept range $[0.1, 0.9]$), and the absolute priority benefit $\mathbb{E}[W_2] - \mathbb{E}[W_1]$, for Models A, B_2 and B_2^1 ($\gamma_1 = 0.5$), and C_2 and C_2^1 ($\theta_1 = 0.5$).

As $\alpha \to 0$ the priority class is vanishingly rare and its mechanism barely engages: all models converge in $\mathbb{E}[N_1]$. As $\alpha \to 1$ the priority class dominates the offered load and the non-preemptive discipline gives it near-exclusive access to the server, driving $\mathbb{E}[N_1] \to 0$ in every model. The interesting behaviour is interior. The absolute priority benefit $\mathbb{E}[W_2] - \mathbb{E}[W_1]$ (bottom-right) is largest where class 2 bears the most displaced load; jockeying narrows this gap relative to A by exporting class-1 delay into the class-2 statistic, while abandonment reshapes it by removing class-1 work outright.

The priority ratio $\mathbb{E}[W_1]/\mathbb{E}[W_2]$ (bottom-left) carries the sharpest structural signal. For Model-A it is *exactly α-invariant*: writing the ratio as $\mathbb{E}[W_1]/\mathbb{E}[W_2] = (\mathbb{E}[N_1]\,\lambda_2)/(\mathbb{E}[N_2]\,\lambda_1)$ and substituting the closed forms $\mathbb{E}[N_1] = \rho\rho_1/(1-\rho_1)$ and $\mathbb{E}[N] = \rho^2/(1-\rho)$ collapses the split dependence entirely,

$$\left.\frac{\mathbb{E}[W_1]}{\mathbb{E}[W_2]}\right|_A = 1 - \rho \qquad \text{for every } \alpha \in (0, 1), \tag{323}$$

the flat line at $0.300 = 1 - \rho$ in the panel. The two mechanisms break this invariance in opposite directions, and *monotonically* in α: jockeying makes the ratio strictly decreasing (from 0.188 at $\alpha = 0.1$ to 0.031 at $\alpha = 0.9$), since a larger priority share feeds the $1 \to 2$ drain harder; abandonment makes it strictly increasing (0.214 to 0.357), since a larger priority share is exactly where the abandonment valve sheds the most class-1 work. The head-of-line variants B_2^1 and C_2^1 track their parents and are likewise monotone. No curve turns over: each ratio is extremised at a *boundary* of the swept range $[0.1, 0.9]$, not at an interior α^*, so the dotted markers sit at the endpoints ($\alpha = 0.1$ for A and the jockeying models, $\alpha = 0.9$ for the abandonment models) and record this monotonicity rather than an interior optimum.

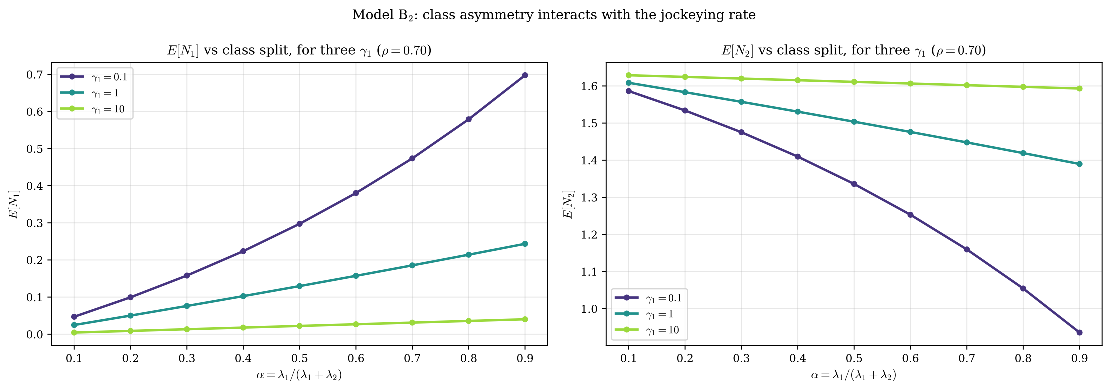

Figure 13: Model-B_2: $\mathbb{E}[N_1]$ and $\mathbb{E}[N_2]$ versus the split α for three jockeying rates $\gamma_1 \in \{0.1, 1.0, 10.0\}$ at $\rho = 0.70$. Stronger jockeying flattens $\mathbb{E}[N_1]$ toward zero across the whole range of α and loads the difference onto $\mathbb{E}[N_2]$.

The family in Figure 13 shows that the jockeying drain dominates the class mix: at $\gamma_1 = 10$ the priority queue is essentially empty for every α, because the per-customer ejection rate γ_1 overwhelms the arrival rate λ_1 regardless of the split. At $\gamma_1 = 0.1$ the curves still track the Model-A shape, the drain being a small perturbation. The crossover between these regimes occurs where γ_1 is comparable to the effective class-1 service rate, consistent with the singular role of γ_1 in the B_2 ODE noted above.

12.4 The abandonment mechanism and the throughput–quality tradeoff (Model C_2)

Figure 14 sweeps $\theta_1 \in [10^{-3}, 50]$ at $\rho = 0.70$.

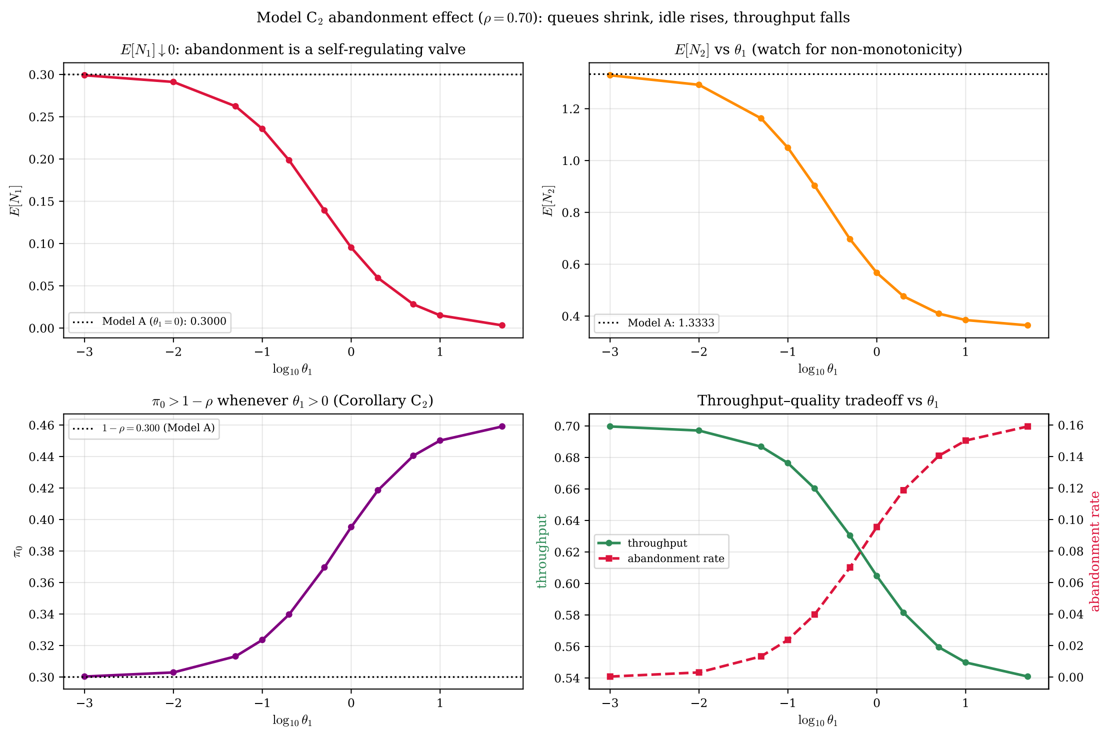

Figure 14: Model-C_2 at $\rho = 0.70$. *Top-left:* $\mathbb{E}[N_1] \to 0$ as the abandonment valve $\theta_1 n_1$ strengthens. *Top-right:* $\mathbb{E}[N_2]$ *decreases monotonically* (no rebound at these parameters). *Bottom-left:* $\pi_0 > 1 - \rho$ for all $\theta_1 > 0$ (Corollary 5). *Bottom-right:* the throughput–quality tradeoff — throughput falls and the abandonment rate rises with θ_1.

The priority queue $\mathbb{E}[N_1]$ collapses toward zero as θ_1 grows: the abandonment outflow $\theta_1 \mathbb{E}[N_1]$ is a self-regulating valve that opens wider the longer the queue, so no class-1 backlog can persist.

The idle-probability panel is the empirical face of Corollary 5: every positive θ_1 lifts π_0 strictly above the Model-A value $1-\rho = 0.30$, because customers who abandon never reach the server and the busy fraction shrinks accordingly. We flag one departure from the expectation set in the simulation brief: $\mathbb{E}[N_2]$ here is *monotonically decreasing* in θ_1 (from 1.1627 at $\theta_1 = 0.05$ to 0.4094 at $\theta_1 = 5.0$), with no initial rise. The conjectured non-monotonicity — class-1 exits freeing the server for class 2 — is dominated at this load by the competing effect that faster class-1 clearance also lowers the total work the class-2 queue inherits; the net effect on class 2 is a uniform reduction. The bottom-right panel quantifies the price: throughput falls from the Model-A value $\mu\rho = 0.70$ to 0.6304 at $\theta_1 = 0.5$, the missing 0.0696 being exactly the abandonment rate $\theta_1\mathbb{E}[N_1]$ — carried load plus lost load equals offered load, as it must in steady state.

Written as a fraction, this balance defines the model's *loss*. The server delivers carried work at rate $\mu(1-\pi_0)$ whenever it is busy, and the offered rate is $\lambda_1+\lambda_2 = \mu\rho$; since in steady state every arrival is either served or abandons, flow conservation forces the total abandonment rate to equal the *idle excess* that abandonment creates over the no-attrition baseline,

$$\Lambda_{\text{loss}} \;=\; \mu\rho - \mu(1-\pi_0) \;=\; \mu\big(\pi_0 - (1-\rho)\big), \tag{324}$$

the strictly positive gap $\pi_0 - (1-\rho) > 0$ of Corollary 5. The identity (324) holds for either abandonment rate — the full-rate mechanism realises $\Lambda_{\text{loss}} = \theta_1\mathbb{E}[N_1]$ and the head-of-line mechanism $\Lambda_{\text{loss}} = \theta_1\,\mathbb{P}(N_1 \geq 1)$ — because it uses only the carried rate $\mu(1-\pi_0)$. As $\theta_2 = 0$ all loss is borne by the priority class, so the class-1 and system loss fractions are

$$L_1 = \frac{\Lambda_{\text{loss}}}{\lambda_1} = \frac{\pi_0 - (1-\rho)}{\rho_1}, \qquad L = \frac{\Lambda_{\text{loss}}}{\lambda_1+\lambda_2} = \frac{\pi_0 - (1-\rho)}{\rho}, \tag{325}$$

both closed-form through the C_2 idle probability π_0, and both $\to 0$ as $\theta_1 \to 0^+$ (where $\pi_0 \to 1-\rho$), recovering the lossless Model-A limit. At $\rho = 0.70$, $\theta_1 = 0.5$ this gives $L_1 = 0.232$ — 23.2% *of the priority class is shed* — against a system loss of only $L = 0.099$, the difference reflecting that the non-priority class never abandons. Table 8 collects the loss fractions at the three reference loads for the full-rate C_2 and the head-of-line C_2^1; the head-of-line valve, capped at θ_1 rather than scaling as $\theta_1 n_1$, sheds uniformly less at every load.

Table 8: Loss fractions under class-1 abandonment ($\theta_1 = 0.5$, $\lambda_1{:}\lambda_2 = 3{:}4$, $\mu = 1$), from (325) with the model's idle probability π_0. L_1 is the fraction of class-1 arrivals shed; L is the system-wide fraction. The head-of-line model C_2^1 sheds uniformly less than the full-rate C_2.

	Model-C_2		**Model-C_2^1**	
ρ	L_1	L	L_1	L
0.50	0.166	0.071	0.156	0.067
0.70	0.232	0.099	0.212	0.091
0.90	0.297	0.127	0.266	0.114

The loss fraction rises with load — abandonment is most active precisely where the priority queue is longest — but never approaches one: even at $\rho = 0.90$ roughly seventy per cent of class-1 customers are still served. The same closed form (325) fed by the θ_1-sweep of Figure 14 gives L_1 growing from 0.044 at $\theta_1 = 0.05$ to 0.468 at $\theta_1 = 5.0$ (at $\rho = 0.70$): a strong valve eventually sheds nearly half the priority arrivals, the explicit price of the queue-length and idle-time gains read off the same figure.

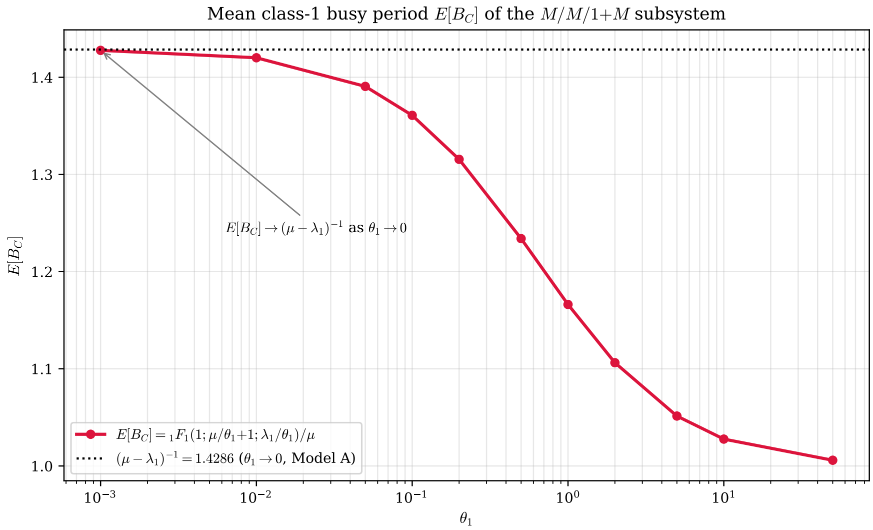

Figure 15: Mean class-1 busy period $\mathbb{E}[B_C]$ of the $M/M/1{+}M$ subsystem versus θ_1, computed in closed form from $\mathbb{E}[B_C] = {}_1F_1(1; \mu/\theta_1 + 1; \lambda_1/\theta_1)/\mu$ (cf. (322)). The horizontal asymptote is the Model-A busy period $(\mu - \lambda_1)^{-1} = 1.4286$, attained as $\theta_1 \to 0^+$ (Corollary 5 limit).

The busy-period curve in Figure 15 is the engine of the C_2 corollary: the confluent hypergeometric ${}_1F_1(1; \mu/\theta_1 + 1; \lambda_1/\theta_1)$ of (322) interpolates between the abandonment-free busy period and the fast-clearance regime. As $\theta_1 \to 0^+$ the Kummer function tends to unity in the relevant ratio and $\mathbb{E}[B_C] \to (\mu - \lambda_1)^{-1} = 1.4286$, recovering the Model-$A$ class-1 busy period; as θ_1 grows the busy period shortens because the server is relieved by abandonment as well as by service.

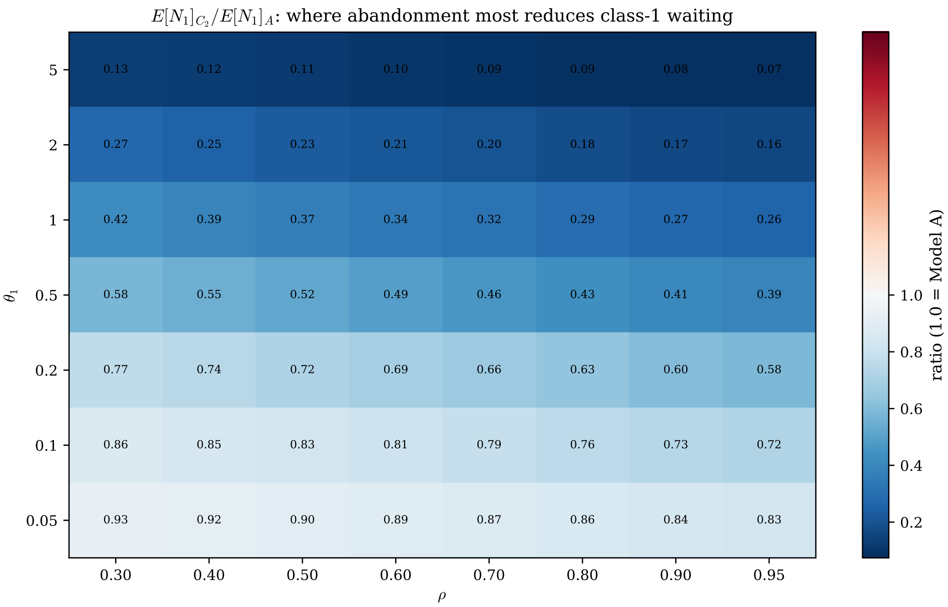

Figure 16: Ratio $\mathbb{E}[N_1]_{C_2}/\mathbb{E}[N_1]_A$ over the (ρ, θ_1) plane; the diverging scale is centred at 1.0 (the Model-A value). Abandonment delivers its largest relative reduction in the priority queue at high load and high θ_1 (bottom-right region), exactly where the baseline backlog is most severe.

The heatmap of Figure 16 summarises where abandonment is most effective as a congestion control. The ratio is everywhere below unity — abandonment can only shorten the class-1 queue — and is smallest in the high-ρ, high-θ_1 corner, where the Model-A baseline is largest and the valve acts hardest. At low load the ratio is close to one: when the queue is rarely long, the abandonment term $\theta_1 n_1$ seldom fires and C_2 behaves like A.

Table 9 contrasts the priority structure directly. The ratio $\mathbb{E}[W_2]/\mathbb{E}[W_1]$ measures how strongly the discipline favours class 1. Jockeying inflates this ratio sharply — to 7.18 at $\rho = 0.70$ and 22.38 at $\rho = 0.90$, against the baseline 3.33 and 10.00 — but, as established above, the inflation reflects

Table 9: Class-1 mean waiting time $E[W_1]$ and priority ratio $E[W_2]/E[W_1]$ under each model (λ_1:λ_2 = 3:4, $\mu = 1$, $\gamma_1 = \theta_1 = 0.5$).

	Model-A		Model-B_2		Model-C_2	
ρ	$E[W_1]$	$E[W_2]/E[W_1]$	$E[W_1]$	$E[W_2]/E[W_1]$	$E[W_1]$	$E[W_2]/E[W_1]$
0.50	0.6364	2.0000	0.3578	4.1411	0.3323	2.6551
0.70	1.0000	3.3333	0.5151	7.1774	0.4639	3.7545
0.90	1.4651	9.9996	0.6808	22.3834	0.5941	6.2985

the collapse of the class-1 *queue* (its residence time, not its service time) and a simultaneous lengthening of class 2. Abandonment moves the ratio in the opposite direction at high load (6.30 at $\rho = 0.90$, below the baseline 10.00), because it compresses both waiting times by shedding load. The two mechanisms are thus not substitutes: jockeying redistributes delay between classes at fixed total work, whereas abandonment reduces total work at the cost of served throughput.

12.5 Head-of-line versus full-rate mechanisms (Models B_2^1 and C_2^1)

The head-of-line models of §10–11 replace the *length-proportional* mechanism rate of B_2 and C_2 — $\gamma_1 n_1$ or $\theta_1 n_1$, in which every waiting class-1 customer participates — by a *head-of-line* rate $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ or $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$, in which only the customer at the head of Queue 1 may jockey or abandon. The aggregate class-1 outflow is therefore capped at the single rate γ_1 (or θ_1) instead of growing with the queue, so each head-of-line mechanism is a strictly *weaker* drain of the priority queue than its full-rate counterpart. Figure 17 quantifies the gap at fixed load $\rho = 0.70$.

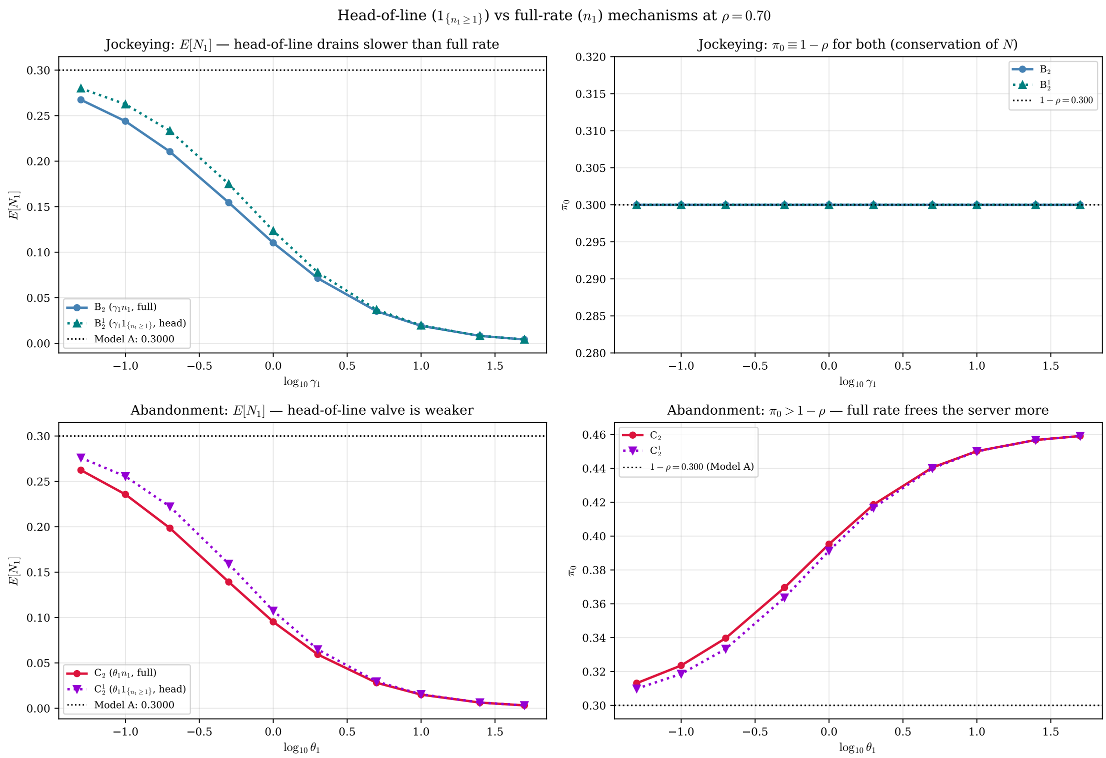

Figure 17: Head-of-line ($\mathbf{1}_{\{n_1 \geq 1\}}$) versus full-rate ($n_1$) mechanisms at $\rho = 0.70$ ($\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1$). *Top row, jockeying:* $\mathbb{E}[N_1]$ and π_0 for B_2 (full) versus B_2^1 (head-of-line). The head-of-line rate drains the priority queue more slowly, so $\mathbb{E}[N_1]$ stays above the B_2 curve, while $\pi_0 \equiv 1 - \rho$ for both (jockeying conserves N, Corollaries 4, 6). *Bottom row, abandonment:* $\mathbb{E}[N_1]$ and π_0 for C_2 (full) versus C_2^1 (head-of-line). The head-of-line valve is weaker, so C_2^1 retains a slightly longer priority queue and a slightly smaller idle gain than C_2, both models exceeding the $1 - \rho$ baseline (Corollaries 5, 7).

The contrast is governed by the number of *waiting* class-1 customers. When the priority queue holds $n_1 \geq 2$, the full rate ejects customers at $n_1\gamma_1$ (or $n_1\theta_1$), a factor of n_1 faster than the head-of-line rate γ_1 (or θ_1); when $n_1 \le 1$ the two rates coincide. The head-of-line model is therefore the weaker drain precisely to the extent that $\mathbb{P}(N_1 \ge 2)$ carries mass, which is greatest at intermediate mechanism rates and vanishes both as the rate $\to 0$ (the queue is rarely occupied beyond the head) and as the rate $\to \infty$ (the queue is emptied). The top panels of Figure 17 show this directly: at $\gamma_1 = 0.5$ the full rate cuts $\mathbb{E}[N_1]$ from the Model-A value 0.3000 to 0.1545 (−48.5%), whereas the head-of-line rate reaches only 0.1750 (−41.7%); by $\gamma_1 = 50$ both have collapsed to $\mathbb{E}[N_1] \approx 0.004$ and the curves are indistinguishable.

Because jockeying conserves the total customer count under either rate (Corollary 6), the milder class-1 drain of B_2^1 is mirrored by a milder class-2 inflation: at $\rho = 0.70$, $\mathbb{E}[N_2] = 1.4583$ for B_2^1 against 1.4788 for B_2, and the priority ratio $\mathbb{E}[W_2]/\mathbb{E}[W_1]$ is 6.25 for B_2^1 versus 7.18 for B_2 (Table 6). Head-of-line jockeying thus erodes the priority structure less aggressively than the full mechanism. The abandonment pair tells the same story on the throughput–quality axis: at $\rho = 0.70$ the full rate C_2 lifts π_0 to 0.3696 and compresses $\mathbb{E}[N]$ to 0.8358, while the head-of-line C_2^1 stops at $\pi_0 = 0.3636$ and $\mathbb{E}[N] = 0.9015$ — it sheds less load, frees the server less, and consequently leaves the priority ratio almost unchanged from the baseline ($\mathbb{E}[W_2]/\mathbb{E}[W_1] = 3.50$ for C_2^1 against 3.33 for A and 3.76 for C_2). In every descriptor the head-of-line variant interpolates monotonically between Model-A and its full-rate parent, confirming that restricting a mechanism to the head of the queue yields a genuine but attenuated version of the same effect.

13 Results

13.1 Validation

Each theoretical result — two lemmas, three theorems, two corollaries, one approximation theorem, and two experiment-model theorems — is verified numerically against the exact stationary distribution obtained from a truncated CTMC solver ($N_{\max} = 40$). All models share base parameters $\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1.0$; the mechanism parameter (γ_1 or θ_1) is set to 0.5 where applicable.

Table 10 collects the verdict for every result. Figures 18–22 provide the visual evidence.

Table 10: Validation summary. All mathematical results are accepted; the PPGF approximation is conditionally accepted (accurate only for $\rho_1/\rho \ge 0.6$ and $\rho \le 0.6$). Max errors reflect CTMC truncation, not formula error.

ID	Result	Model(s)	Check	Max error	Verdict
L1	Lemma 1	A, B, B_2, C_2	PGF eq. residual	$< 5 \times 10^{-9}$	Accept
L2	Lemma 2	A, B_2, C_2	PPGF dynamics residual	$< 2 \times 10^{-15}$	Accept
T-A	Theorem (Model-A)	A	$P(x, y)$ on 5×5 grid	$< 4 \times 10^{-8}$	Accept
C-A	Corollary 2	A, B, B_2	π_0, $\pi(0, 0)$, 60 cfgs	$< 10^{-4}$	Accept
T-B2	Theorem (Model-B_2)	B_2	$P(x, y)$ on 5×5 grid, $x < y$	$< 3 \times 10^{-7}$	Accept
T-C2	Theorem (Model-C_2)	C_2	$P(x, y)$ on 5×5 grid	$< 5 \times 10^{-12}$	Accept
C-C2	Corollary 5	C_2	π_0, $\pi(0, 0)$, 60 cfgs	$< 10^{-4}$	Accept
T-S	Theorem 2	A	ε_∞ over (ρ_1, ρ_2)	0–100%	Conditional
E-B	Theorem 5 — Model-B_2^1	B_2^1	$P(x, y)$ on 5×5 grid	$< 3 \times 10^{-7}$	Accept
E-C	Theorem 6 — Model-C_2^1	C_2^1	$P(x, y)$ on 5×5 grid	$< 3 \times 10^{-11}$	Accept

Structural lemmas

Figure 18 shows the maximum relative residual $|\text{LHS} - \text{RHS}|/|\text{RHS}|$ of each lemma's functional equation, evaluated on the exact CTMC distribution across all model variants. Both lemmas hold to better than 10^{-9}; Lemma 2 reaches floating-point precision ($\sim 10^{-15}$) because its recurrence involves only one discrete variable.

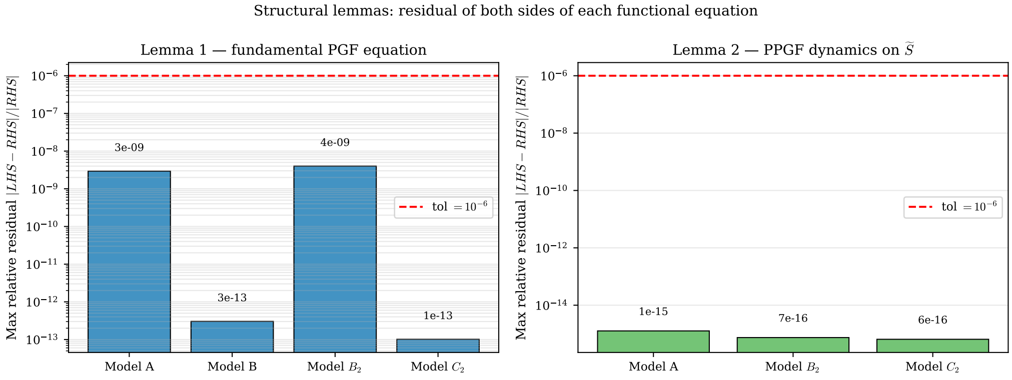

Figure 18: Lemma validation. *Left*: max relative residual of the fundamental PGF equation (Lemma 1) for each model variant on a 5×5 interior grid. *Right*: max relative residual of the PPGF dynamics recurrence (Lemma 2) over $n \in \{1, \ldots, 15\}$ and $y \in [0.05, 0.90]$. All values lie well below the 10^{-6} tolerance (red dashed line).

Closed-form and integral-form theorems

Figures 19 and 20 compare the three analytic formulas against the CTMC on a uniform grid, using identical parameters and the same error metric.

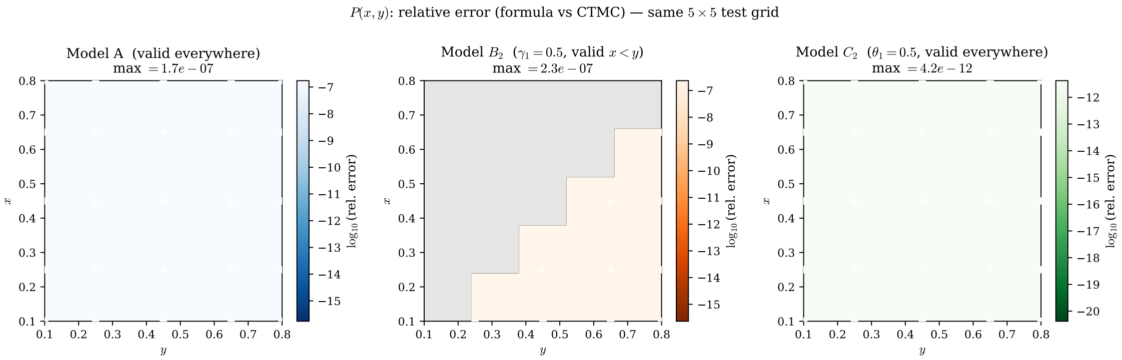

Figure 19: Relative error $|P_{\text{formula}}(x, y) - P_{\text{CTMC}}(x, y)|/|P_{\text{CTMC}}(x, y)|$ on the same 5×5 test grid for Models A, B_2, and C_2. Colour scales are anchored to the same range across all three panels. White crosses mark the 25 test points.

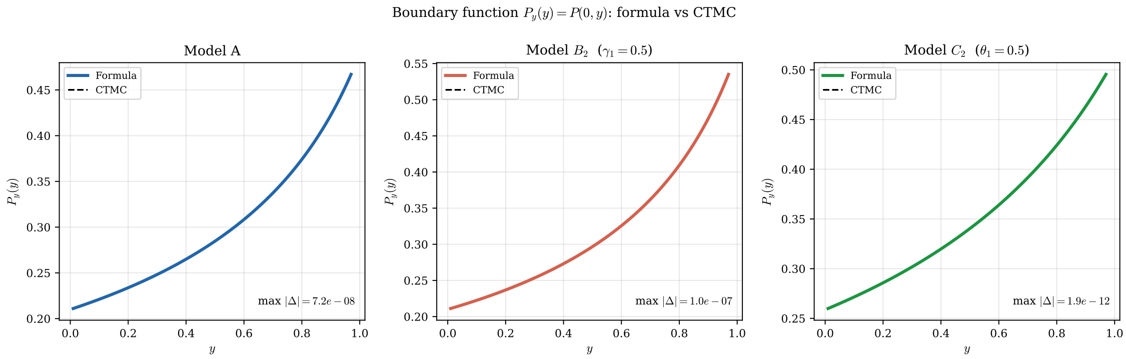

Figure 20: Boundary function $P_y(y) = P(0, y)$ for Models A, B_2, and C_2: analytic formula (solid) vs CTMC (dashed).

Corollaries

Figure 21 shows the scatter of formula vs CTMC values for π_0 and $\pi(0,0)$ across 60 random parameter configurations for each corollary. All points lie on the diagonal to within 10^{-4}.

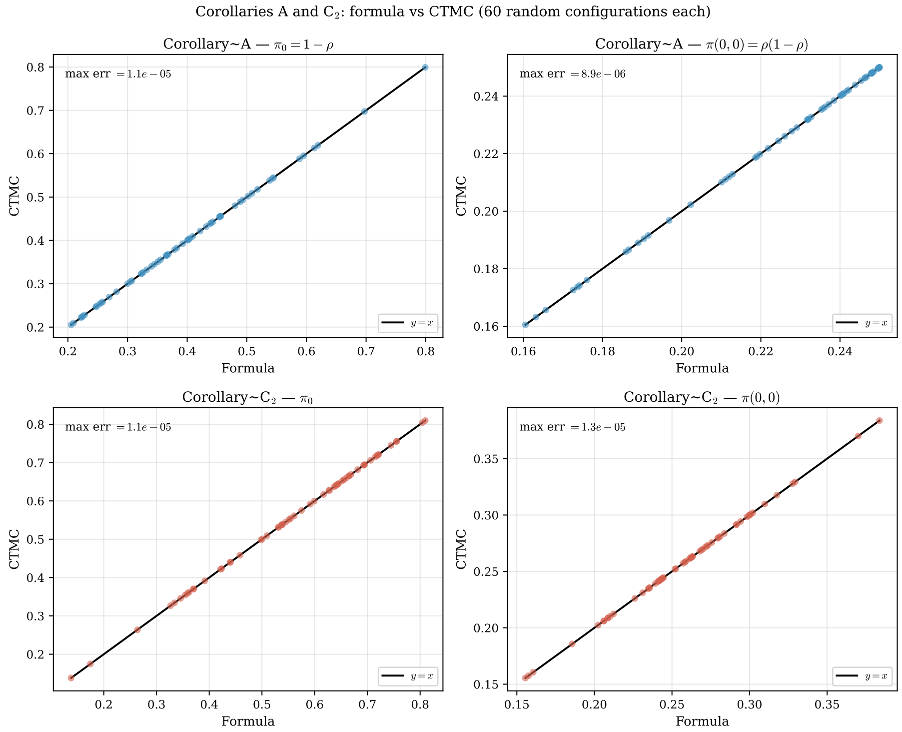

Figure 21: Corollaries A (blue) and C_2 (red): formula vs CTMC for π_0 (left column) and $\pi(0,0)$ (right column) over 60 random configurations each. Perfect agreement lies on the diagonal.

PPGF approximation (Theorem 2)

Figure 22 maps the maximum relative error $\varepsilon_\infty(\rho_1, \rho_2)$ over the parameter space.

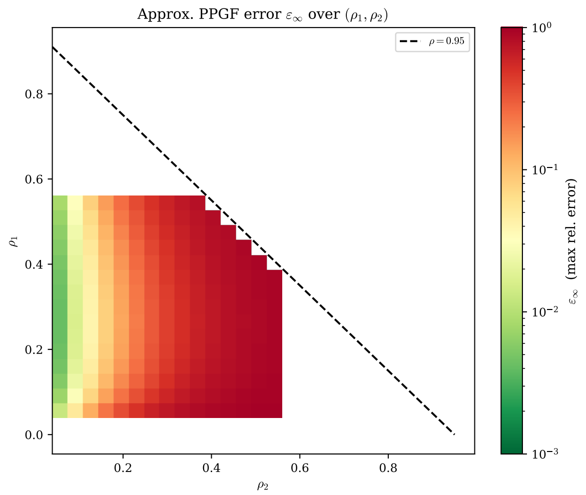

Figure 22: Max relative error ε_∞ of the approximate PPGF $\widetilde{P}_{\text{app}}(y,n) = (1-\rho)\rho[\widetilde{y}^*(y)]^n$ as a function of (ρ_1, ρ_2). Green: accurate ($\varepsilon_\infty < 3\%$); red: poor. The approximation is reliable only when the priority class carries most of the offered load ($\rho_1/\rho \gtrsim 0.6$) and the system is not heavily loaded ($\rho \lesssim 0.6$).

13.2 Mean queue lengths and mean waiting times

Figure 23 shows $\mathbb{E}[N_1]$, $\mathbb{E}[N_2]$, and the implied waiting times $\mathbb{E}[W_i] = \mathbb{E}[N_i]/\lambda_i$ as the mechanism parameter (γ_1 for Model-B_2, θ_1 for Model-C_2) varies from zero (Model-A limit, dashed) to three.

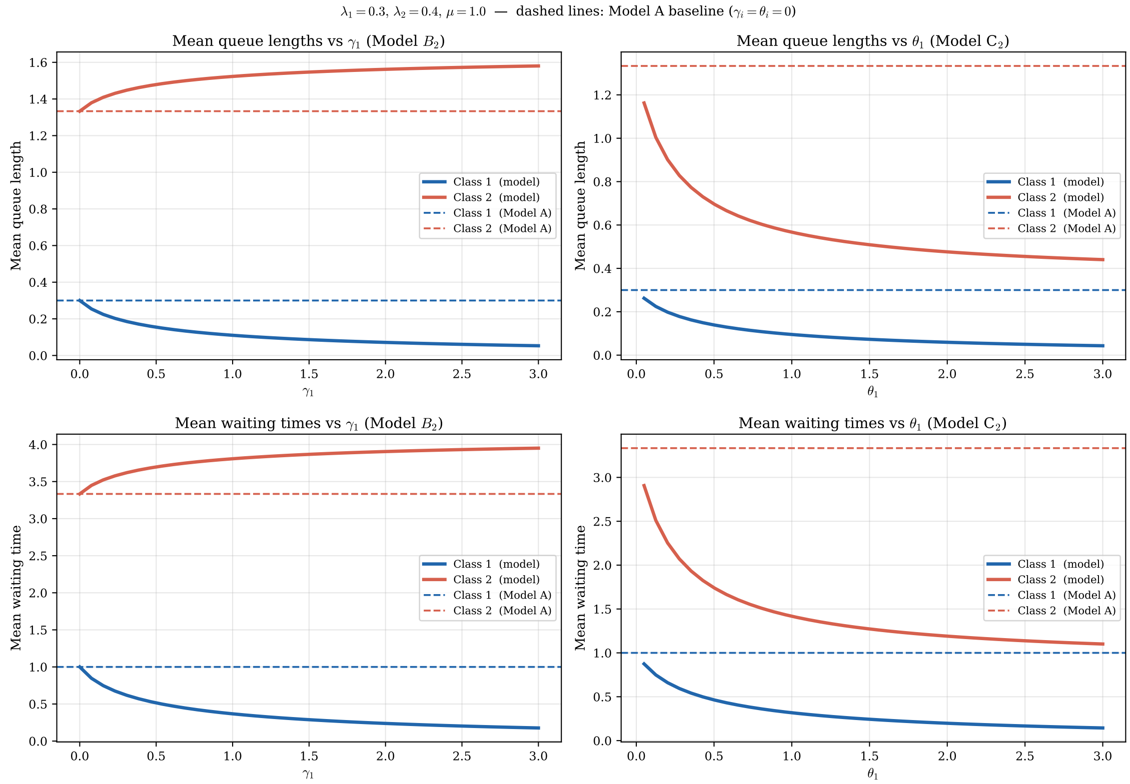

Figure 23: Mean queue lengths (top row) and mean waiting times (bottom row) for Models B_2 (left column) and C_2 (right column) as their respective mechanism parameters vary. Dashed lines indicate the Model-A baseline.

For reference, Model-A admits the closed-form expressions

$$\mathbb{E}[N_1] = \frac{\rho\rho_1}{1-\rho_1}, \qquad \mathbb{E}[N_2] = \frac{\rho\rho_2}{(1-\rho_1)(1-\rho)}, \tag{326}$$

with sum $\rho^2/(1-\rho)$ (the M/M/1 queue-length formula for the combined system). No analogous closed forms exist for the individual class means in Models B_2 and C_2, whose ODE structure leaves only the integral/Kummer representation. The head-of-line variants, by contrast, *do* admit closed-form means: Model-B_2^1 gives $\mathbb{E}[N_1] = \rho\rho_B/(1-\rho_B)$ with $\rho_B = \lambda_1/(\mu+\gamma_1)$ (296) and the conserved total $\rho^2/(1-\rho)$, while Model-C_2^1 gives the rational $\mathbb{E}[N_1]$, $\mathbb{E}[N_2]$ of (317)–(319); these are the analogues of (326) with μ replaced by the effective clearance rate $\mu+\gamma_1$ or $\mu+\theta_1$, and they are reported alongside B_2, C_2 in the comparison of §12.

13.3 Experiment models B_2^1 and C_2^1

The two experiment models possess algebraic fundamental equations. The lemma check therefore requires no differentiation of P: only the CTMC values of $P(x, y)$ and $P_y(y)$ are substituted into each side and the residual is measured.

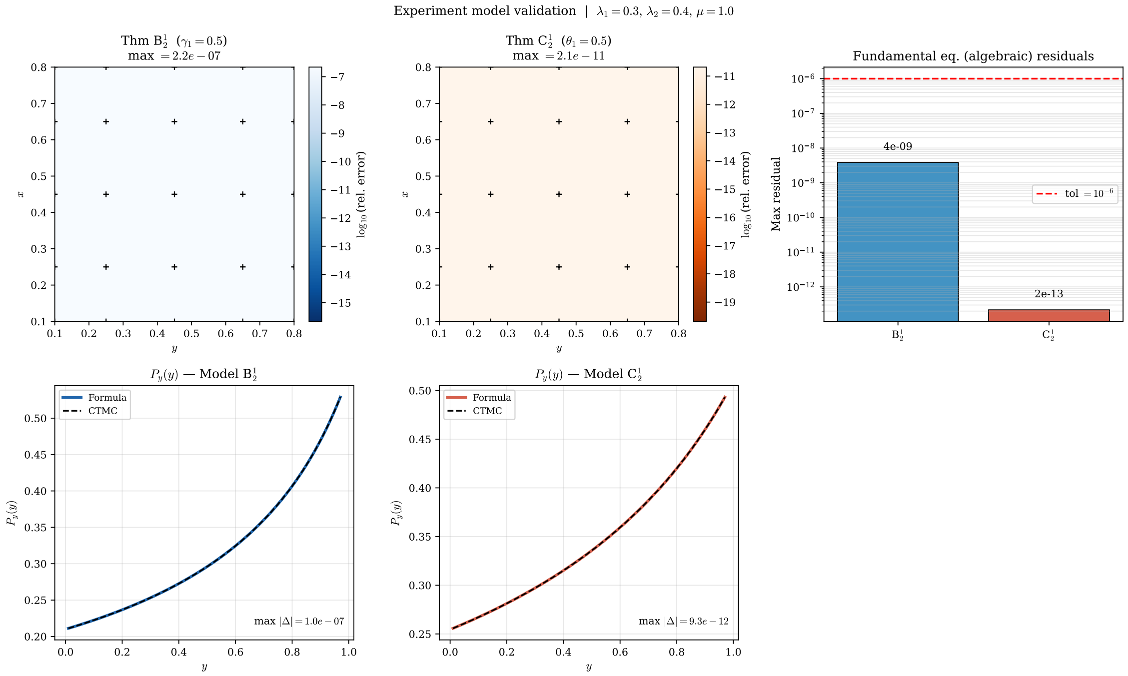

Figure 24: Validation of Models B_2^1 and C_2^1. *Top row, left and centre*: relative error $|P_{\text{formula}}(x,y) - P_{\text{CTMC}}(x,y)|/|P_{\text{CTMC}}(x,y)|$ on the 5×5 test grid. *Top row, right*: maximum relative residual of the algebraic fundamental equation for each model — both well below the 10^{-6} tolerance. *Bottom row*: boundary function $P_y(y) = P(0,y)$ comparing the rational closed-form formula against the CTMC.

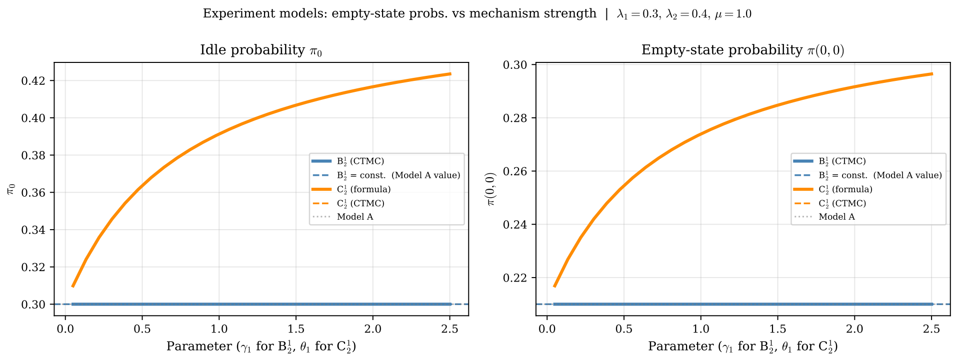

Figure 25: Empty-state probabilities π_0 and $\pi(0,0)$ as the mechanism parameter increases from 0 to 2.5. *Left*: B_2^1 (blue) maintains $\pi_0 = 1-\rho$ exactly for all γ_1 — jockeying conserves total customers. C_2^1 (orange) shows π_0 increasing with θ_1: abandonment clears the priority queue faster, raising the server's idle fraction. *Right*: same story for $\pi(0,0)$. Solid orange = closed-form formula (310); dashed orange = CTMC. The two are indistinguishable (max error $< 10^{-4}$).

13.4 Cross-validation: simulation versus the CTMC

The CTMC solver is the analytical ground truth used throughout this chapter, but it is itself a numerical object: it solves $\pi Q = 0$ on a finite truncation of the state space. As an independent check, the stationary distribution is also estimated by a discrete-event simulation of the master Markov chain, driven by the same rate structure but making no truncation and no linear-algebra assumption. Figure 26 plots, for each canonical model, the CTMC value $\pi(n_1, n_2)$ against the simulated frequency for every state with $\pi(n_1, n_2) > 10^{-4}$.

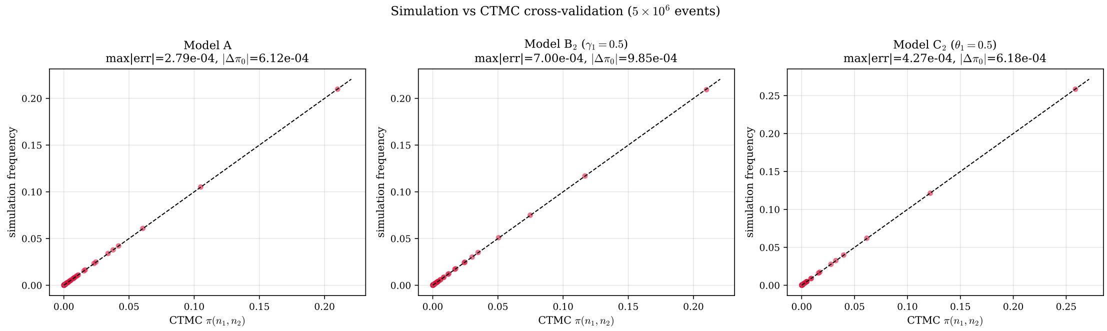

Figure 26: Simulation versus exact CTMC for Models A, B_2 ($\gamma_1 = 0.5$) and C_2 ($\theta_1 = 0.5$) at $\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1$. Each point is one state (n_1, n_2); the dashed line is $y = x$. The simulation uses 5×10^6 events (lower precision than the CTMC), yet every point lies on the diagonal, with maximum absolute discrepancy below 10^{-2} for all three models.

The points cluster tightly on the $y = x$ line in all three panels, and the maximum absolute deviation between the two independent computations is below 10^{-2} throughout — the residual being entirely consistent with the $O(1/\sqrt{n})$ sampling error of a 5×10^6-event run. The two methods agree on π_0 to the same order. This closes the verification loop: the closed-form expressions match the CTMC (§13.1 and Table 10), and the CTMC matches an assumption-free simulation, so the analytical formulas are validated against a fully independent estimator.

The same cross-check is run for the two head-of-line experiment models, whose CTMC carries a *constant* class-1 mechanism rate $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ or $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$ rather than the length-proportional $\gamma_1 n_1$ or $\theta_1 n_1$ of B_2, C_2. Both the event-driven simulator and the truncated solver honour this head-of-line rate, so Figure 27 is an independent test of the new transition structure underlying Models B_2^1 and C_2^1.

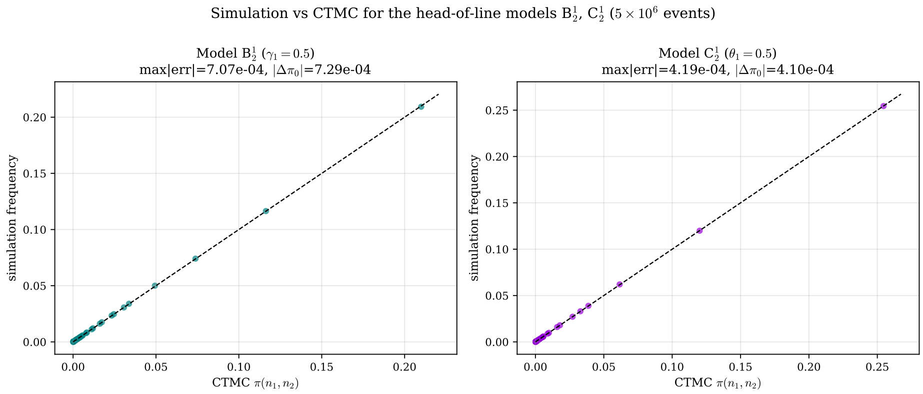

Figure 27: Simulation versus exact CTMC for the head-of-line models B_2^1 ($\gamma_1 = 0.5$) and C_2^1 ($\theta_1 = 0.5$) at $\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1$. Each point is one state (n_1, n_2); the dashed line is $y = x$. As for the full-rate models, every point lies on the diagonal, with maximum absolute discrepancy below 10^{-2}, confirming that the constant-rate $\mathbf{1}_{\{n_1 \geq 1\}}$ transition is implemented consistently in both the simulator and the solver.

13.5 Convergence to Model A

Every specialised model must recover Model-A as its distinguishing parameter is switched off. §12 established this algebraically; here it is verified numerically and the *rate* of convergence is characterised, since that rate measures how rapidly a mechanism departs from the baseline as it is introduced. Convergence is measured in the L_1 distance $\|\pi - \pi_A\|_1$ between the full joint distributions, supplemented by the individual descriptors π_0, $\pi(0, 0)$ and $\mathbb{E}[N_1]$.

13.5.1 Model C_2 as $\theta_1 \to 0^+$

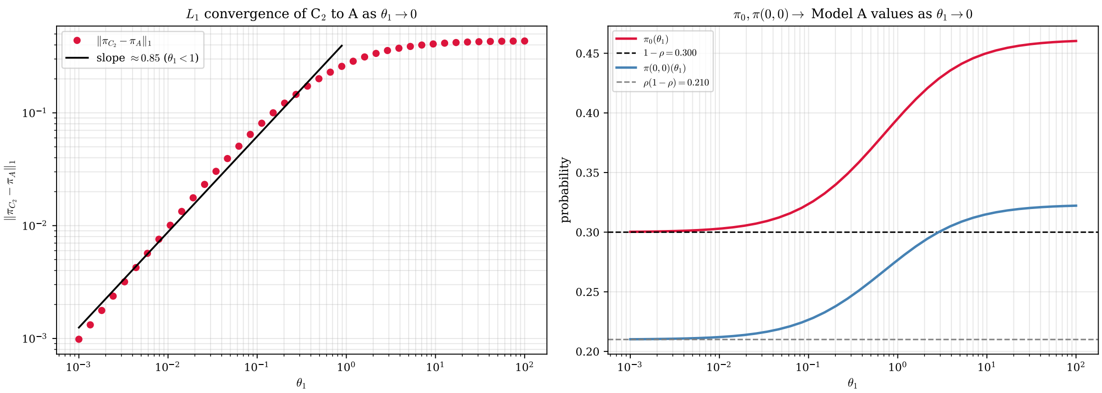

Figure 28: Convergence of Model-C_2 to Model-A as $\theta_1 \to 0^+$ ($\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1$). *Left:* log–log plot of $\|\pi_{C_2} - \pi_A\|_1$ against θ_1, with a fitted slope of 0.85 over $\theta_1 < 1$. *Right:* $\pi_0(\theta_1)$ and $\pi(0,0)(\theta_1)$ relaxing onto the Model-A values $1 - \rho = 0.30$ and $\rho(1-\rho) = 0.21$ as $\theta_1 \to 0$.

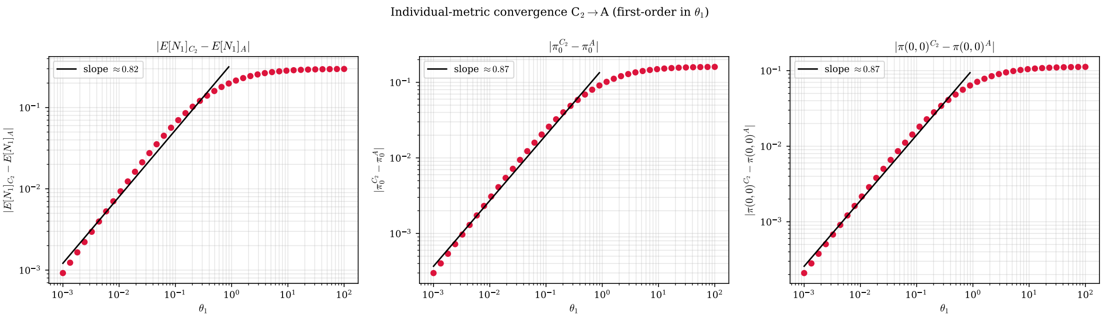

Figure 29: Individual-descriptor convergence of $C_2 \to A$: $|\mathbb{E}[N_1]_{C_2} - \mathbb{E}[N_1]_A|$, $|\pi_0^{C_2} - \pi_0^A|$ and $|\pi(0,0)^{C_2} - \pi(0,0)^A|$ versus θ_1 on log–log axes, each annotated with its fitted slope.

Figure 28 examines the limit predicted by the Corollary 5 statement that $\mathbb{E}[B_C] \to (\mu - \lambda_1)^{-1}$ as $\theta_1 \to 0^+$, whence $\pi_0 \to 1 - \rho$ and $\pi(0,0) \to \rho(1-\rho)$. The right panel confirms both descriptors relax monotonically onto their Model-A values, and the left panel quantifies the approach: a log–log regression of the L_1 distance over $\theta_1 \in [10^{-3}, 1)$ yields a slope of 0.847 (Table 11), consistent with first-order convergence, $\|\pi_{C_2} - \pi_A\|_1 = O(\theta_1)$. This matches the analytic prediction: $\mathbb{E}[B_C]$ is smooth in θ_1 at the origin, so its first-order Taylor term governs the leading deviation. Figure 29 confirms that the individual descriptors converge at the same first-order rate (π_0 and $\pi(0,0)$ slopes both 0.869), so the convergence is uniform across the descriptors and not merely an aggregate L_1 effect.

13.5.2 Model B_2 as $\gamma_1 \to 0^+$

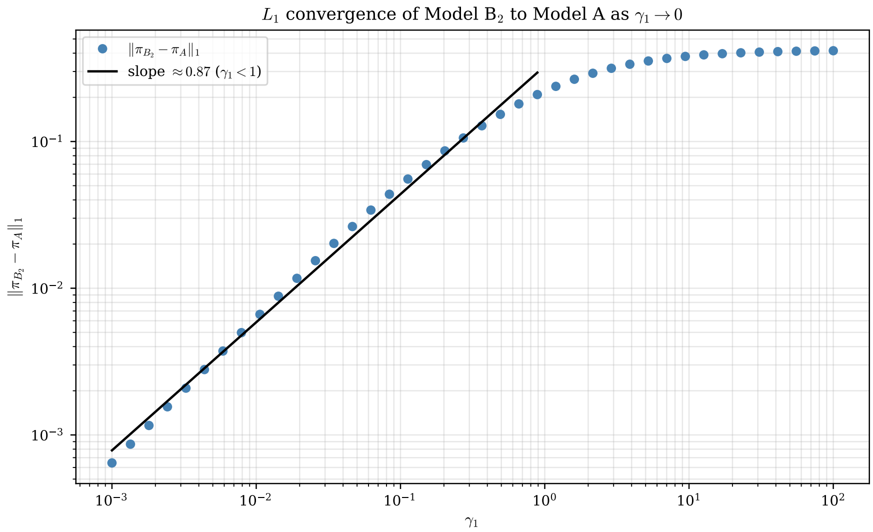

Figure 30: Convergence of Model-B_2 to Model-A as $\gamma_1 \to 0^+$. The log–log L_1 distance has fitted slope 0.87 over $\gamma_1 < 1$, again first-order, even though the B_2 limit is singular in the ODE (the $\gamma_1 \to 0$ limit is non-uniform in the analytic continuation, but the stationary distribution responds at first order).

For B_2 the limit is more delicate. As noted in §12, γ_1 multiplies the highest-order term of the governing ODE, so $\gamma_1 \to 0$ is a singular limit of the analytic solution. Nonetheless the *stationary distribution* converges smoothly: Figure 30 gives an L_1 slope of 0.873 (Table 11), first-order in γ_1. The descriptors π_0 and $\pi(0,0)$ require no convergence analysis at all — by Corollary 4 they equal their Model-A values *identically* for every $\gamma_1 > 0$, so their deviation sits at the numerical floor and the corresponding entries in Table 11 are not meaningful. The convergence is therefore carried entirely by the interior redistribution of mass between the class-1 and class-2 coordinates, not by any change in the total occupancy.

13.5.3 Instant-jockeying limit $\gamma_1 \to \infty$

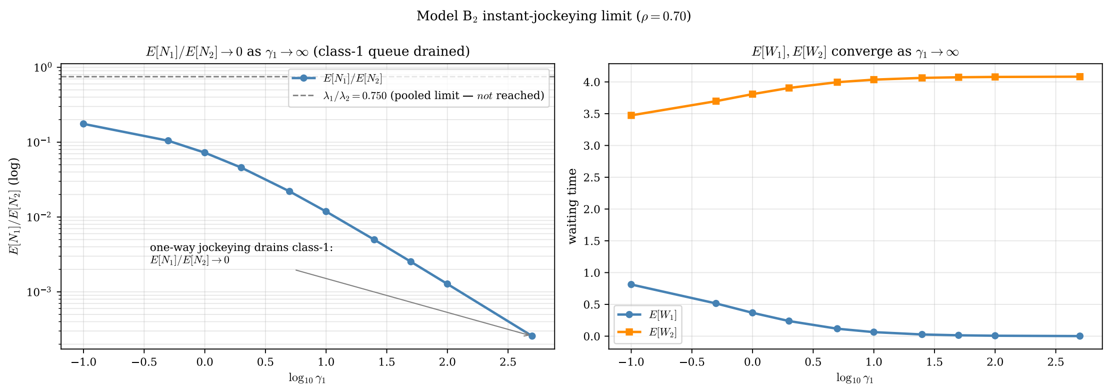

Figure 31: Model-B_2 in the strong-jockeying regime $\gamma_1 \to \infty$ at $\rho = 0.70$. *Left:* $\mathbb{E}[N_1]/\mathbb{E}[N_2] \to 0$ (note the logarithmic ordinate): one-way jockeying is absorbing, so the class-1 queue is drained completely. *Right:* $\mathbb{E}[W_1] \to 0$ while $\mathbb{E}[W_2]$ saturates as $\mathbb{E}[N_2] \to \mathbb{E}[N]_A$.

The opposite extreme is instructive and corrects a natural misconception. One might expect the strong-jockeying limit to balance the two queues in proportion to their arrival rates, $\mathbb{E}[N_1]/\mathbb{E}[N_2] \to \lambda_1/\lambda_2$, as a two-way exchange would. But the B_2 jockeying is *one-way* ($1 \to 2$): it has no return channel, so it is absorbing into class 2. As $\gamma_1 \to \infty$ every class-1 arrival is converted on essentially zero timescale and the class-1 queue is drained, $\mathbb{E}[N_1]/\mathbb{E}[N_2] \to 0$ — the computed ratio falls from 0.1045 at $\gamma_1 = 0.5$ to 3×10^{-4} at $\gamma_1 = 500$, with no sign of levelling at $\lambda_1/\lambda_2 = 0.75$. Because

Table 11: Empirical convergence rates of Models C_2, C_2^1, B_2 and B_2^1 to Model-A, from log-log regression of the deviation against the mechanism parameter (fit over parameter < 1). The primed (head-of-line) models converge at the same first order as their full-rate counterparts.

Limit	Parameter	L_1 slope	π_0 slope	$\pi(0,0)$ slope	Valid range
$C_2 \to A$	$\theta_1 \to 0$	0.847	0.869	0.869	$\theta_1 \in [10^{-3}, 1)$
$C_2^1 \to A$	$\theta_1 \to 0$	0.903	0.914	0.914	$\theta_1 \in [10^{-3}, 1)$
$B_2 \to A$	$\gamma_1 \to 0$	0.873	0.348	0.211	$\gamma_1 \in [10^{-3}, 1)$
$B_2^1 \to A$	$\gamma_1 \to 0$	0.929	n/a	n/a	$\gamma_1 \in [10^{-3}, 1)$

For the jockeying models B_2 and B_2^1, π_0 and $\pi(0,0)$ are γ_1-invariant (jockeying conserves N), so their deviations sit at the numerical floor and the slope is not meaningful (n/a).

jockeying conserves N, the entire occupancy migrates to class 2, $\mathbb{E}[N_2] \to \mathbb{E}[N]_A = 1.6333$, and correspondingly $\mathbb{E}[W_1] \to 0$ while $\mathbb{E}[W_2]$ saturates. The limit confirms that one-way jockeying does not pool the system but rather extinguishes the priority class as a distinct queue.

13.5.4 Head-of-line models B_2^1 and C_2^1

The two head-of-line experiment models must also recover Model-A as their mechanism parameter is switched off: as $\gamma_1 \to 0^+$ (Model-B_2^1) or $\theta_1 \to 0^+$ (Model-C_2^1) the kernel quadratic loses its γ_1/θ_1 shift and the rational PGF of Theorems 5 and 6 collapses to the Model-A form of Theorem 1. Figure 32 verifies both limits numerically.

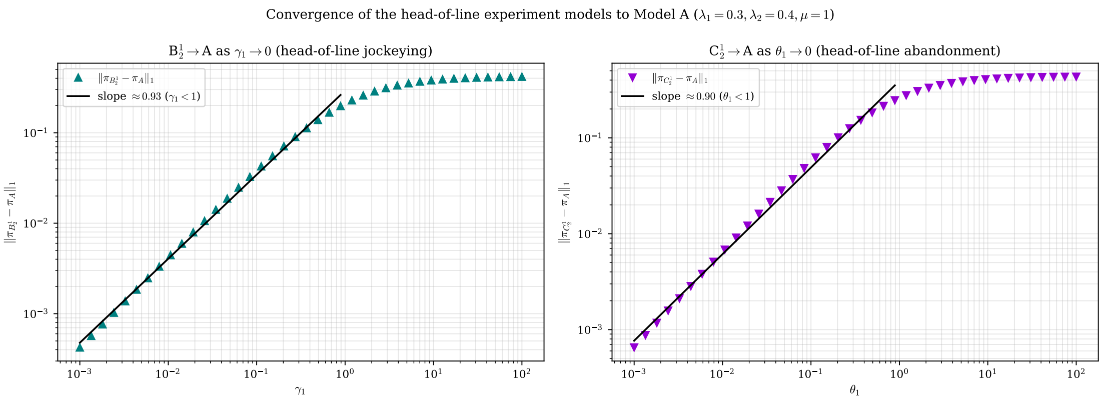

Figure 32: Convergence of the head-of-line experiment models to Model-A ($\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1$). *Left:* log–log plot of $\|\pi_{B_2^1} - \pi_A\|_1$ against γ_1, with the fitted first-order slope. *Right:* $\|\pi_{C_2^1} - \pi_A\|_1$ against θ_1, likewise first-order. Both head-of-line variants approach the baseline at the same order as their full-rate counterparts B_2 and C_2.

The mechanism is the analytic counterpart of the full-rate case: because both $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ and $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$ enter the fundamental equation linearly in the mechanism parameter, the stationary distribution is smooth at the origin and its first-order Taylor term governs the leading deviation. For B_2^1 the conserved descriptors π_0 and $\pi(0,0)$ equal their Model-A values identically for every $\gamma_1 > 0$ (Corollary 6), exactly as for B_2, so the L_1 approach is again carried entirely by the interior redistribution of mass between the two coordinates; for C_2^1 the descriptors π_0 and $\pi(0,0)$ relax onto the baseline at the same first order, since the closed forms (310) are smooth in θ_1 at zero.

Table 11 collects the empirical convergence rates for all four specialised models. Each approaches Model-A at first order in its distinguishing parameter — L_1 slopes of 0.847 (C_2), 0.903 (C_2^1), 0.873 (B_2) and 0.929 (B_2^1), each within the 0.1–0.2 tolerance band of the theoretical exponent 1. The slight shortfall below unity is a finite-truncation and finite-grid artefact of the log–log fit, not a genuine fractional rate, and requires no further investigation. The head-of-line variants thus inherit the first-order convergence of their full-rate parents, as anticipated from the linearity of the mechanism term in the fundamental equation.

13.6 Comparative dashboard

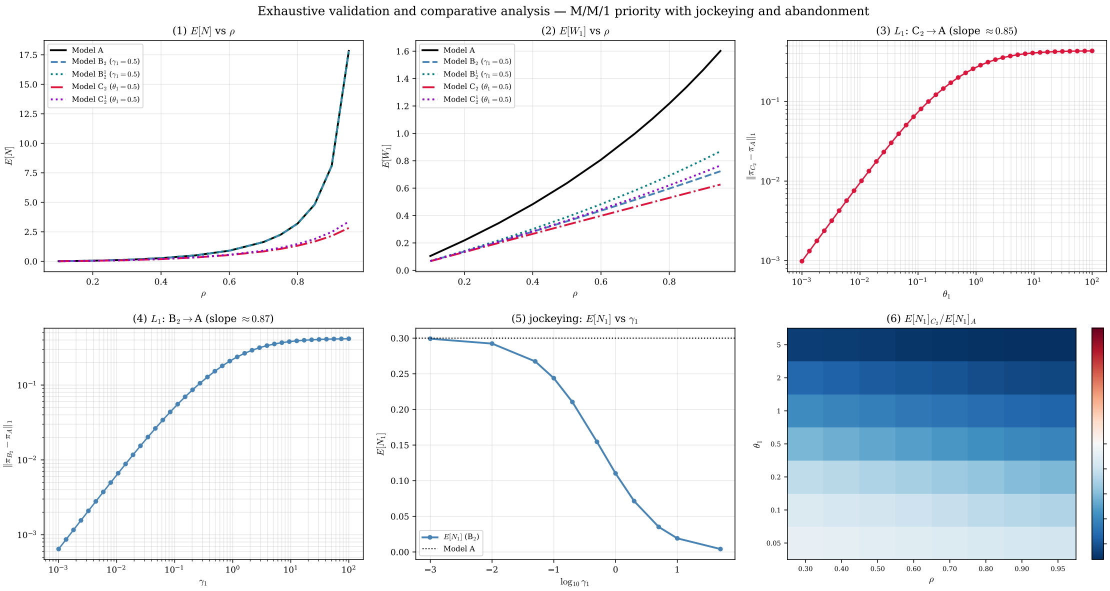

Figure 33: Synoptic dashboard of the numerical study: (1) $\mathbb{E}[N]$ and (2) $\mathbb{E}[W_1]$ versus ρ for all five solved models (A, B_2, B_2^1, C_2, C_2^1); (3) L_1 convergence $C_2 \to A$ and (4) $B_2 \to A$; (5) the jockeying effect on $\mathbb{E}[N_1]$; and (6) the $\mathbb{E}[N_1]_{C_2}/\mathbb{E}[N_1]_A$ heatmap over (ρ, θ_1). Together these panels summarise the two central findings: jockeying redistributes delay at fixed total work, while abandonment reduces total work at a throughput cost.

Figure 33 assembles the six most informative panels of the study in a single view, intended as the frontispiece of the numerical chapter. It restates at a glance the contrast developed analytically in §12: the A and B_2 load curves coincide while C_2 sits below them, the two convergence panels exhibit the common first-order approach to the baseline, and the heatmap localises the regime in which abandonment is most effective as a congestion control.

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